

AUTOMATED VEHICLE SYSTEM POWERED BY SOLAR ENERGY

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FALL 2025 INTAKE

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ABSTRACT

AUTOMATED VEHICLE SYSTEM POWERED BY SOLAR ENERGY

This project presents the design and implementation of an automated electric vehicle system powered by solar energy. The main objective of the project is to improve the sustainability and automation capabilities of small electric vehicles used inside King Abdulaziz University campus. The system combines an autonomous navigation platform with a renewable energy charging system to reduce human operation and improve energy efficiency.

The autonomous system was designed using computer vision, LiDAR sensing, and real-time control algorithms to enable obstacle detection, path following, and automatic stopping functions. The renewable energy subsystem uses a 100W monocrystalline solar panel, an MPPT charge controller, and a 12V Lithium Iron battery to provide efficient solar charging for the vehicle.

The project implementation included both hardware and software integration. ROS2-based navigation and sensor processing were implemented for autonomous operation, while solar charging experiments were conducted to evaluate the charging efficiency of the renewable energy system. Experimental results showed that the MPPT charger achieved an efficiency close to the required target under normal operating conditions, while the autonomous system successfully performed obstacle detection and vehicle control functions.

The final prototype demonstrates the feasibility of integrating autonomous driving technologies with renewable energy systems in low-speed electric vehicles. The project also supports Saudi Arabia Vision 2030 goals in sustainability, renewable energy, and smart transportation systems.

Index Terms — Autonomous Vehicles, Solar Energy, MPPT Charge Controller, LiDAR, ROS2, Electric Vehicles, Renewable Energy, Computer Vision, Nav2.

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CHAPTER – 1 INTRODUCTION

1.1 BACKGROUND

King Abdulaziz University has a fleet of Battery-powered golf carts. Which presents an opportunity for improvements in sustainability and driving systems. This project consists of two main improved systems, the first is an autonomous system that can be implemented on Electrical Vehicles (EV) which serves as a case study of automation and providing a foundation for managing similar assets. The second improved system is the power system of the EV, which includes adding solar energy harvester which will be providing the main power supply. Implementing these improvements will contribute to Saudi Arabia 2030 Vision of renewable energy sources and automations goals [1].

The project requires an understanding of several technical subsystems. The driving subsystem which addresses forward and backward movements, The steering system provides rotation and direction, as well as the energy system that powers the vehicle. It is important to understand these subsystems to ensure safe and reliable operation.

A significant challenge with Autonomous EV is ensuring fully accurate behaviors. This must not be tolerated as the cost of a small error will be enormous as the lives of humans are at stake. The safety risks in automation arise primarily from sensor inaccuracies, algorithmic misjudgments, actuation failures, and cybersecurity vulnerabilities. These risks can lead to incorrect environmental interpretation, delayed or unintended reactions, especially in mixed-traffic scenarios where human-driven and automated vehicles coexist. The authors emphasize that while automation promises reduced human error, it also introduces new technical and ethical risks that must be rigorously managed to ensure safe adoption of autonomous mobility systems [2].

Therefore, this project aims to upgrade KAU's existing electric vehicles into smart, self-driving systems powered by solar energy. By applying basic principles of control systems, power electronics, and automation, the project also aims to support Saudi Arabia's 2030 Vision goals for clean energy and advanced technology. This work will improve the

performance and independence of campus transportation and serve as a model that can be upscaled for future automated electric vehicles in the Kingdom.

1.2 PROBLEM STATEMENT

The problem to be addressed is the need for improvements in EV which will consist of an Automation system and a power system of renewable solar energy. Current electric carts depend on manual control and external charging, reducing efficiency and sustainability. This project aims to develop a self-sustaining automated cart system, powered by solar energy as a prototype that can later be scaled for larger applications such as golf carts and campus vehicles.

1.3 PROJECT OBJECTIVES

This project aims to improve EVs by constructing an autonomous system implemented with solar energy, it will consist of two systems, autonomous system and renewable energy system. The objectives are divided into higher-level strategic objectives, which define the long-term global impact, and lower-level technical objectives, which represent specific, short-term engineering goals.

1.3.1 High Level Objectives

The high-level objectives represent the overall vision of the project for the long term. They include

- Supporting Saudi Arabia 2030 vision by adopting renewable energy systems as the main charging source for the vehicle
- Contributing to Vision 2030 goals by introducing autonomous driving functions that reduce human involvement and improve safety [3].
- Improving KAU's EVs fleet by constructing an Autonomous system and involving a renewable energy source, reducing work power amount & costs to promote resources efficiency.

1.3.2 Lower-Level Objectives

The lower-level objectives are translating the vision into an actionable step. It is divided into two phases.

Autonomous system

- Implement an autonomous navigation system that allows the vehicle to follow predetermined safe routes.
- Incorporate automatic stop functionality that halts the vehicle when an obstacle is detected, ensuring safe autonomous operation
- The EV reliably activates and deactivates the system in response to user inputs.
- The EV movements in left and right directions matches the map outlay

Renewable energy system

- Integrate solar panels as the primary charging system for the battery, ensuring sustainable energy generation suitable for periodic vehicle operation

1.4 PROJECT DESIGN REQUIREMENTS

1.4.1 Design Specifications

For the design specifications, we divided it into two systems since the automation system requires different specifications from the renewable energy system. In the **automation system** we will dive into the specifications related to speed, safety and accuracy which are the main components of an autonomous system, as for the **renewable energy system** we need to mention separate systems values as EV upscaling changes the battery values which change the required renewable energy system specifications to make it work, in this system we will mention EV battery voltage, solar panel rating energy, charger efficiency.

Automation System:

- **Vehicle Speed:** can travel in speed range of **0 – 25 Km/h**
- **Destination Accuracy:** The vehicle should reach the selected destination within a 1-meter radius on the generated map, with an overall accuracy range of 95%–100%.

- **Accurate decisions to stop:** accuracy of decision 100%
- **Reaction time to stop:** reaction time will be in a range of 0 – 0.25 seconds
- **Safety:** EV stops in a range of 3.6 to 5.5 meters from **25 km/h**. It is calculated in the table below, based on the Queensland government's safety rules [4] .

Table 1: calculated stopping distance for each speed after the reaction time

Stopping Distance	
<i>Speed (KM/H)</i>	<i>Calculated Stopping Distance Range (M)</i>
0	0.00 m
5	0.15 – 0.22 m
10	0.59 – 0.88 m
15	1.32 – 1.98 m
20	2.34 – 3.52 m
25	3.66 – 5.49 m

Renewable Energy System:

As mentioned above, the renewable energy system really depends on the EV battery system to begin with, so we made a Table 2 below which has 3 different specifications for this system depending on the EV batteries voltage to cover at least 1 hour of energy per week. The table below will specify the batteries voltage, the solar panel voltage and rating needed to charge the batteries and the charger efficiency of the solar panels charging capabilities.

Table 2: Renewable Energy System, Different Batteries Specifications [5].

System Voltage	Solar Panel Rating	Solar Panel Voltage	Charger Efficiency
12 V DC	100 -150 W	18 V or Greater than	75-85 %
24 V DC	200 - 250 W	36 V or Greater than	85-88 %
48 V DC	300 - 400 W	72 V or Greater than	90-95 %

1.4.2 Constraints

Constraints are necessary in each project to maintain realistic limits and ensure no project scope of creep happens.

Automation System

- **Component Voltage Constraint:** All electronic components must be powered at their specific, datasheet-rated voltages.
- **Processing Constraint:** Control and navigation algorithms must be executed in real time with a total system delay of **less than 1 second**.

- **Sensor Range Constraint:** Obstacle detection sensors must reliably identify objects within at least **6 meters**.
- **Safety Constraint:** The vehicle must automatically stop if a human is detected.
- **Mechanical Constraint:** Added automation hardware (sensors, control units) must not affect the mechanical system.

Renewable Energy System

- **Power Constraint:** The solar system must generate enough energy to fully recharge the battery for at least 1 hour of use per week.
- **Weight Constraint:** The combined weight of the solar panels, battery, and controller must not exceed **20 kg**.
- **Voltage Constraint:** The charging system must match the selected operating voltage (12 V, 24 V, or 48 V DC).
- **Efficiency Constraint:** Solar charging efficiency must remain **above 80%** under standard sunlight conditions.
- **Safety Constraint:** The power circuit must include an emergency shut-off switch.

1.4.3 Engineering Standards

Safety is an essential aspect of any engineering project, and especially in our project of implementing an autonomous system, safety is a must, and we need standards to achieve, to make sure our system meets known engineering standards, so we will use several key engineering standards to guide our design. For any professional autonomous vehicle to be exported to sell, it must ensure the following safety principles of ISO26262 are implemented [6]. Which is a simplified fault-response plan implemented to ensure safe operation in case of sensor or control malfunctions, the vehicle must enter a safe state by stopping the motor, alerting the operator, and waiting for manual reset. Additionally, we will ensure that all electrical enclosures meet the IP26 protection rating to safeguard against solid objects and water intrusion [7]. Finally, as we integrate new upgrades such as the solar panel, we will mount it safely following the best practices from UL 1703 [8]. By combining these standards together, we are ensuring that each component works seamlessly to create a safe and reliable electric golf cart.

1.4.4 Assumptions

In designing projects, we cannot design it to handle all conditions, so we need to assume some things to make the work simpler and focus on most cases, not special cases, and to add scale for the project.

Our assumptions for the project are as follows:

- The EV will be used on KAU campus in a clear path for it without major potholes.
- Existence of the car's main components (motors, controller, batteries)
- Fully operating mechanical system of the car
- The car will use its fully charged cycle once a week.
- EV left outside so Solar panels receive enough sunlight for full weekly battery charging
- Testing and operation will not occur during severe weather conditions such as heavy rain, sandstorms, or prolonged shading.
- The EV autonomous system will be tested during clear daylight hours 8:00 AM – 4:00 PM under sunlight to ensure reliable sensor and camera performance.
- Steering movements to run the autonomous depends on EV
- All system devices will operate under secure and reliable network conditions.
- Normal power supply will be used for testing the autonomous system, while solar power will be used when testing the integrated system.

1.4.5 Project Deliverables

The final project will deliver a fully operational prototype of an **electric vehicle (EV)** equipped with an **autonomous driving system** powered by **solar energy**. The system will combine advanced control, sensing, and renewable energy technologies to demonstrate a safe, efficient, and sustainable mode of transport. The project outcomes focus on both the technical performance and reliability of the integrated systems, ensuring that the vehicle operates autonomously and maintains full battery charge through solar power. The deliverables consist of:

- Solar charging system with panels, charge controller, and battery pack.
- An autonomous driving system using sensors and AI to stop when facing humans and avoid static obstacles.
- Final report, presentation, and demonstration video documenting design and results.

CHAPTER – 2 LITERATURE REVIEW

2. LITERATURE REVIEW

This chapter reviews previous studies and technologies relevant to enhancement of electric vehicles, with a particular focus on automation and solar energy integration. These two domains represent the core aspects of the project's performance upgrade phase.

The automation section explores existing research and development in autonomous driving technologies, including sensor integration, path following algorithms, and obstacle detection methods applied to small scale electric vehicles and golf carts.

The solar panel section reviews literature on renewable energy utilization in EVs, and the practical feasibility of integrating solar panels into low-speed electric vehicles. This section emphasizes the potential of solar power charging to extend driving range and promote environmental sustainability.

2.2. Automation

2.2.1. Control System Integration for an Electric Golf Cart using RC

This study presents a comprehensive approach to converting an electric golf cart into an autonomous test platform designed for educational and research purposes. Conducted at Eastern Michigan University, the project focuses on integrating control systems, sensors, and actuators to enable partial and later full automation of a Club Car golf cart.

The research emphasizes the integration of multiple subsystems through a PID control system to achieve precise digital control over steering, braking, and throttle mechanisms. The automation process began by mounting servo motors to replicate driver inputs mechanically. Specifically, the Teknic integrated servo motors were used to control the steering and braking systems, while an MCOR4 throttle potentiometer was used for acceleration control. These components were interfaced through an Arduino Uno microcontroller, which received analog signals from a remote controller system (RC), processed them, and generated digital Pulse Width Modulation (PWM) outputs to drive the servos.

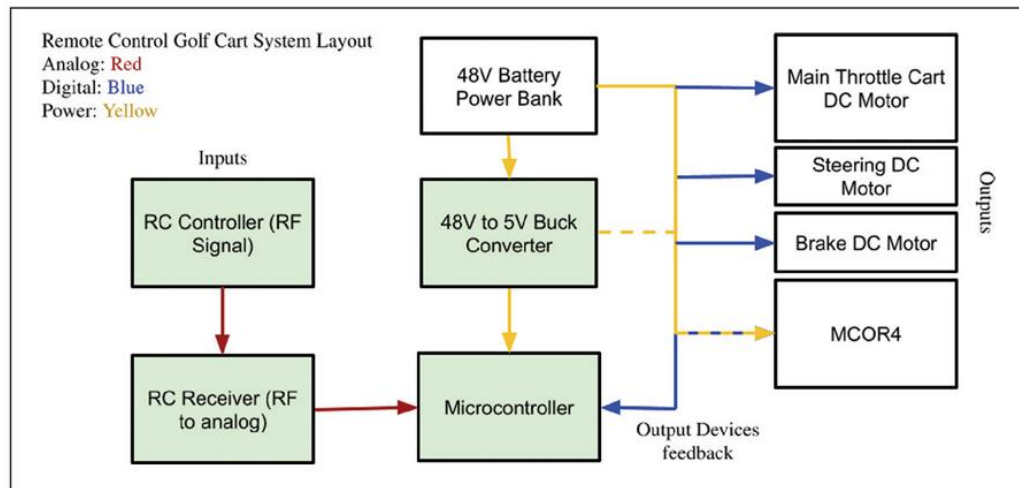


Figure 1 - integration between sensors, microcontroller, servos, and power systems [9] .

To address practical limitations, the researchers simulated the control algorithms in a virtual environment before implementation. These simulations enabled testing of analog-to-digital communication, servo feedback, and PID tuning, ensuring stable system response prior to real-world deployment. The project also incorporated remote control capability as an intermediary phase between manual operation and full autonomy, preparing the foundation for future Level 4 autonomous functionality as defined by the SAE. [9]

2.2.2. Improvement of the break and steering system for the golf-cart

This project enhanced a golf cart's steering, braking, and throttle systems to prepare for autonomous operation. The team used two Arduino Mega controllers, one managing steering and the other braking and throttle, and added an Arduino Uno for emergency braking. Each subsystem used feedback loops with sensors and encoders to improve control precision and response time. The steering motor was re-positioned directly on the steering shaft to reduce mechanical play, and the MCOR4 electronic throttle was used for smoother acceleration control. Although the system remained remotely operated, it successfully achieved semi-autonomous actuation (Level 2 automation) and established a safe, modular control foundation for future autonomous integration [10].

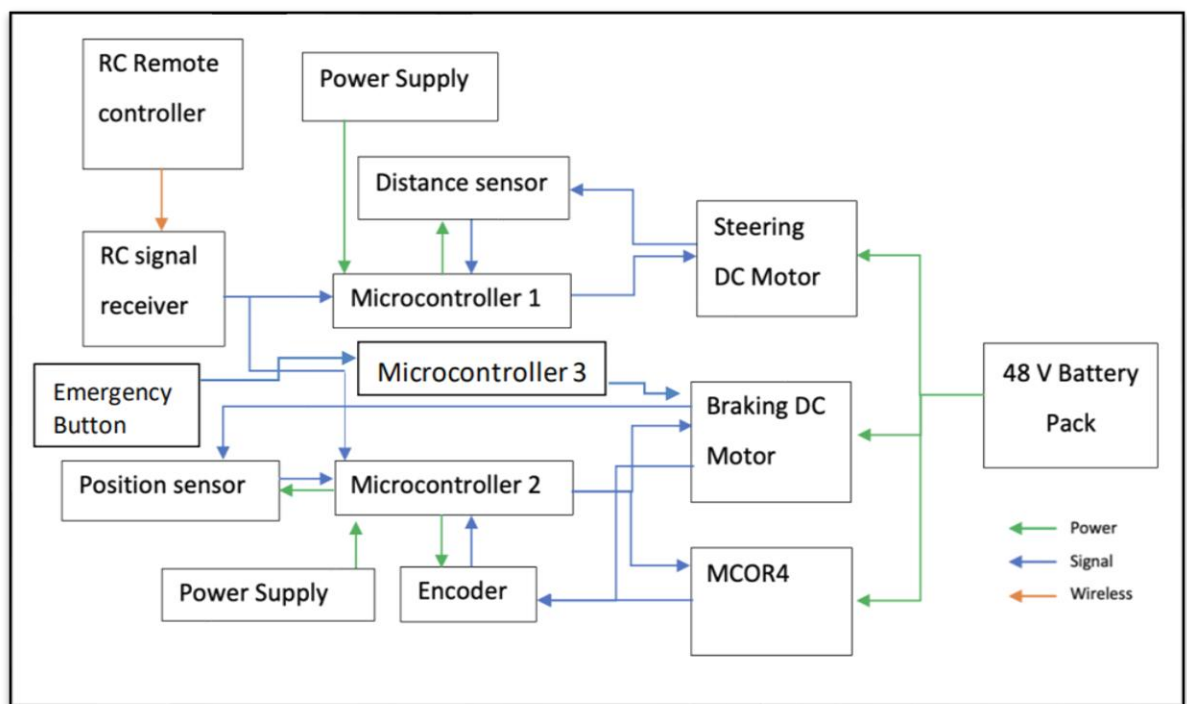


Figure 2 - Concept Design of the Control System Using Three Microcontroller [10].

2.2.3. Development A Self Driving Golf Cart Using Kinect Sensor (ROS)

This study developed a fully autonomous golf cart that leveraged the Microsoft Kinect v2 sensor for environmental perception and the Robot Operating System (ROS) as the control framework. The Kinect provided 3D depth data and RGB images for obstacle detection and distance estimation, processed using OpenCV and ROS nodes to construct a real-time map of the environment. The control architecture integrated path planning, localization, and obstacle avoidance modules, allowing the golf cart to follow predefined routes without human input. Motor actuation was implemented through an Arduino microcontroller connected to ROS via a serial interface, translating ROS commands into PWM signals for speed and steering control. The prototype successfully navigated indoor and outdoor test tracks, proving the feasibility of low-cost autonomous navigation using Kinect and ROS [11].

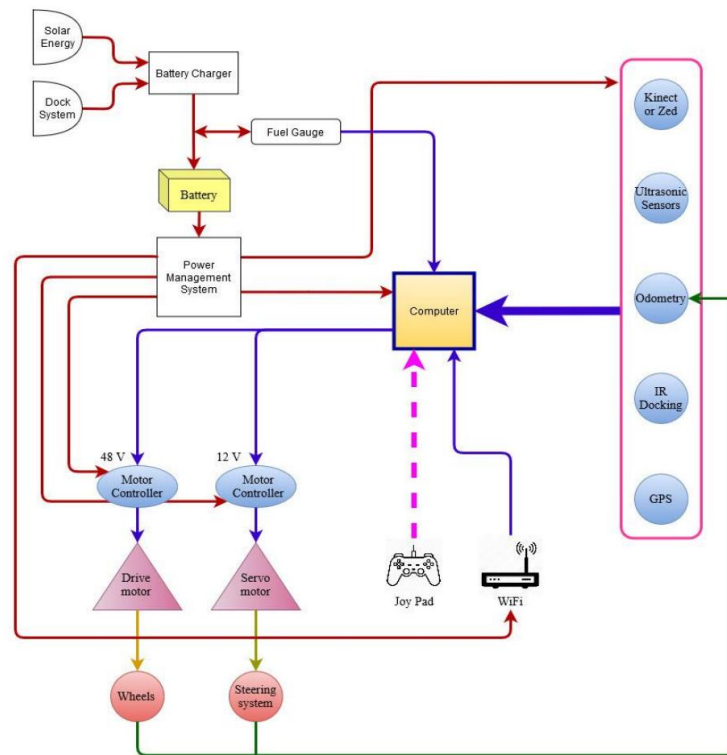


Figure 3 - High Level diagram of hardware

2.2.4. Sensor-Fusion for Autonomous Navigation using ArduPilot and ROS

This study proposes a robust autonomous navigation system for an outdoor vehicle, specifically an orchard spraying robot. The project's core challenge is reliable navigation and obstacle avoidance in complex environments where GPS may be unreliable. The paper's solution is a "companion computer" architecture that fuses data from multiple sensors to achieve a high degree of accuracy and safety [12].

The system architecture is a perfect example of a co-pilot design. It uses a Pixhawk 4 flight controller, which runs the ArduPilot (ArduRover) firmware, to handle all real-time vehicle control. A Raspberry Pi 4B is used as the high-level companion computer, which runs the Robot Operating System (ROS). This architecture allows the Raspberry Pi to perform computationally expensive tasks, like sensor fusion, while the Pixhawk reliably executes the low-level driving and steering commands.

For perception and localization, the system does not rely on a single sensor. It fuses data from a 2D LiDAR for reliable obstacle detection and a T265 visual odometry camera to track the vehicle's position relative to the ground. This data is processed on the Raspberry Pi using an Extended Kalman Filter (EKF), allowing the robot to maintain a precise position estimate even in areas where GPS is lost. This study demonstrates that a combined ArduPilot and ROS system is a powerful and feasible solution for autonomous path planning and real-time obstacle avoidance.

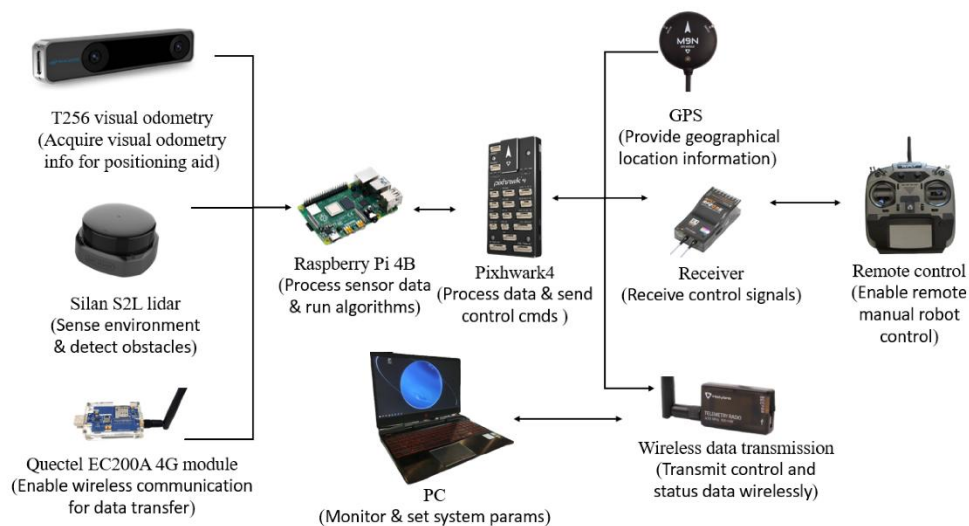


Figure 4 - Block diagram of system hardware

2.3. Solar Panel

2.3.1. Component Selection and System Architecture for Solar Charging

This guide reviews the fundamentals of solar panels, identifying four essential components: the solar panel, the battery, a charge controller, and an inverter. It differentiates between panel technologies (monocrystalline and polycrystalline). A central focus is placed on the charge controller, which is categorized into two main types: Pulse Width Modulation (PWM) and Maximum Power Point Tracking (MPPT). The MPPT controller is identified as the more advanced solution, offering up to 30% greater efficiency by converting excess panel voltage into additional charging current, thereby maximizing the energy harvested [13].

2.3.2. Implementation and Safety of a 12V Solar Charging Circuit

A 12V solar charging system is built around a solar charge controller, which connects the solar panel to the battery [14]. A panel's power rating, like 100W, is based on perfect lab conditions that are almost never met in the real world [15]. Factors like the panel's angle, heat, and shade lower its performance. This means a 100W panel will likely only produce about 300 to 600 watt-hours of energy in a normal day [15]. It is important to use this realistic number when designing a solar setup. The charge controller is also a key safety device. It watches the battery's voltage and automatically cuts off the power when the battery is full. This "auto cut-off" feature is essential because it prevents overcharging, which can permanently damage the battery [14].

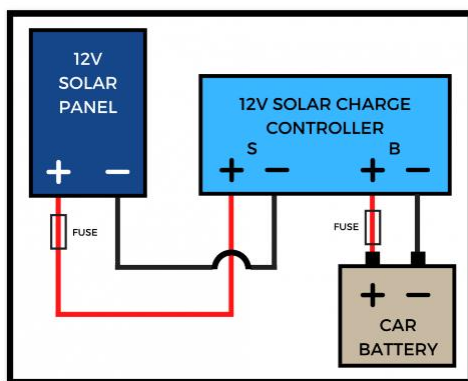


Figure 5 - Block diagram of the circuit



Figure 6 - Block Diagram of Circuit

CHAPTER – 3 PROJECT DESIGN

3.1. ALTERNATIVE DESIGNS

To select the best path forward, we will now propose three distinct alternative designs. The table below serves as an alternative matrix, summarizing the various technologies we researched for Power and Autonomous subsystems. The three alternatives we present are logical combinations of these components. A design is not limited to one item per row; for instance, an autonomous system might combine multiple sensors like LiDAR and GPS to achieve its goal.

It is important to note that this section focuses on the high-level system architecture (the types of components and how they connect), not on specific component specifications. Our goal is to design a system that is scalable and can be applied to different vehicles. The exact part values, such as the battery voltage or solar panel wattage, will be defined later in the 3.2 Baseline Design section, where we will tailor the architecture to our specific project vehicle.

Table 3 - Alternative Matrix

<i>Systems</i>	<i>Subsystems</i>	<i>Alternative 1</i>	<i>Alternative 2</i>	<i>Alternative 3</i>	<i>Alternative 4</i>	<i>Alternative 5</i>
Power system	Renewable energy	Flexible Solar panels	Sturdy Solar panels	Regenerative Brakes	-	-
	Battery	Lithium Iron Phosphate (LiFePO ₄) Batteries	Absorbed Glass Mat (AGM) Batteries	Lead-Acid Batteries	-	-
	Charge Controller	MPPT	PWM	-	-	-
Autonomous System	Microcontroller	Arduino Uno	ESP32	Raspberry pi pico	Arduino Mega	Pixhawk
	Single-Board Computer	Raspberry Pi	NVIDIA Jetson Nano	NVIDIA Orin	-	-
	Autonomous Behavior	Follows specific tracks or markers	Waypoints/ GPS navigation	-	-	-
	Software	AI/Computer Vision	ROS	Arduover	-	-
	Camera	Digital Cameras	Depth Cameras	Thermal Cameras	-	-
	Sensors	LiDAR	Ultrasonic	GPS	IR sensor	IMU sensor

3.1.1. Alternative 1:

Alternative 1 is a high-performance, AI-driven platform. The design is built on two core subsystems: an efficient, 100% solar-based Power System and an advanced Autonomous System. This architecture uses a powerful companion computer running the Robot Operating System (ROS) to handle AI and perception, which then sends commands to a dedicated, real-time autopilot for vehicle control.

The primary autonomous processing is handled by the NVIDIA Jetson Nano, which runs the main ROS core. It fuses data from the LiDAR and Digital Camera to perform AI-based object classification and 3D path planning. This smart brain sends navigation commands to the Pixhawk, which acts as the low-level, real-time pilot that safely controls the vehicle's steering and driving mechanisms. The power system is based on a high-efficiency MPPT controller and LiFePO₄ batteries to meet the high-power demands of the Jetson Nano and ensure complete solar autonomy.

Hardware Components:

- Flexible Monocrystalline Solar Panels
- Lithium Iron Phosphate (LiFePO₄) Batteries
- MPPT Solar Charge Controller
- NVIDIA Jetson Nano
- Pixhawk
- LiDAR Sensor
- Digital Camera
- GPS Module

3.1.1.1. Power System:

This Power System is designed to make the vehicle entirely self-sufficient, relying completely on solar energy. It features flexible monocrystalline solar panels connected in series to an MPPT Solar Charge Controller. The use of MPPT technology is critical for maximizing energy harvesting from the panels in all light conditions. The controller manages the flow of energy to a bank of the batteries. This advanced battery technology is chosen over lead-acid because its high charging efficiency, low weight, and long cycle life are essential for reliably powering the high-demand autonomous components, such as the NVIDIA Jetson Nano.

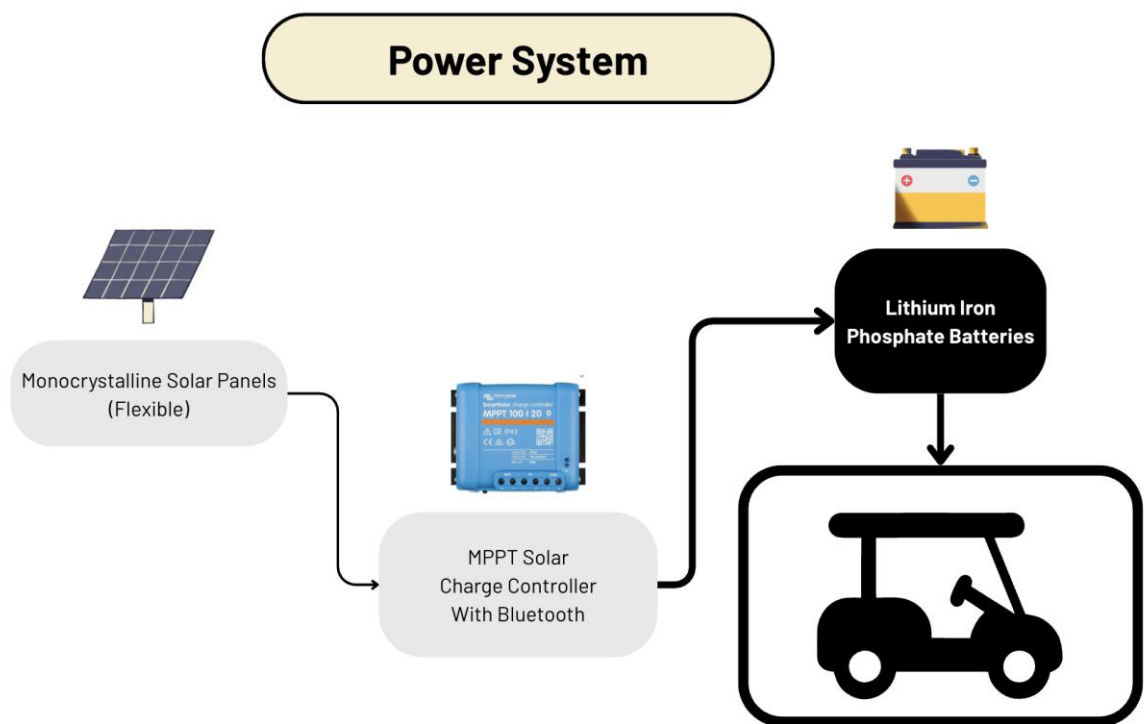


Figure 7 - Power System for Alternatives 1

3.1.1.2. Autonomous System:

The Autonomous System is a high-performance, AI-driven solution based on the Robot Operating System (ROS). This architecture uses two brains: a main computer for thinking and a real- controller for acting.

The primary brain is the NVIDIA Jetson Nano, which runs the main ROS core. It serves as the co-pilot, handling all complex perception tasks. It receives data from the Digital Camera to run AI-based computer vision models like object classification and data from the LiDAR sensor to build a 3D map and detect obstacles.

The Jetson Nano fuses this sensor data to make smart navigation decisions, which it sends as simple commands to the Pixhawk. The Pixhawk acts as a real-time pilot and low-level controller. It receives its own GPS data for precise global localization and translates the Jetson's commands into PWM signals that operate the vehicle's steering and driving/braking mechanisms.

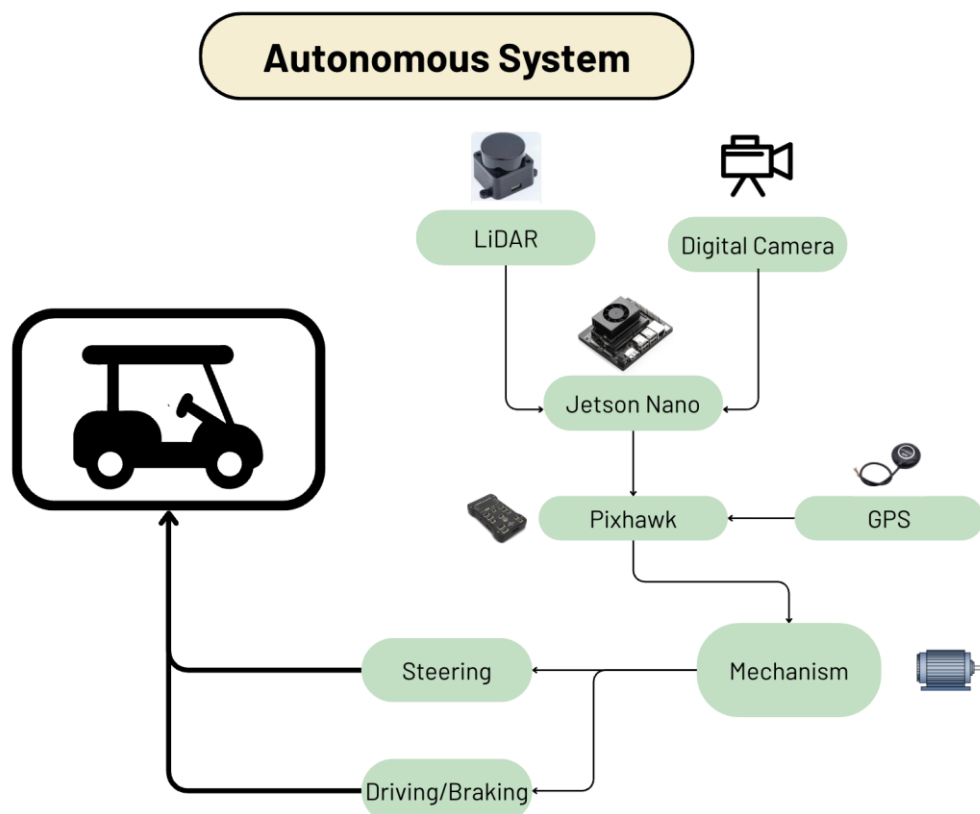


Figure 8 - Autonomous System for Alternative 1

3.1.1.3. Completed Block Diagram

As shown in the block diagram, this architecture is powered by a Lithium Iron Phosphate (LiFePO₄) battery, which is charged exclusively by the Solar Power System (MPPT Controller). The battery powers the Autonomous System, which uses the Jetson Nano running ROS, processing data from the LiDAR and Digital Camera to think. It then sends navigation commands to the Pixhawk, which acts as the pilot, using its own dedicated GPS to execute real-time control over the vehicle's steering and driving mechanisms.

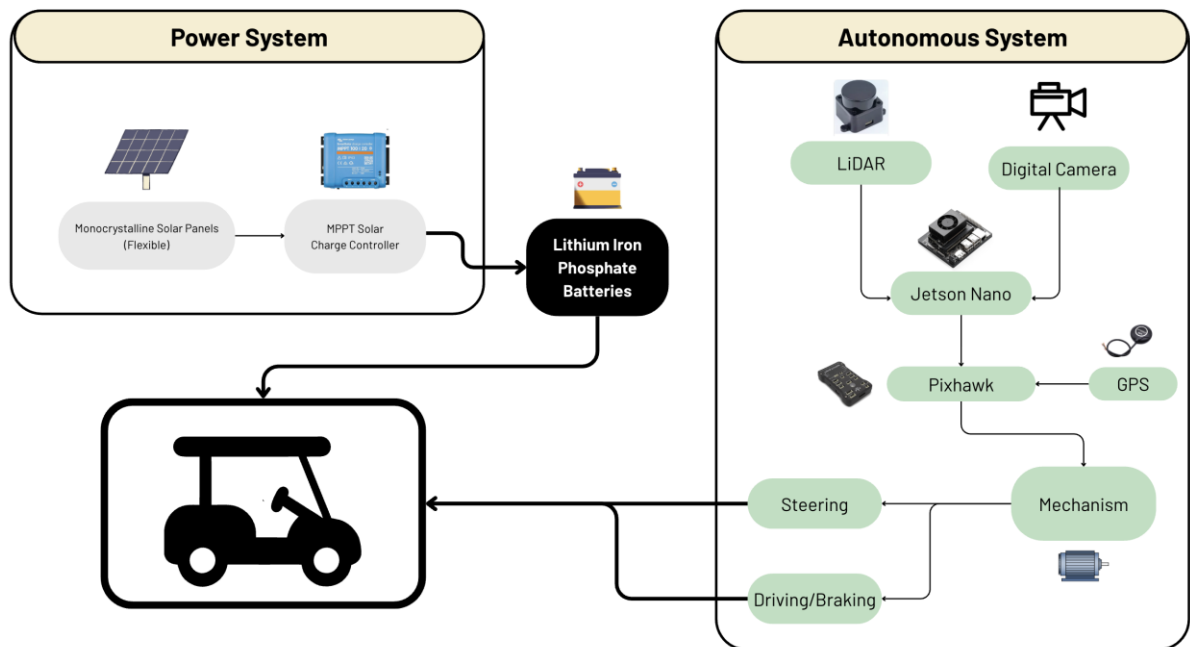


Figure 9 - Completed Block Diagram for Alternative 1

3.1.1.4. Cost Estimation

The table below shows the main components used in this design and their estimated prices. The price was categorized in ranges to satisfy all different power systems, the total cost of this design is 4030SAR, which makes it an expensive option that meets the project goals and has high performance and safety, as well as this system being the most used one in automotive movement in places such as robotics, factories and cars.

Table 4 - Cost Estimation for Alternative 1

<i>Component</i>	<i>Price range (SAR)</i>	<i>Image</i>
Flexible Solar Panel Monocrystalline	335 – 1300	
MPPT	80 – 300	
LiFePO4 Lithium Battery	360	
Pixhawk	250	
M8N GPS	100	
RPLIDAR A1	380	

NVIDIA Jetson Orin Nano Developer Kit	1100	
25Kg Steel Servo	50	
USB Camera	190	
Total	4030	

3.1.2. Alternative 2:

Alternative 2 is a balanced design that achieves advanced autonomous features without the high complexity of ROS. It is built on two subsystems: a cost-effective, off-grid Power System and a companion computer Autonomous System.

This architecture uses a Pixhawk as the real-time pilot to handle all driving and navigation. A Raspberry Pi is used as the co-pilot, running simple Python and OpenCV scripts to process sensor data. This design is a trade-off, using a low-cost Lead-Acid battery to reduce the project's total cost, while still using a high-efficiency MPPT controller and LiDAR to ensure the solar and safety systems are reliable.

Hardware Components:

- Flexible Monocrystalline Solar Panels
- Lead-Acid Batteries
- MPPT Solar Charge Controller
- Raspberry Pi

- Pixhawk
- LiDAR Sensor
- Digital Camera
- GPS Module

3.1.2.1. Power System:

This Power System is designed to be a cost-effective, off-grid solution. It uses flexible monocrystalline solar panels connected in series to an MPPT Charge Controller. The use of an MPPT controller is a critical design choice to maximize the energy harvested from the panels, which is necessary for a 100% solar-powered vehicle. To reduce the total project cost, the system stores its energy in the Lead-Acid batteries. While less efficient and heavier than lithium, this battery technology is reliable for a low-frequency use-case.

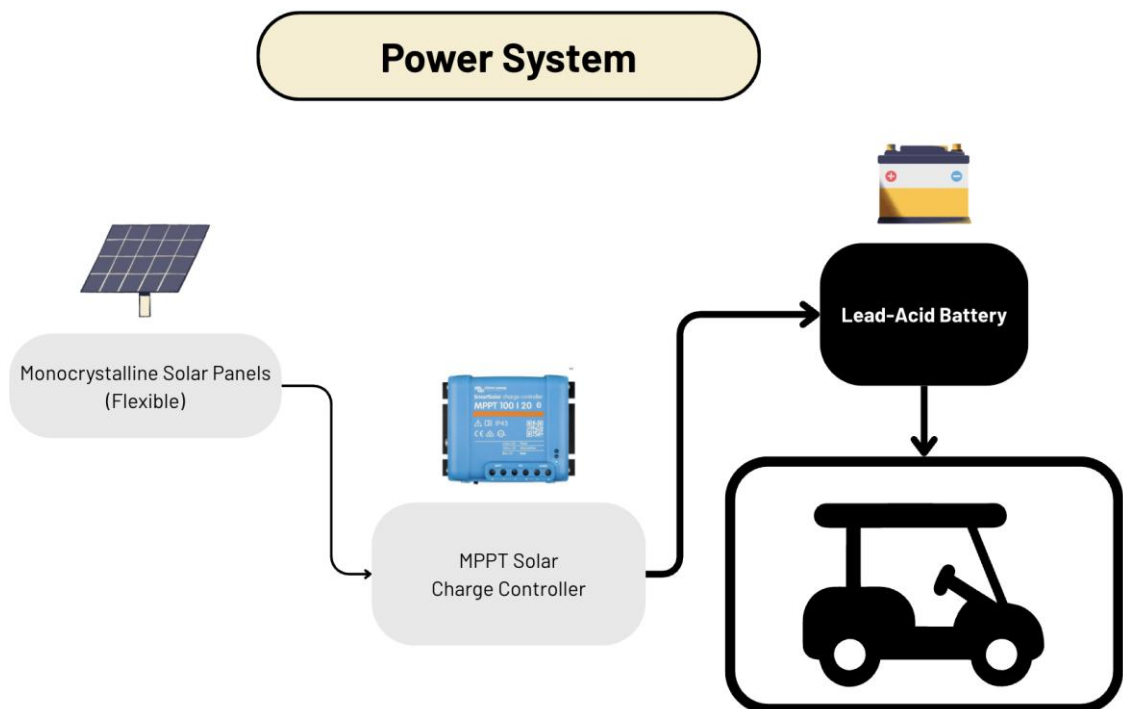


Figure 10 - Power System for Alternatives 2

3.1.2.2. Autonomous System:

The Autonomous System is designed using a companion computer architecture, which separates the high-level thinking from the real-time driving. The Pixhawk, running ArduRover firmware, acts as the primary pilot. It connects directly to the GPS module and its sole mission is to reliably follow its pre-loaded GPS waypoint path. The Raspberry Pi acts as the co-pilot, or smart brain, handling all advanced perception. The LiDAR and Digital Camera are connected to the Raspberry Pi, which runs custom Python scripts with OpenCV to perform sensor fusion and classification. This allows the system to execute an intelligent safety logic: when the LiDAR detects an object, the Raspberry Pi uses the camera to classify it. If the object is identified as a human, the Pi sends an immediate stop command to the Pixhawk for maximum safety. If the object is identified as static, the Pi's script calculates a short detour path and sends temporary steering override commands to the Pixhawk to navigate around the obstacle before resuming the main GPS mission.

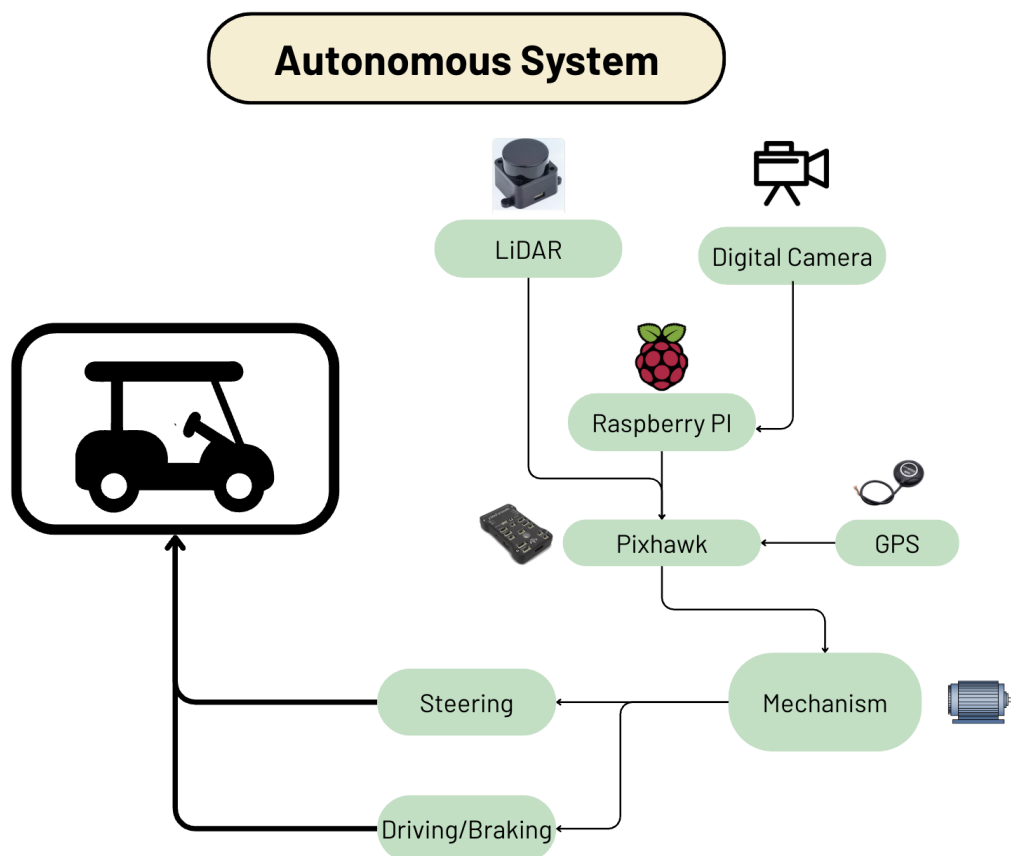


Figure 11 - Autonomous System for Alternative 2

3.1.2.3. Completed Block Diagram

As the block diagram illustrates, this architecture is powered by a Lead-Acid battery, which is charged exclusively by the Solar Power System via an MPPT controller. The battery provides power to the entire Autonomous System. This system is built on a companion computer model: the Raspberry Pi acts as the high-level brain, using the LiDAR and Digital Camera to run Python scripts for obstacle classification. It then sends smart navigation commands to the Pixhawk, which functions as the real-time pilot. The Pixhawk uses its own dedicated GPS to follow the primary waypoint path and execute control over the vehicle's steering and driving mechanisms.

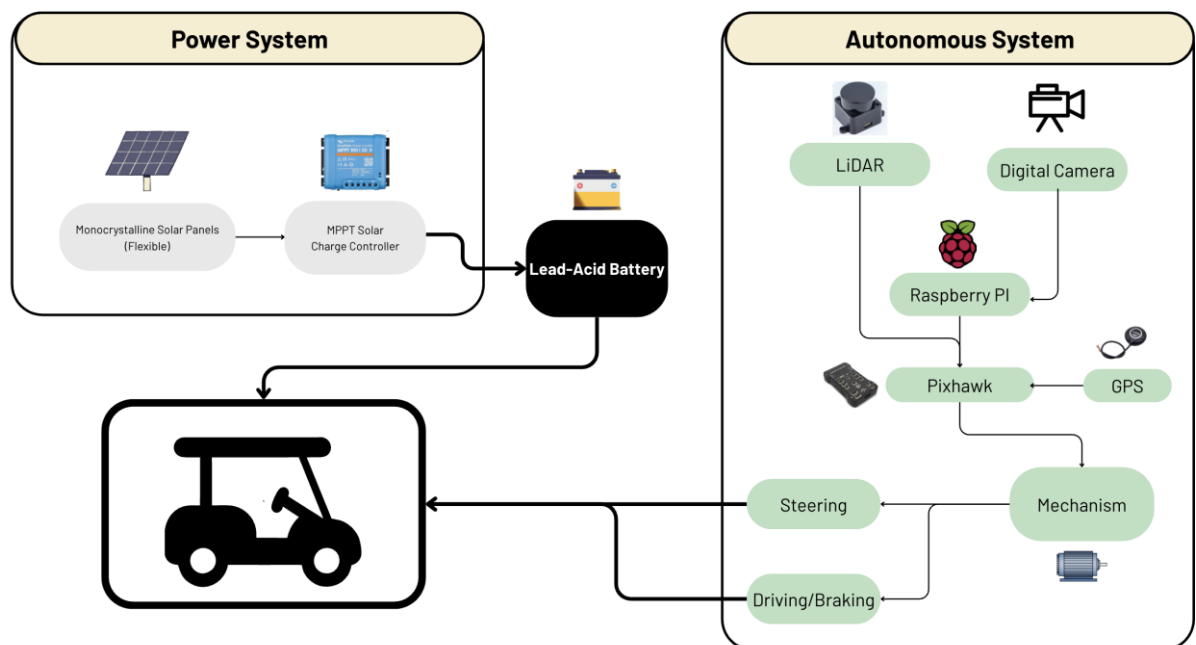


Figure 12 - Completed Block Diagram for Alternative 2

3.1.2.4. Cost Estimation

The table below shows the main components used in this design and their estimated prices. The price was categorized in ranges to satisfy all different power systems. The total cost of this design is 3470 SAR, which makes it an affordable option that meets the project goals and has moderate performance and safety.

Table 5 - Cost Estimation for Alternative 2

<i>Component</i>	<i>Price range (SAR)</i>	<i>Image</i>
Flexible Solar Panel Monocrystalline	335 – 1300	
MPPT	80 – 300	
Lead-Acid Battery	100 – 300	
Pixhawk	250	
M8N GPS	100	

RPLIDAR A1	380	
Raspberry PI	400 – 600	
Digital Camera	190	
25Kg Steel Servo	50	
Total	3470	

3.1.3. Alternative 3:

Alternative 3 is the low-cost design. This solution is engineered to meet the autonomous and solar-power goals using the most cost-effective and simple components available. The entire system is controlled by a single Arduino Mega microcontroller, which is responsible for running all navigation, safety, and control code written by the team.

This design uses GPS waypoint navigation to follow predetermined paths and an array of Ultrasonic sensors for basic obstacle detection. To minimize the budget, the power system uses a simple PWM charge controller and standard Lead-Acid batteries.

Hardware Components:

- Flexible Monocrystalline Solar Panel
- Lead-Acid Batteries
- PWM Solar Charge Controller

- Arduino Mega
- GPS Module
- Ultrasonic Sensors

3.1.3.1. Power System:

This Power System is designed to be the lowest-cost solution for achieving 100% solar charging. It uses flexible monocrystalline solar panels, which feed into a PWM Solar Charge Controller. Unlike an MPPT controller, the PWM version is a simpler device and is less efficient at harvesting energy, especially in variable light. This controller charges the Lead-Acid batteries. While this power system is very heavy and has significant efficiency losses in both the controller and the batteries, it meets the 100% solar requirement at a minimal budget.

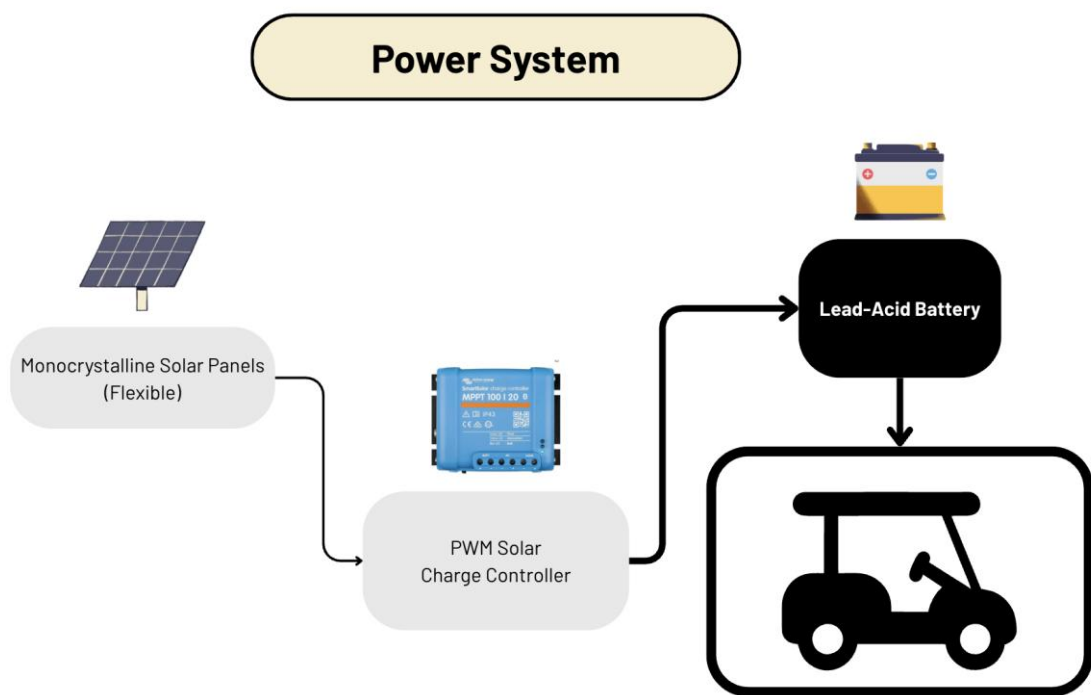


Figure 13 - Power System for Alternatives 3

3.1.3.2. Autonomous System:

The Autonomous System is a solution controlled entirely by an Arduino Mega. This single microcontroller is responsible for all real-time navigation and safety logic, which must be programmed by the team. To follow a predetermined path, the system uses a standard GPS module to get its location. The Arduino's code must continuously fuse this data and calculate the correct steering angle to aim the vehicle toward the next waypoint in its stored list. For safety, an array of Ultrasonic sensors is used for basic obstacle detection. If an object is detected within the predefined range, the Arduino immediately commands the vehicle's mechanisms to stop.

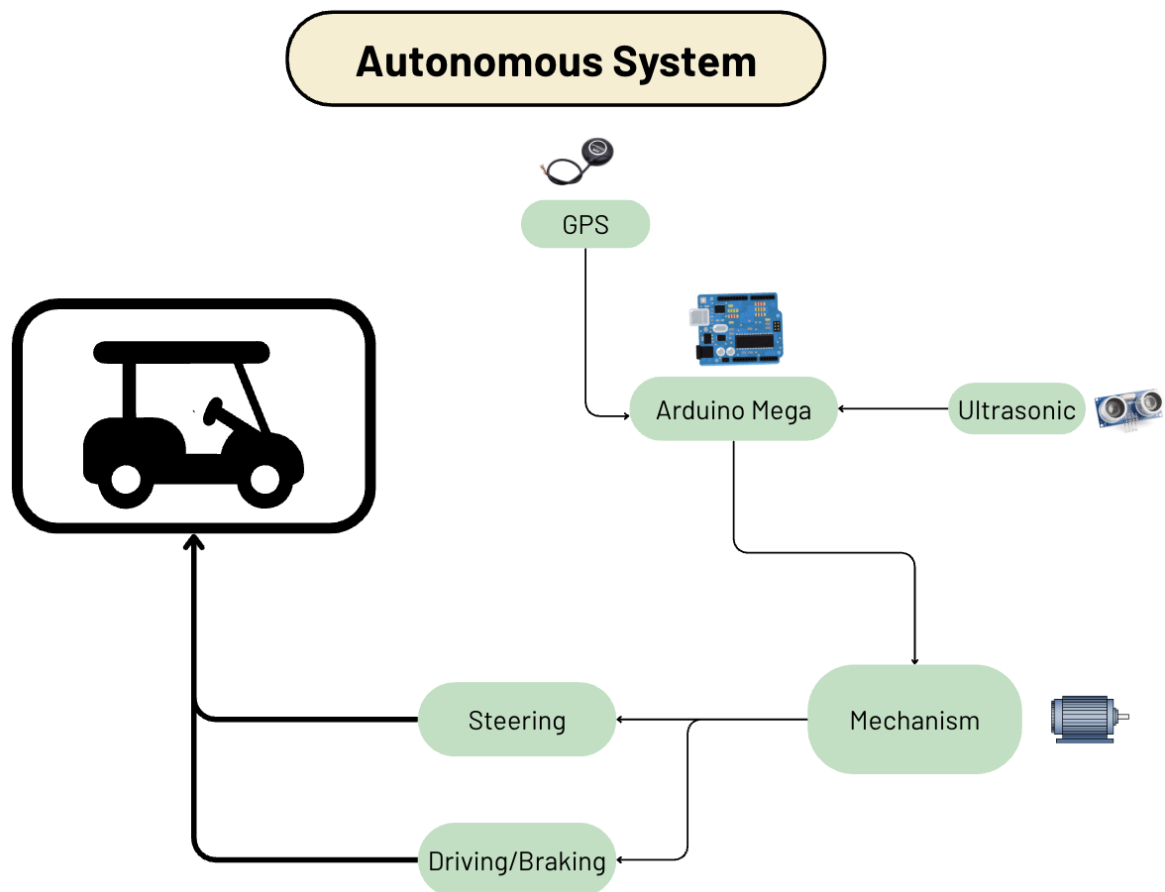


Figure 14 - Autonomous System For Alternative 3

3.1.3.3. Completed Block Diagram

As the block diagram shows, this is the simplest architecture. The Power System uses a PWM controller to charge the Lead-Acid battery from the solar panels. This battery then powers both the main vehicle drive and the entire Autonomous System. This autonomous system is centralized on a single Arduino Mega, which acts as the only brain. The Arduino reads data from the GPS to navigate its path and from the Ultrasonic sensors for safety, directly controlling the vehicle's steering and driving mechanisms.

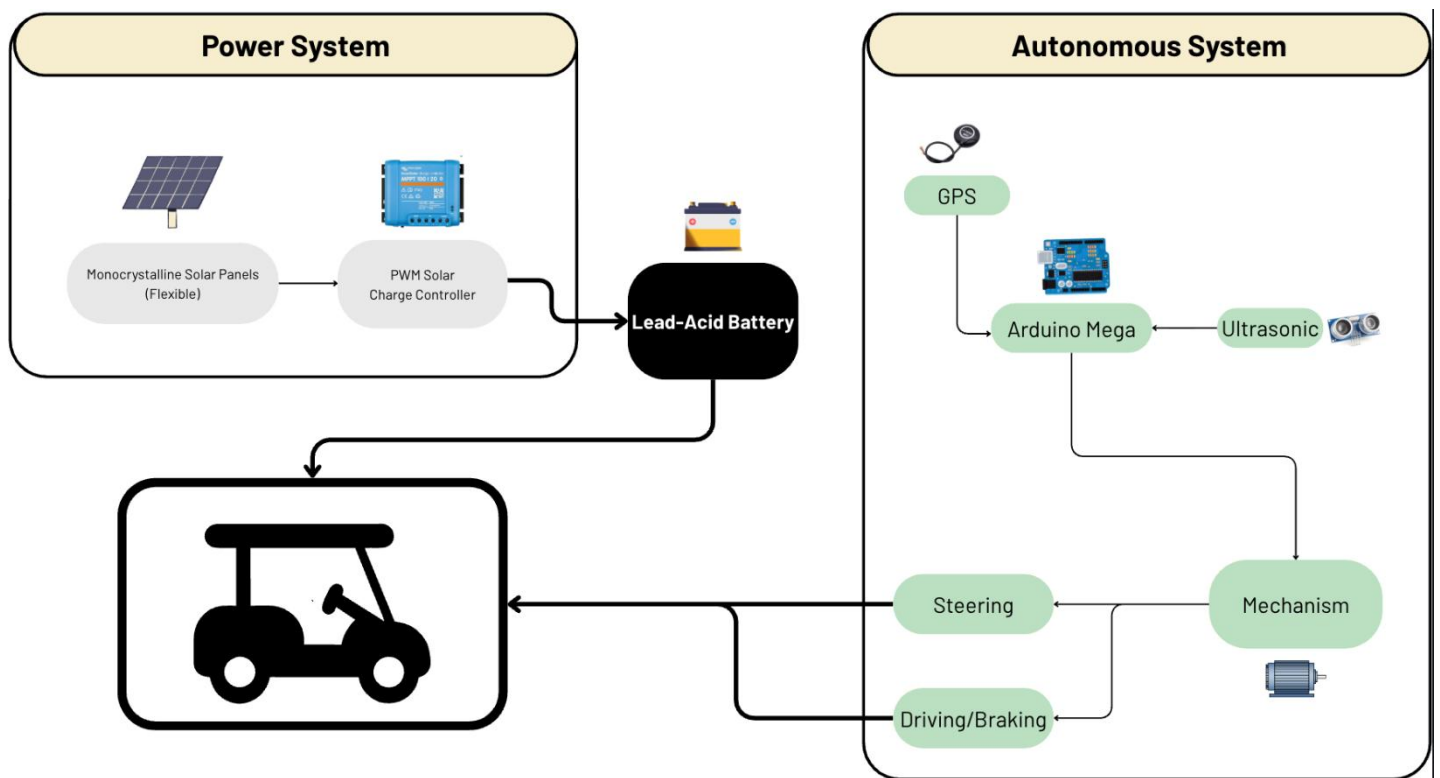


Figure 15 - Completed Block Diagram for Alternative 3

3.1.3.4. Cost Estimation

The table below shows the main components used in this design and their estimated prices. The price was categorized in ranges to satisfy all different power systems. The total cost of this design is 605 SAR, which makes it an affordable and cheap option that still meets the project goals of autonomous and solar energy although it lacks very much in stability and accuracy of autonomous movement. Which might cause safety issues.

Table 6 - Cost Estimation for Alternative 3

<i>Component</i>	<i>Price range (SAR</i>	<i>Photo</i>
Solar Panels	335 – 1300	
PWM Charging controller	30	
Lead-Acid Battery	100	
Arduino mega	80	
GPS Module	25	
Ultrasonic Sensor	15	
Motor Driver	20	
TOTAL	605	

3.1.4. Advantages and Disadvantages

Table 7 - Advantages and Disadvantages

Alternatives	Advantages	Disadvantages
Alternative 1	• Highest Performance Potential • Most "Intelligent" System • Industry-Standard Software	• Difficult to implement • Highest Financial Cost
Alternative 2	• Lower cost than AI-based system • Balanced complexity vs. performance • Most Efficient Power System	• GPS & Vision Limitations • Slow Reaction Time
Alternative 3	• Lowest financial cost. • Sensors (GPS/Ultrasonic) are not blinded by sunlight. • Very easy to implement	• Low accuracy • Unsafe and unreliable • Inefficient Power system • Programming takes too much time.

3.1.5. Baseline Selection

After evaluating all three alternatives, **Alternative 1** was selected as the baseline design. This choice prioritizes performance, reliability, and safety over initial cost and complexity. The architecture, which integrates the NVIDIA Jetson Nano with the Robot Operating System (ROS), offers the most capable and scalable solution. While other options are more affordable, they lack true AI-driven perception. Alternative 1 enables advanced computer vision, allowing the system to not only detect but also classify obstacles, distinguishing, for instance, between a pedestrian and a static object. This intelligent capability ensures safer autonomous behavior and offers a flexible platform that can be easily scaled for larger applications, making Alternative 1 the optimal choice for our design.

3.2. BASELINE DESIGN

After selecting Alternative 1 from the three different designs above, we are now tasked to create the baseline design based on that choice. Our goal is to make a prototype that meets the project objectives and specifications of Alternative 1. To achieve this, we selected an EV with a **12V battery system**, which we will apply our baseline design onto to form the prototype and later evaluate its performance and record the results in detail.

We are tasked to specify each hardware component of the design with their specified datasheet information's that will help us in formatting the prototype, then we take the alternative design block diagram and expand it to include more details while also explaining each blocks exact job in the baseline design, moving forward to the circuits schematic which will show the entire components with each connection including the wiring details, then we proceeds to workflow of our design which will explains how each part in the design communicate to achieve the project goals either by power, control or logical flow. then moving forward to the inputs and outputs section of the baseline design which will cover the higher-level view of the prototype as in what can be expected of inputs to the prototype to reach certain desired outputs, finishing off with relevant engineering standards and the cost estimation for this baseline design, describing what engineering standards are met and how did we implement them in our baseline design after developing the circuits, block diagrams, flowchart and explaining in details how each hardware components works. Reaching the final point of estimating a budget for our design including the detailed hardware components prices, that will be bought to engineer the prototype.

After this chapter we will start the implementation phase to test our prototype.

3.2.1. Hardware Specification

1- Solar Panel

Table 8 - Solar Panel Specification

<i>Part name</i>	<i>Solar panel</i>	
Type	Flexible Monocrystalline	
Nominal power	100 W	
Nominal voltage	12 V system (≈ 18 V open circuit)	
Dimension	$1010 \times 670 \times 3$ mm	
Weight	1.5kg	

2- Charge Controller

Table 9 - Charge Controller Specification

<i>Part name</i>	<i>MPPT 75 15</i>	
Max PV open-circuit voltage	75V	
Battery voltage	12/24V	
Rated charge current	15 A	
Dimension	$100 \times 113 \times 40$ mm	
Peak efficiency	$\sim 98\%$	

3- Battery

Table 10 - Battery Specification

<i>Part name</i>	<i>Batteries</i>	
Battery type	LiFePO ₄ Lithium battery	
Nominal voltage	12.8V	
Rated capacity	25Ah	
Max charge voltage	~ 14.4 V ± 0.2 V	
Discharge cut-off voltage	~ 10 V	

4- Flight Controller

Table 11 - Flight Controller Specifications

<i>Part Name</i>	<i>Pixhawk 2.4.8</i>	
Processor	STM32F765 (Cortex-M7) @ 216 MHz, 2 MB Flash + 512 KB RAM	
Sensors	ICM-20689 accel/gyro, IST8310 magnetometer, MS5611 barometer	
Supply voltage	7–16V	

5- Power Module

Table 12 - Power Module Specification

<i>Part Name</i>	<i>Pixhawk Power Module</i>	
Input voltage	7.4V to 25.2V	
Regulated output	~5.3V	
Rated current	60A	

6- GPS Module

Table 13 - GPS Module Specification

<i>Part Name</i>	<i>NEO-M8N</i>	
Position accuracy	2 m	
Max update rate	~5.3V	
Supply voltage	1.65V – 3.6V	

7- LiDAR

Table 14 - LiDAR Specifications

<i>Part Name</i>	<i>RPLIDAR A1 (2D Lidar)</i>	
Scanning degree range	360° scanning 2D laser scanner	
Scanning distance range	0.15m to 12m	
Sampling rate	8000 times/s	
Supply voltage	5V DC	

8- DC-DC Converter

Table 15 - DC-DC Converter Specifications

<i>Part name</i>	<i>DC-DC Converter</i>	
Input voltage	12V	
Output voltage	19V	
Rated current	5A	

9- Onboard Computer

Table 16 - Onboard Computer Specifications

<i>Part name</i>	<i>NVIDIA Jetson Orin Nano</i>	
SoM: GPU	Ampere architecture with 1024 CUDA cores + 32 tensor cores	
CPU	6-core Arm Cortex-A78AE v8.2 64-bit	
Memory	8 GB LPDDR5, 128-bit, 68 GB/s bandwidth	
Power	15W	
Carrier board connectivity	2× MIPI CSI-2 22-pin, 4× USB 3.2 Gen2, Gigabit Ethernet, DisplayPort, 40-pin header	

10- Servo Motor

Table 17 - Servo motor Specifications

<i>Part name</i>	<i>25 Kg Steel Servo (High-Torque)</i>	
Operating voltage range	4.8V ~7.4V	
Motor type	Brushless	
Dimension	54x20x46mm	
Weight	78.3g	
Torque	25kg.cm	

11- Camera

Table 18 - Camera Specifications

<i>Part name</i>	<i>USB Camera</i>	
Operating voltage	5V DC	
Connecting Port Type	USB2.0 High Speed	
Resolution & Frame rate	1280x800 @ 120fps MJPEG	
Sensor	OmniVision OV9281	

12- Motor Controller

Table 19 - Motor Controller Specifications

<i>Part name</i>	<i>RC ESC</i>	
Current / Burst Current	60A / 320A	
Voltage	12 V	
Motor Type	Brushed motor	
Battery	2-3 cells Lipo.4-9 cells NiMH	

13- Voltage Regulator

Table 20 - Voltage Regulator Specifications

<i>Part name</i>	<i>UBEC</i>	
Input Voltage	5.5V-26V	
Output Voltage	6 V	
Continuous output current	10 A	

3.2.2. Detailed Block diagram

This block diagram shown in figure 16 illustrates the overall operation of the autonomous vehicle system. The process begins with the solar panel, which provides power to the charge controller and stores energy in the main battery. From there, the electrical power is distributed to different subsystems to ensure reliable and stable operation.

The battery supplies energy to the Jetson Orin Nano through a DC-DC converter, to the Pixhawk through a power module, and to the vehicle's motors and steering servo through the ESC and UBEC circuits. The Jetson Orin Nano functions as the main processing unit, receiving visual and LiDAR data to analyze the environment and make navigation decisions. These high-level commands are then sent to the Pixhawk, which handles real-time control of the motors and steering mechanisms.

Through this coordinated structure, the system achieves efficient power management, intelligent perception, and precise motion control, enabling fully autonomous vehicle operation.

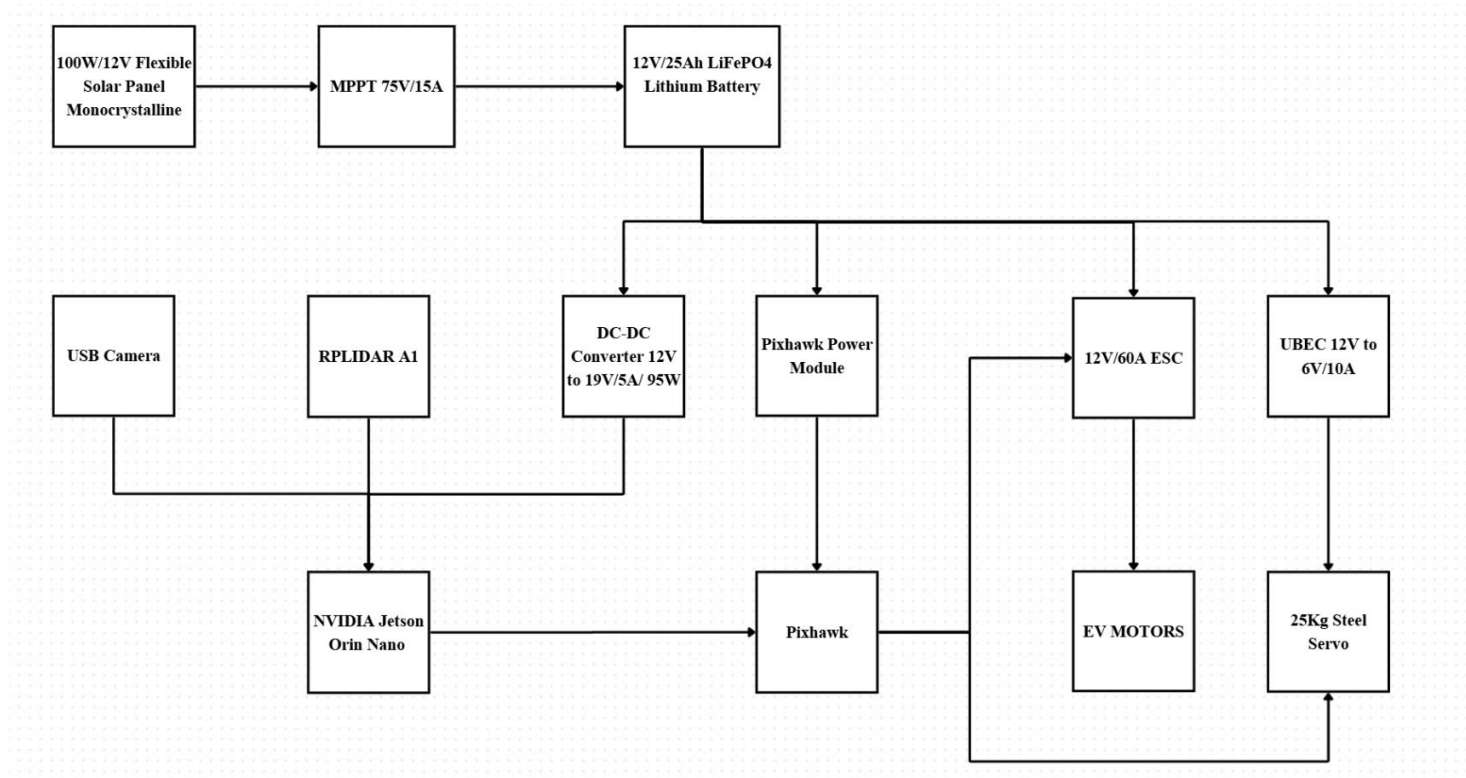


Figure 16 – Baseline Design Block Diagram

Power System

Figure 17 illustrates the design of the vehicle's off-grid power system. The system is designed for 100% solar autonomy. Energy is captured by the **100W/12V Monocrystalline Solar Panel**, which provides input power. This power is then regulated by a high-efficiency **MPPT 75V/15A Solar Charge Controller**, which optimizes the energy harvested and safely charges the **12V/25Ah LiFePO4 Lithium Battery**. The battery acts as the central power hub, storing the energy and providing a stable 12V supply to the entire system.

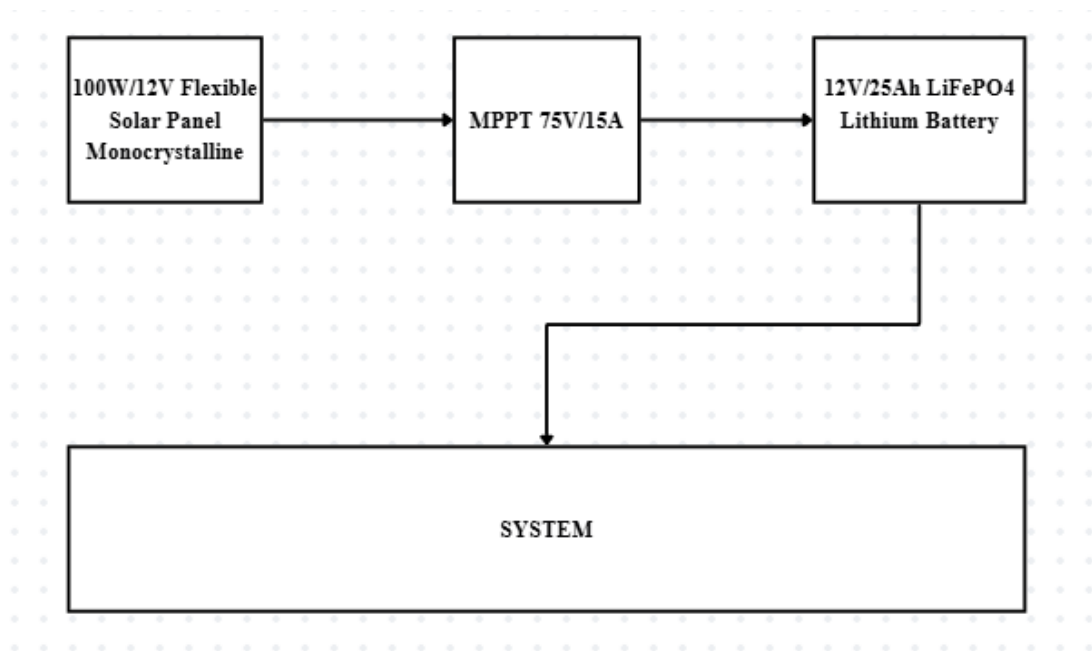


Figure 17 - Power System

Power Distribution

Figure 18 shows the vehicle's power distribution strategy, which is critical for system stability. The central **power supply** (the 12V LiFePO4 battery) provides power to four parallel and isolated circuits. This design separates components to prevent electrical interference. A **12V-to-19V Converter** powers the sensitive Jetson Orin Nano, while the **Pixhawk Power Module** provides clean 5V power to the Pixhawk. The high-current components are powered separately: a **12V/60A ESC** runs the main EV motors, and a **10A UBEC** provides 6V power to the steering servo. This isolation is essential to prevent motor noise from crashing the autonomous control system.

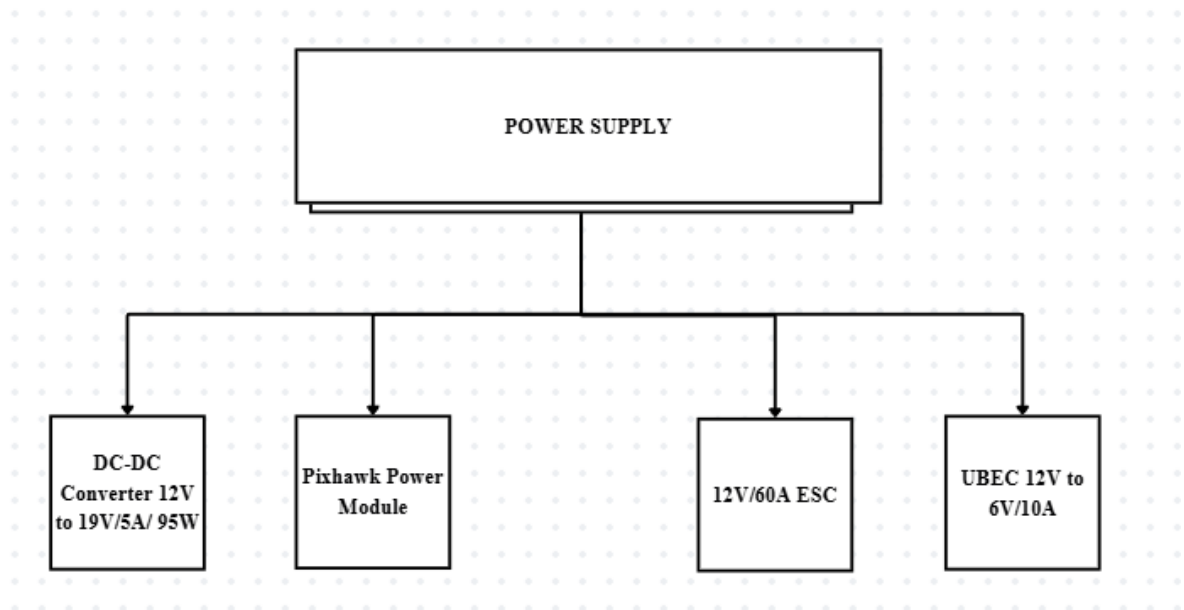


Figure 18 - Power Distribution

Co-Pilot Subsystem

Figure 19 details the co-pilot unit of the autonomous system. The **NVIDIA Jetson Orin Nano** serves as the central brain, running the high-level ROS navigation and AI-driven perception scripts. It receives its power from a dedicated **12V-to-19V/5A DC-DC Converter** to ensure a stable 19V supply. All the advanced senses connect directly to the Jetson: the **USB Camera** provides the video feed for AI object classification, and the **RPLIDAR A1** provides the 360-degree distance data for obstacle detection and mapping.

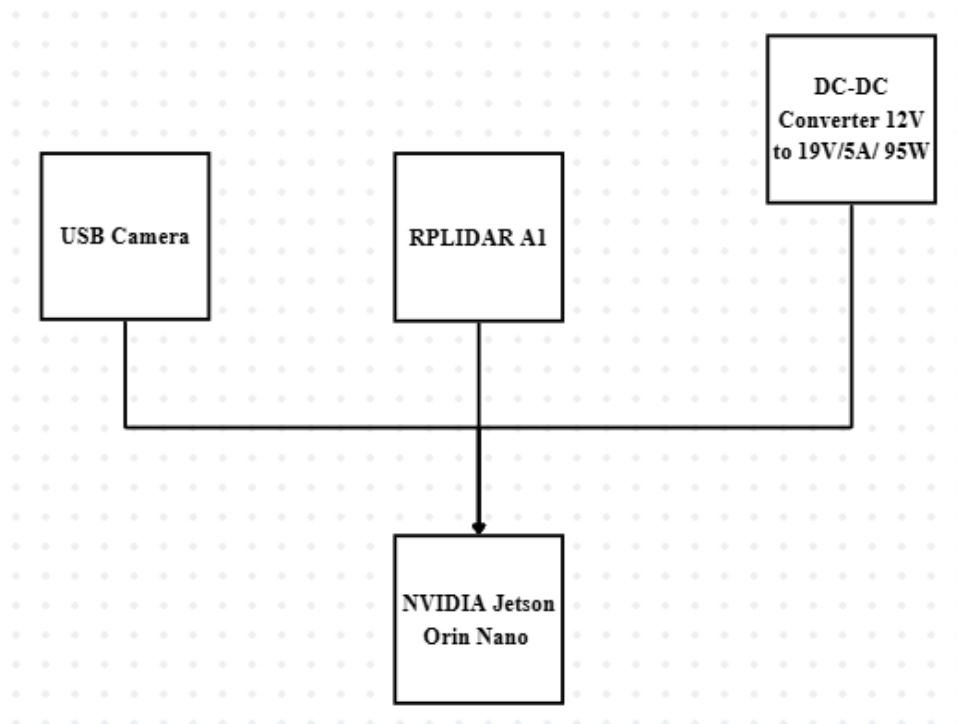


Figure 19 - Co-Pilot Subsystem

Pilot Subsystem

This diagram shows the connections for the Pixhawk and the actuator subsystems. The Pixhawk's brain is powered by its own dedicated Pixhawk Power Module. The Pixhawk then sends two separate control signals to the actuators. First, it sends a throttle signal to the 12V/60A ESC, which manages the high-current power for the main EV Motors. Second, it sends a steering signal to the 25Kg Steel Servo. Crucially, the servo receives its own separate power from a high-current 10A UBEC (12V to 6V). This isolation ensures the servo's high power draw cannot affect the Pixhawk.

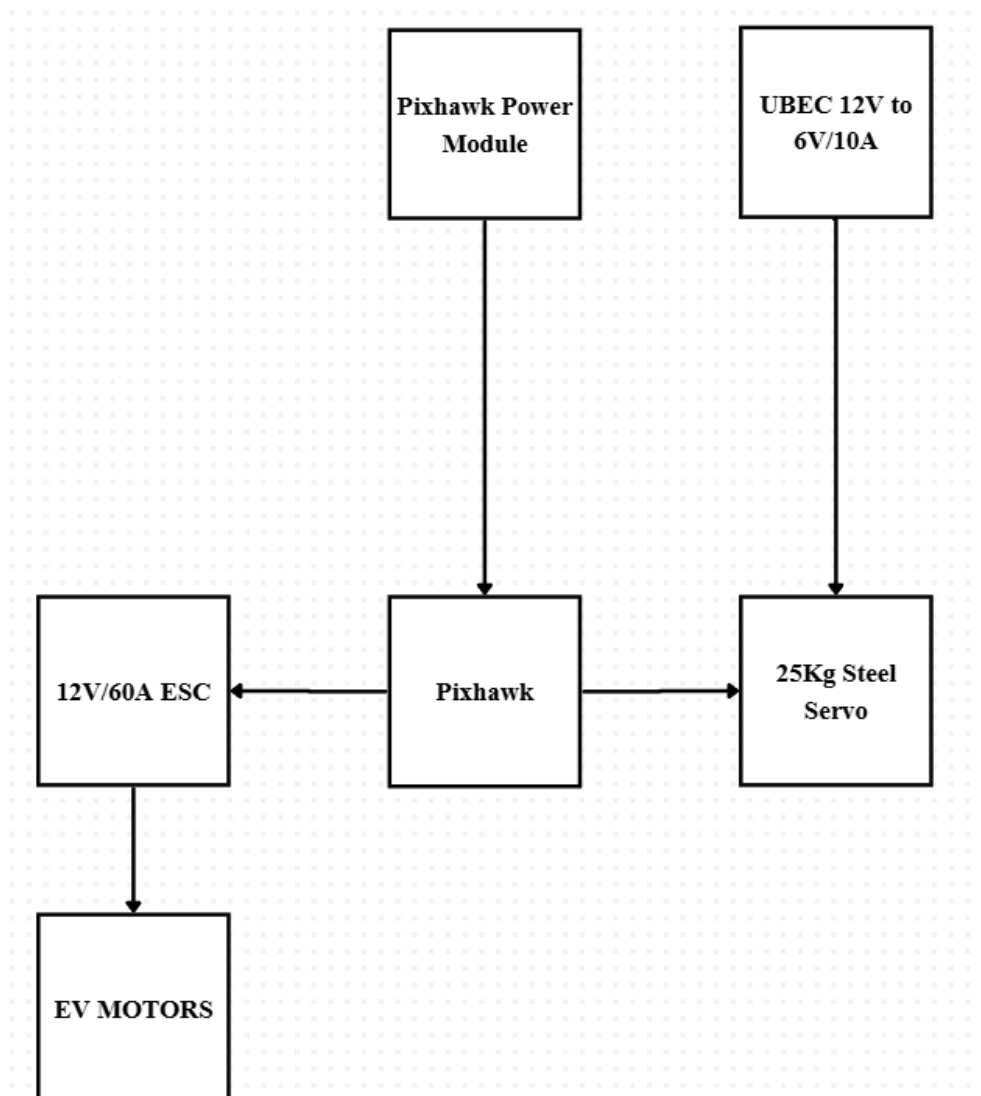


Figure 20 - Pilot Subsystem

3.2.3. Circuit Schematic

This schematic details the precise wiring and power architecture of the baseline design, which a simple block diagram overlooks. The Power System is centered on the 12V LiFePO4 Battery, which is charged from the 100W solar panel via the MPPT 75V/15A controller. The system's stability is achieved by splitting the battery's 12V output into four isolated, parallel power circuits: a 12V-to-19V step-up converter for the Jetson Nano, a Pixhawk Power Module for the Pixhawk's 5V rail, a high-current 12V/60A ESC for the motors, and a 10A UBEC for the 6V steering servo. This isolation prevents the motors and servo from causing the Pixhawk and Jetson to crash.

The Autonomous System data connections are also specific. The Pixhawk handles all real-time navigation and control, connecting directly to the GPS module via its UART port. The Jetson Nano runs ROS and manages all perception, connecting to the LiDAR and USB Camera via its USB ports. The two controllers communicate via a USB cable, which connects one of the Jetson Nano's USB host ports to the Pixhawk's Micro-USB port to send and receive MAVLink commands. Finally, the Mechanism shows that the Pixhawk's MAIN OUT 1 signal pin controls the ESC (which drives both motors in parallel), while the AUX OUT 1 signal pin controls the steering servo.

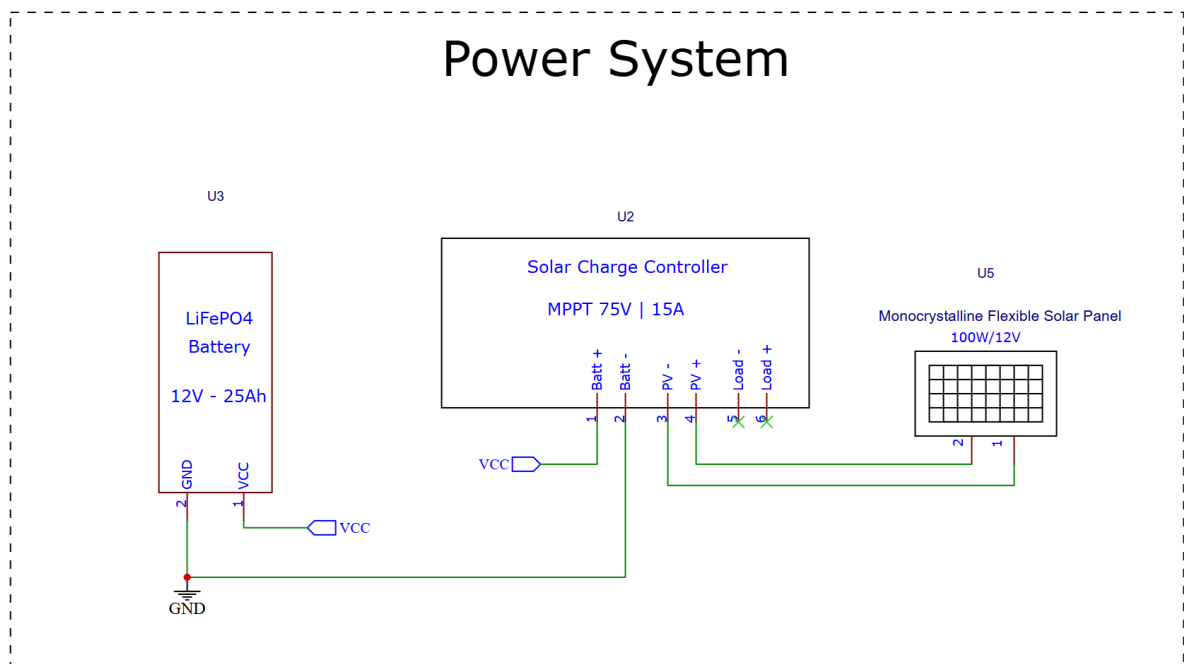


Figure 21 – Power System Schematic

Jetson Orin Nano "Co-Pilot"

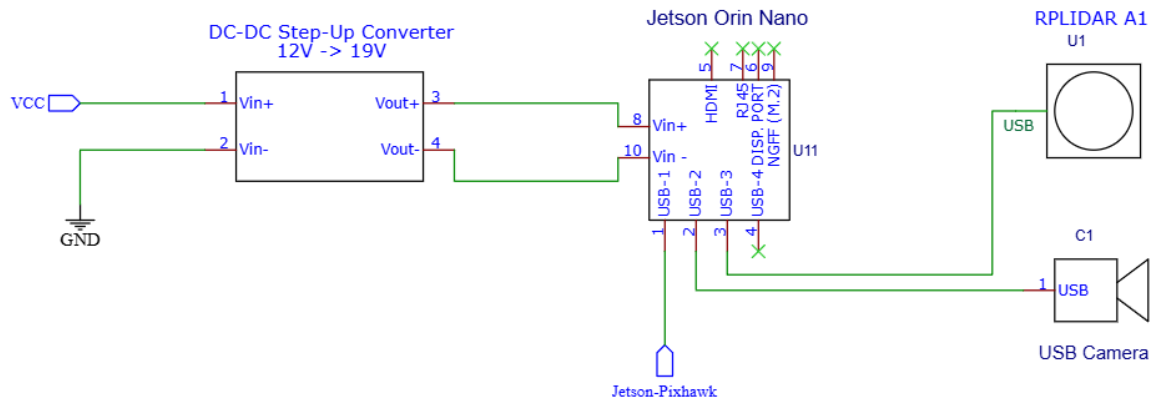


Figure 22 - Co-Pilot Schematic

Pixhawk "Pilot"

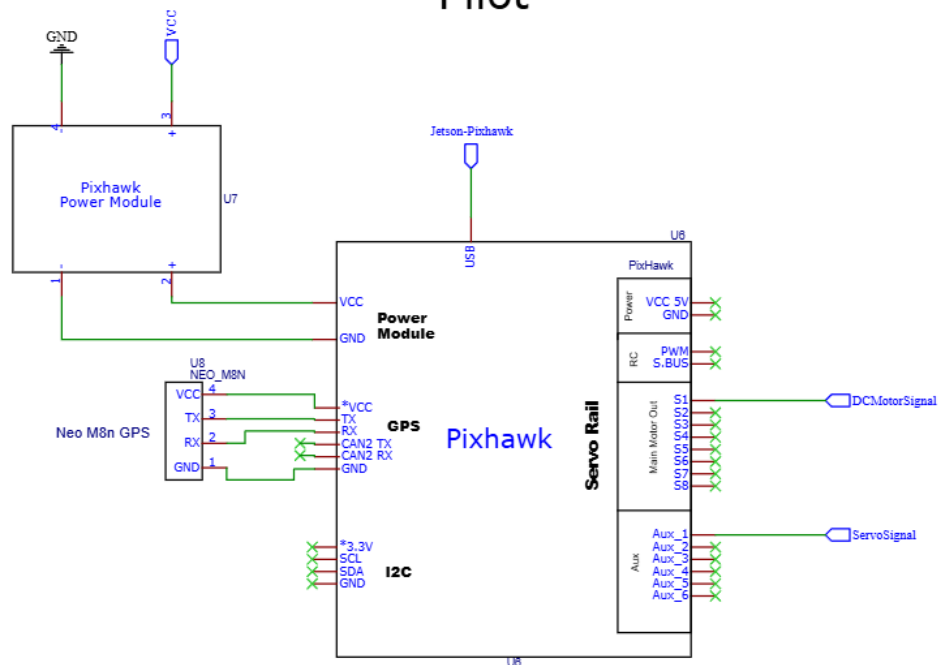


Figure 23 - Pilot Schematic

Mechanism

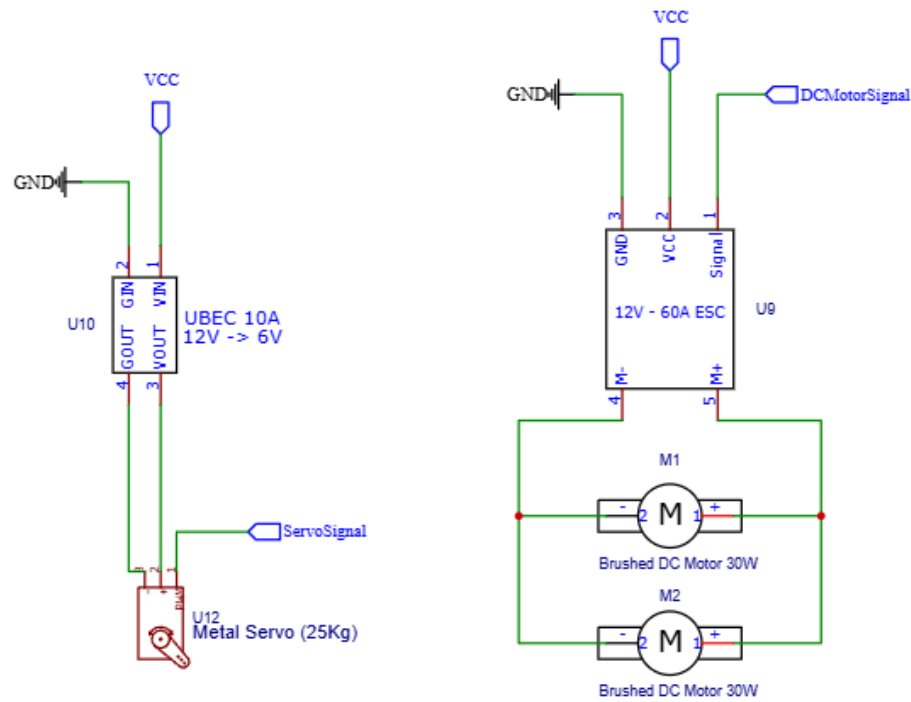


Figure 24 - Mechanism Schematic

3.2.4. Workflow

In this section we will sketch a flowchart to illustrate how the system works showing how the supplement of power flows in our system, the connections between each component, and what type of connection it is, the signals and outputs to be made during execution, and how the decisions of the signals and outputs are determined. But before we proceed to the flowchart we will construct a logical table to find a function for each output necessary in our project to ensure that we covered each logical decision in our system specifically the autonomous system, this table will show each all possible logical scenarios that might happen when we run the autonomous system, this is an extra step to ensure safety in our design.

3.2.4.1 Logical Decisions:

Below in Table 21 all those logical decisions in our system will be shown. The first four columns show all possible inputs to our microcontroller, which is the jetson nano, to cover all cases where the inputs might be 1 or 0, we must make sixteen rows in Table 21 which will cover all possible scenarios. We have four different inputs that need to be considered in the logic decision, first the **CAMERA** giving it 1, means it working and 0 the exact opposite, the same idea with our the second input **LiDAR**, and our last two **HUMAN** and **OBESTCAL** inputs they are considered as data inputs to the jetson having the value 1 means they appear as a data and 0 they don't. On the other hand, we have outputs which are what we want the jetson nano to perform if its encounter with any of the possible 16 inputs, starting with **DRIVE** giving it a 1 will make the EV follow the path, then **STOP** giving it 1 means that car autonomous movement will stop either for intended reasons such as a human appears Infront of the car or an emergency stop for some kind of malfunction happing in the hardware, the last output will be **AVOID** giving it a 1 will cause the EV to strive out of the path to not to be stuck Infront of an inanimate. Finally, giving any output a zero means it will not be the output of the microcontroller, in all sixteen scenarios there must be a **maximum** of one output having the 1 to make sure all scenarios are covered and guarantees no overlapping of outputs. In the end we will look at each scenario and decide which output we want our microcontroller to perform.

Table 21 - Logical Table

INPUTS				OUTPUTS		
CAMERA	LIDAR	HUMAN	OBESTCAL	DRIVE	STOP	AVOID
0	0	0	0	0	1	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	0	1	1	0	1	0
1	1	0	0	1	0	0
1	1	0	1	0	0	1
1	1	1	0	0	1	0
1	1	1	1	0	1	0

After filling out the Table 7, we can determine each logical functions for the three outputs which are shown below, some small details need to be explained to understand the logical equations, the functions for each output will help with the programming side of the autonomous system while also explaining the reasons for each decision taken by the microcontroller.

Logical functions:

$$DRIVE = CH'O'$$

$$STOP = C' + H + L'O$$

$$AVOID = CLH'O$$

C stands for CAMERA
L stands for LiDAR
H stands for HUMAN
O stands for OBESTCAL

3.2.4.2 FLOWCHART:

After defining the logical functions for each output of our system in all scenarios, we can proceed to construct and explain our flowchart which will show the flow of our system including the power flow, devices connections, each input connection and the decisions made to get outputs to control our autonomous system.

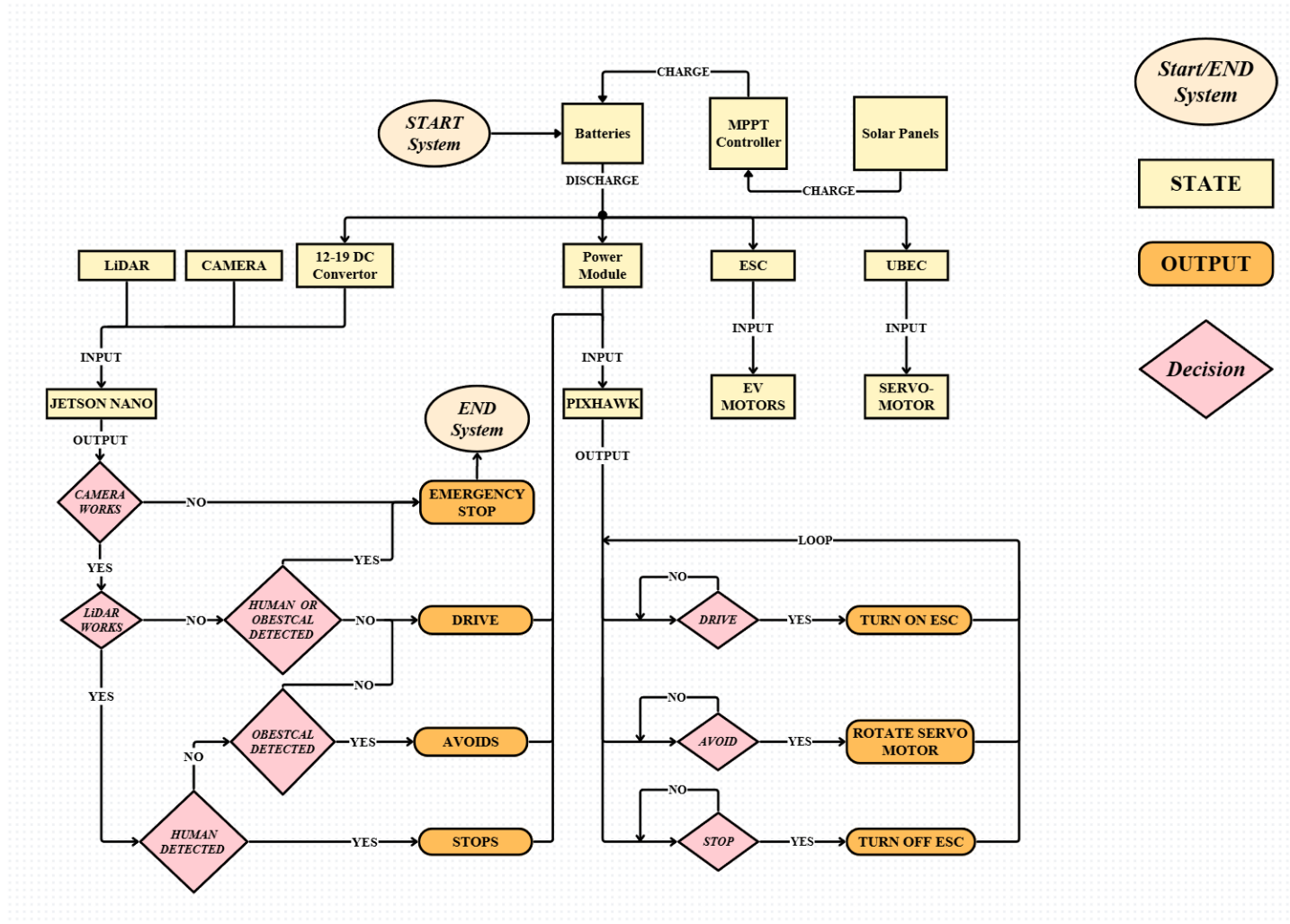


Figure 25 - Flowchart

Flowchart is shown in figure 25 above, to start explaining the flowchart, we divided the flowchart into three major compartments power flow, logical flow, input and outputs flow. Starting with a first major compartment of our chart which is the power, beginning with the **BATTERIES** which will be charged by the **SOLAR PANELS** through an **MMPT CHARGE CONTROLLER**, then from the batteries the power will flow to **FOUR** different voltage controllers, which will control the powering voltage flowing through our hardware parts, **JETSON NANO**, **PIXHAWK**, **SERVO-MOTOR** and **EV MOTORS**.

Moving on the second major compartment which is the logical flow, this compartment consists of logical decisions as **YES** or **NO** this will decide which correct output our system must produce in case of different scenarios, we have two continuous loops of logical decisions that needs to be made, one is a complex loop that we solved and figured out by our logical functions for each output above, and the other is quite simple, starting with the **JETSON NANO** logical flow which is the complex one it can be seen that there are **three outputs** which will come out of the **JETSON NANO** and go to the **PIXHAWK** as signals to decide one the three commands of action that will be taken by the **PIXHAWK** continues loop of decision making, there one last output coming out of the **JETSON NANO** which is the emergency stop, this output was made to stop the system ensuring safety in case of a malfunction in one of the parts.

Third and last major compartment, which is the inputs and outputs flow. Starting with the inputs the **LiDAR** and **CAMERA** will act as data input to our **JETSON NANO** to produce the output signal of **DRIVE, STOP and AVOID** these signals will act as an input to the **PIXHAWK** which will produce three output commands to control the EV, these commands will act as an input to **ESC** and **SERVO-MOTOR** which will produce the final output needed in this system which is drive forward, stop, and avoid. The **ESC** will control the voltage of the **EV MOTORS** which produces two movements that drive forward and stop, the **SERVO-MOTOR** will control avoiding movement of the EV.

After explaining the flowchart in three different major compartments, combining them together will give you the system needed to run the Automated Vehicle System power by Solar Energy.

3.2.5. System Input / Outputs and operating instructions

The system operates fully autonomously, integrating data from multiple sensors and user inputs to make intelligent driving decisions. Table 22 summarizes the system's inputs and outputs, outlining the key parameters and their functions where inputs provide environmental and user data, and outputs deliver control actions and feedback. The overall data flow is illustrated in Figure 26, which presents the black box diagram showing how external signals and user commands interact with the vehicle's control and monitoring subsystems.

Table 22 - Inputs and Outputs of the system

<i>Inputs</i>	
<i>Parameter</i>	<i>Description</i>
Solar Energy	Powers the vehicle through the solar charging system.
GPS Satellite Data	real-time positioning for route navigation and destination tracking.
LiDAR Data	Detects obstacles and measures surrounding distances.
Visual Image	Captured by onboard cameras for object detection.
User Destination Selection	The operator selects the desired destination
Emergency Stop Command	Allow the operator to immediately stop the vehicle for safety
<i>Outputs</i>	
<i>Parameter</i>	<i>Description</i>
Steering Control	Adjusts wheel direction based on the navigation algorithm
Throttle Control	Control motor speed according to the planned trajectory and obstacle data

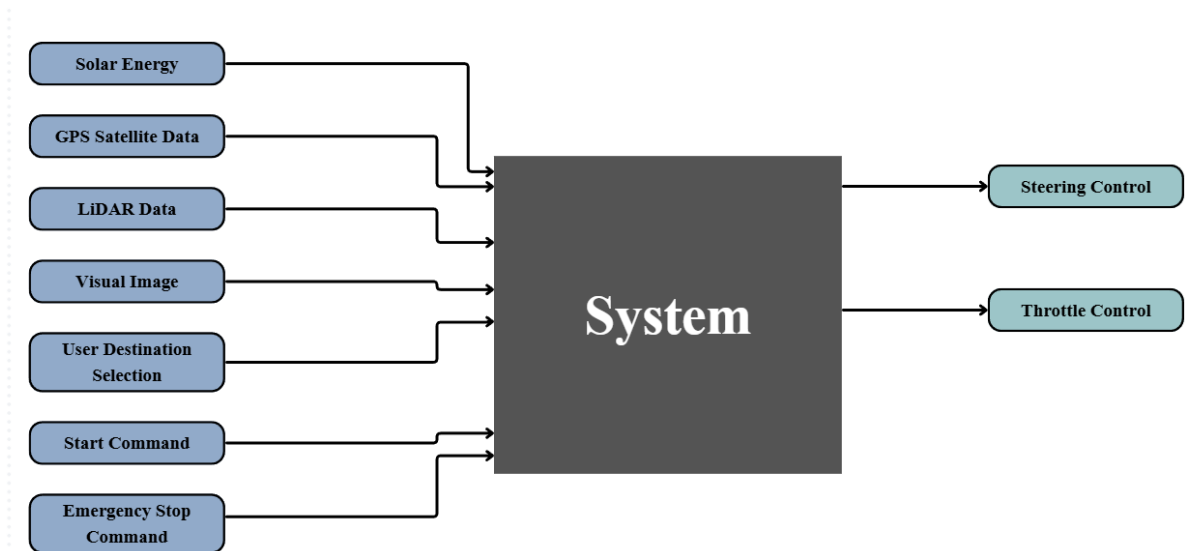


Figure 26 - Black Box Diagram for the System

Operating Instructions

1. Turn on the vehicle's main power switch.
2. The operator chooses the desired destination using buttons.
3. The vehicle will now autonomously navigate the GPS path. It will use its AI and LiDAR to identify and react to obstacles, such as stopping for a human or planning a path around a static object.
4. The operator can stop the vehicle at any time by pressing the physical Emergency Stop button.

3.2.6. Relevant Engineering Standard

The table below outlines how each selected standard is practically implemented in the baseline design. It highlights the specific role of every standard in guiding component protection, control system reliability, and safe solar integration, ensuring that each subsystem follows proper engineering practices during assembly and operation.

Table 23 - Baseline Design's Relevant Engineering Standard

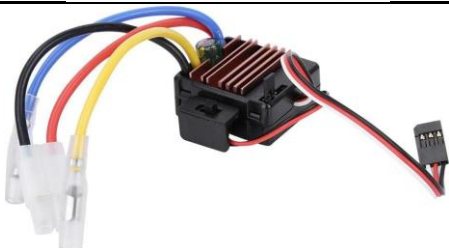
<i>Standard</i>	<i>Title / Domain</i>	<i>Application in Baseline Design</i>
ISO 26262	Road Vehicles – Functional Safety	Used in the control logic between Jetson Orin Nano and Pixhawk to ensure the system safely stops in case of sensor or control failure.
IEC 60529 (IP26)	Degrees of Protection (IP Code)	Electrical components such as the battery, MPPT, and converters are enclosed to protect against dust and water.
UL 1703	Flat-Plate Photovoltaic Modules and Panels	Followed in mounting and wiring the solar panel to ensure safe insulation and fire protection.

By applying these standards, the automated vehicle system achieves international compliance in both safety and performance. ISO 26262 ensures functional safety in control decisions, UL 1703 guarantees reliable solar panels stability, IEC 60529 provides environmental protection towards electrical components. Together, these standards create a strong foundation for a safe, sustainable, and professionally engineered autonomous solar-powered vehicle.

3.2.7. Cost Estimation

Table 24 - Baseline Cost Estimation

<i>Component</i>	<i>Quantity</i>	<i>Total Cost (SAR)</i>	<i>Image</i>
100W/12V Flexible Solar Panel Monocrystalline	1	335	
MPPT 75V/15A	1	275	
12V/25Ah LiFePO4 Lithium Battery	1	360	
Pixhawk	1	250	
Pixhawk Power Module	1	30	
M8N GPS	1	100	
RPLIDAR A1	1	380	

DC-DC Converter 12V to 19V/5A/ 95W	1	80	
NVIDIA Jetson Orin Nano Developer Kit	1	1100	
25Kg Steel Servo	1	50	
USB Camera	1	190	
12V/60A ESC	1	140	
UBEC 12V to 6V/10A	1	100	
Total	3390		

CHAPTER – 4 IMPLEMENTATION

4.1. IMPLEMENTATION OVERVIEW

This chapter explains project was implemented practically. The implementation was divided into hardware and software parts, then both parts were integrated and tested on the prototype vehicle.

The hardware implementation included the power system, steering system, throttle system, motor driver, sensors, and control circuit. The software implementation included the ROS2 setup, sensor processing, navigation logic, and communication between the components.

During implementation, some changes were made from the baseline design to simplify testing and improve control. The Jetson Orin Nano was replaced with a laptop, the Pixhawk was replaced with an Arduino, the ESC was replaced with a BTS motor driver, and an optical flow sensor has been added to the system. These changes are explained and justified in the following section.

4.2. BASELINE DESIGN ADJUSTMENT AND JUSTIFICATION

During the implementation stage, some changes were made to the original baseline design due to component availability and practical testing needs. The Jetson Orin Nano was replaced with a laptop because it was more available and easier to use for running the software environment. The Pixhawk was also replaced with an Arduino because the navigation approach changed from GPS-based navigation to a map-based navigation system, so the project no longer required the Pixhawk for waypoint control. In addition, the ESC was replaced with a BTS motor driver because it was easier to integrate with the Arduino, more available, and provided better control performance for the DC motors in the prototype. An optical flow sensor was also added to the motor system to provide feedback about the vehicle speed, which helped the system monitor the actual motor movement during testing and control

Table 25 - Final Design Changes

<i>Baseline Design Component</i>	<i>Final Implemented Component</i>	<i>Justification</i>
----------------------------------	------------------------------------	----------------------

Jetson Orin Nano	Laptop: Intel Core i7 16 GB RAM	More available and easier to use during implementation
Pixhawk	Arduino Uno	Navigation algorithm changed from GPS-based navigation to map-based navigation
12V/60A ESC	BTS7960 Motor Driver	• Easier to integrate with Arduino • More available than the selected ESC • Provides better practical control for the DC motors
No optical flow sensor in baseline design	Optical flow sensor added	• Provides speed feedback from the motors and allow the system to monitor the actual vehicle speed • Navigation algorithm changed from GPS-based navigation to map-based navigation

4.2.1. Old Baseline Design

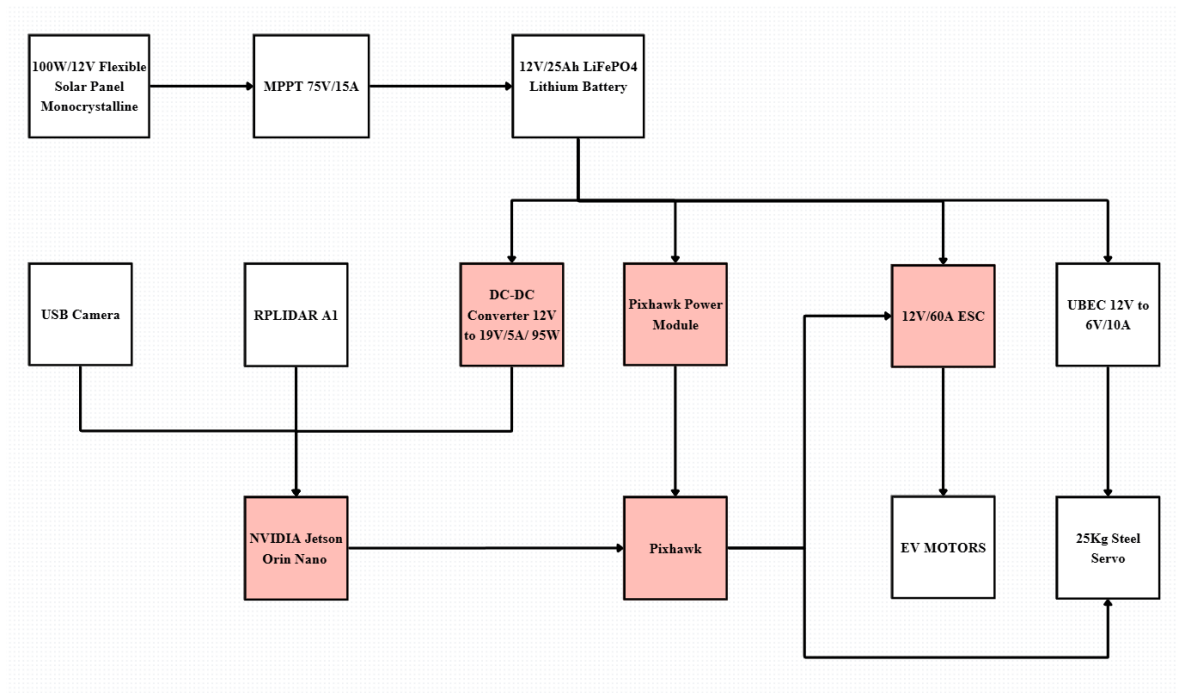


Figure 27 - Old Baseline Design

4.1.1. New Baseline Design

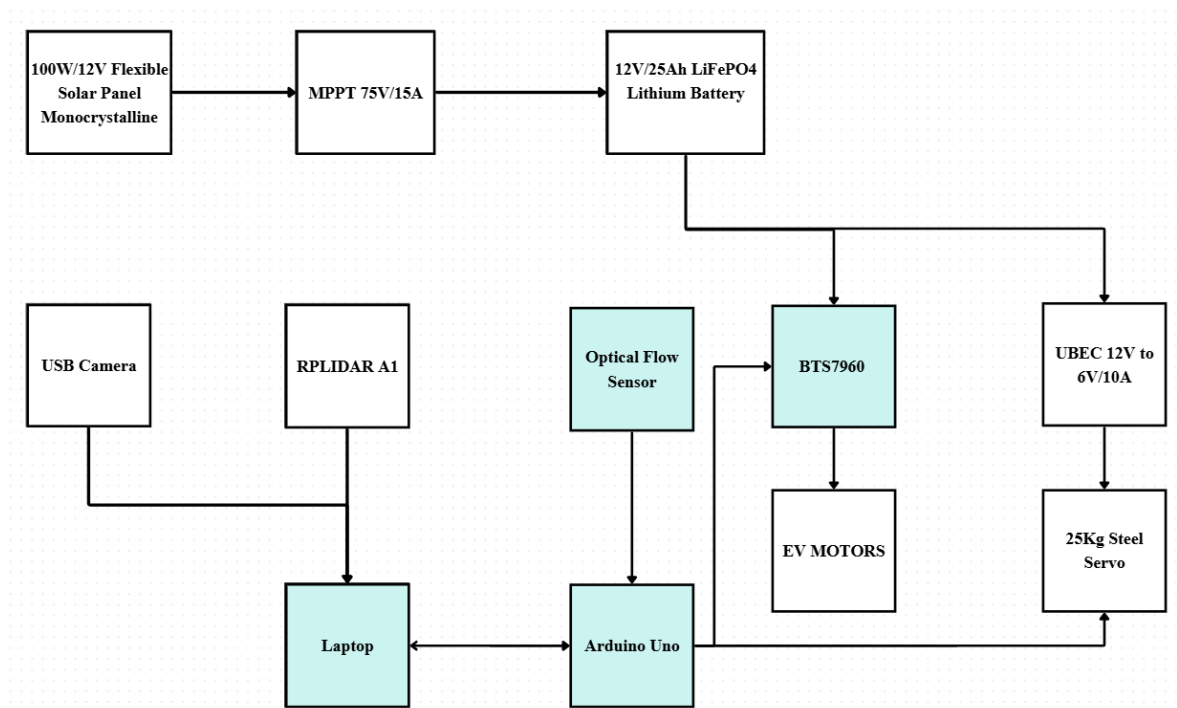


Figure 28 - New Baseline Design

4.2. HARDWARE IMPLEMENTATION

The hardware implementation focused on assembling the physical components of the prototype and connecting them in a way that supports stable operation. This included the power system, steering system, throttle system, motor driver circuit, sensors, and control unit. Each hardware subsystem was implemented and tested separately first, then connected together during the system integration stage.

4.2.1. *Steering System implementation*

The steering system was implemented by connecting a servo motor to the vehicle steering shaft. Since the shaft requires high torque to rotate, the servo was not connected directly. Instead, a gear mechanism was designed to increase the torque transferred from the servo to the steering shaft. During early testing, the servo was able to rotate the steering shaft, but it struggled under load and did not provide smooth steering response. Therefore, the gear mechanism was added to double the torque delivered to the steering shaft without changing the servo motor, which improved the steering smoothness and response speed.

The mechanism was designed using **SolidWorks**, and the required parts were fabricated using **PETG** 3D printing. A custom 3D-printed mount was used to fix the servo close to the steering shaft and keep the gears properly aligned. Two gears were also printed: a 15-teeth gear mounted on the servo motor and a 30-teeth gear mounted on the steering shaft.

The gear ratio is calculated as:

$$Gear\ Ratio = \frac{30}{15} = 2:1$$

This means the torque applied to the steering shaft is approximately doubled, while the steering shaft rotates at half the servo speed. This helped the servo move the steering shaft more effectively and reduced the load on the servo during steering.

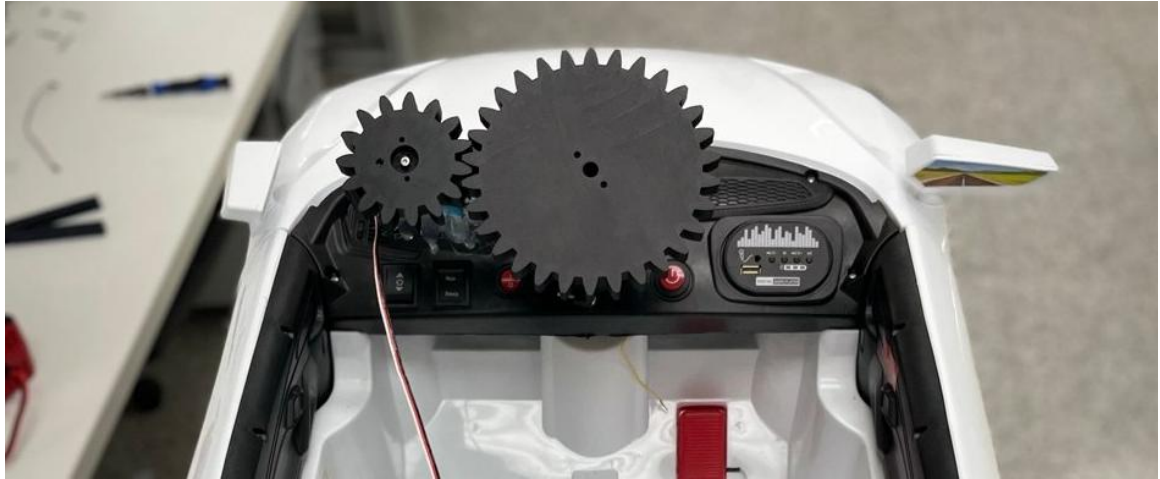


Figure 29 - Steering Gears

The figures show the steering mechanism design and assembly process. The 3D-printed bracket holds the steering shaft and provides a fixed base for the servo motor. The smaller servo gear drives the larger steering shaft gear, converting the servo rotation into controlled left and right steering movement. After assembly, the system was tested by sending control signals from the Arduino to confirm correct steering operation.

PETG was selected for the printed parts because it provides better strength and durability than PLA, making it more suitable for mechanical parts exposed to load and vibration during testing.



Figure 30 - Steering Shaft

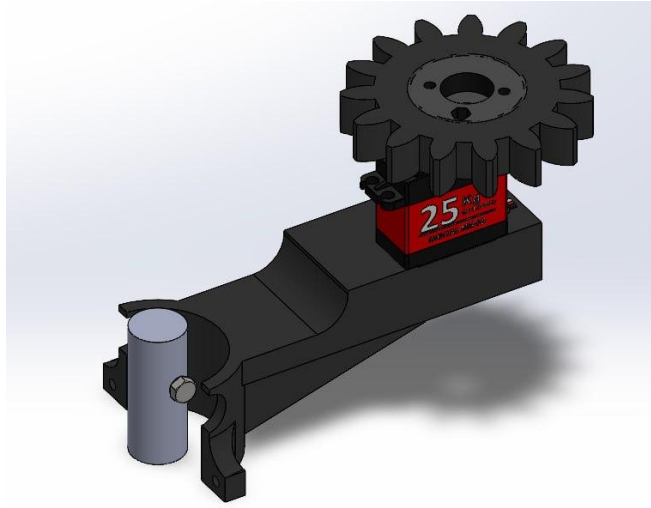


Figure 31 - Steering Assembly: Step 1

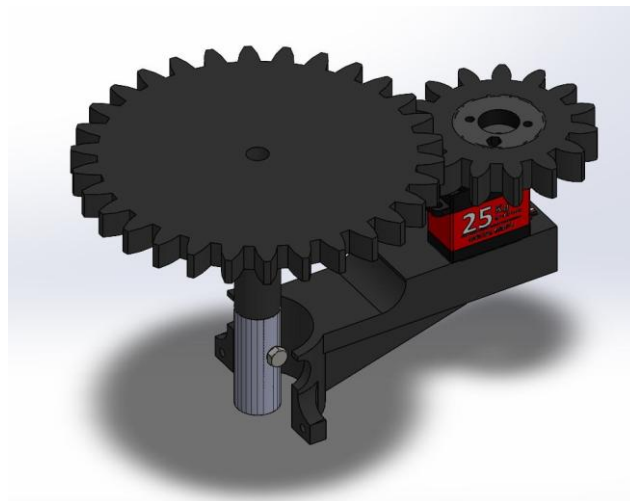


Figure 32 - Steering Assembly: Step 2

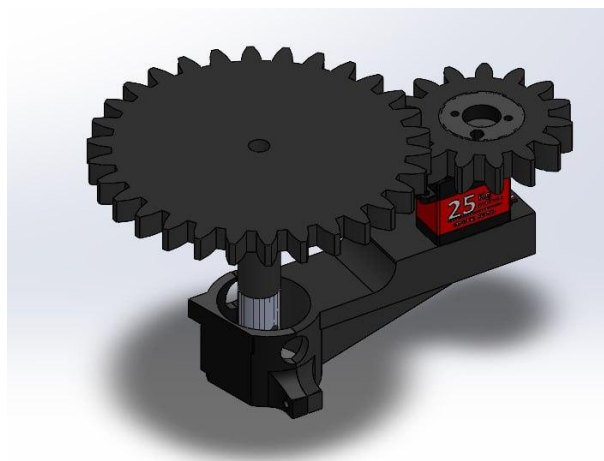


Figure 33 - Final Assembly of the Steering Gears

4.2.2. Throttle System Implementation

The throttle system was implemented to control the forward movement and speed of the prototype vehicle. In the final design, two 30 W DC motors were used as the main driving motors. The Arduino receives movement commands from the laptop and sends control signals to the BTS motor driver. The BTS driver then controls the two DC motors by adjusting the motor speed and direction. The system was first tested at low speed to confirm correct motor rotation, then gradually tested on the prototype vehicle to ensure stable movement.

4.2.3. Sensor Installation and Integration

The sensor system was installed to support mapping, obstacle detection, and visual feedback during autonomous operation. The main sensors used in the prototype were a LiDAR and a wide-angle camera. The LiDAR was used to scan the surrounding area and provide distance data for mapping and obstacle detection, while the wide-angle camera was installed at the front of the vehicle to provide a wider view of the driving path.

To improve the physical installation, custom **3D-printed** mounts were designed and used for both the LiDAR and the camera. These mounts helped keep the sensors fixed in stable positions and reduced unwanted movement during testing. The LiDAR was mounted in a suitable position to scan the surrounding area clearly, while the camera was mounted at the front of the vehicle facing the driving direction.

Both sensors were connected to the laptop, where the ROS2 system receives and processes their data. Before full system integration, the LiDAR and camera were tested separately to confirm that the scan data and camera feed were working correctly.

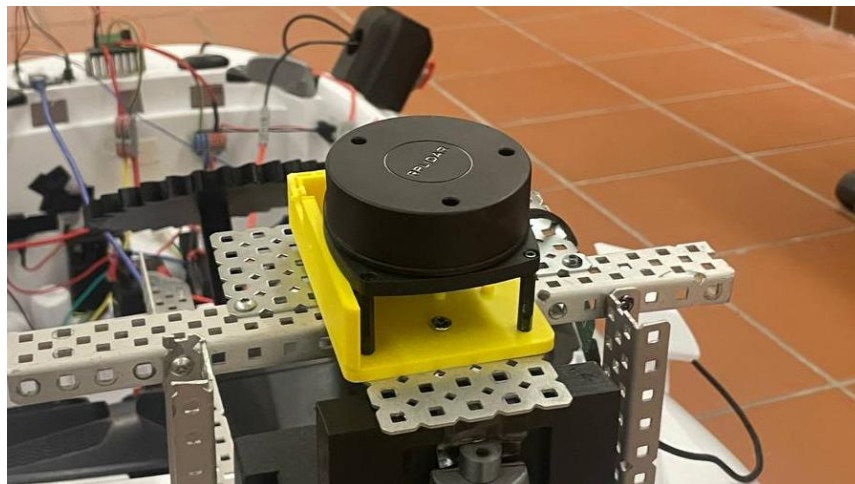


Figure 34 - 3D-Printed LiDAR Mount Installed on the Prototype

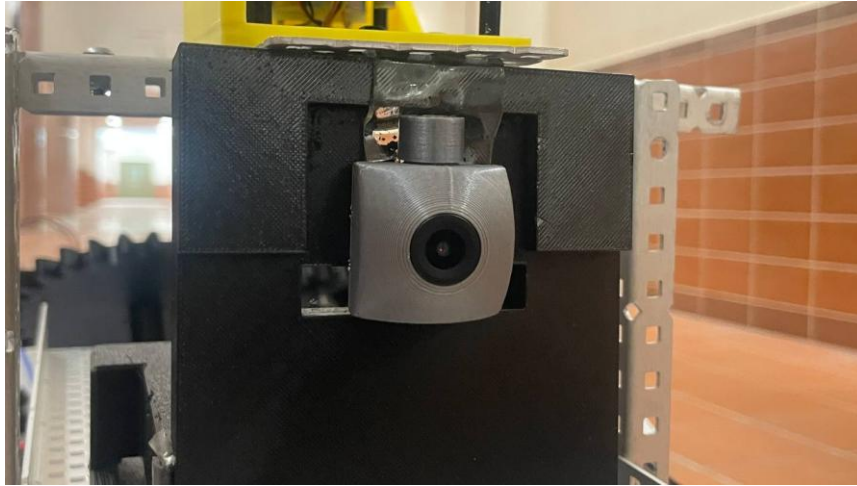


Figure 35 - 3D-Printed Wide-Angle Camera Mount Installed on the Prototype

4.1.1. Power System implementation

The power system was implemented to supply and charge the prototype vehicle. It consists of the solar panel, MPPT charge controller, battery, fuse, and main power switch. In this implementation, the solar panel was used as a charging station rather than being permanently mounted on the vehicle. This means the vehicle battery can be connected to the solar charging setup when charging is needed, while the vehicle operates using the stored energy from the battery. The solar panel harvests energy from sunlight and sends it to the MPPT charge controller. The MPPT controller regulates the charging process and safely charges the battery. The battery then acts as the main power source for the prototype during operation.

A fuse was added between the battery and the load side to protect the circuit from excessive current or short circuits. In addition, a main safety switch was installed to allow the system to be turned ON and OFF during testing and operation. This made the power system safer and easier to control, especially during hardware debugging and maintenance.

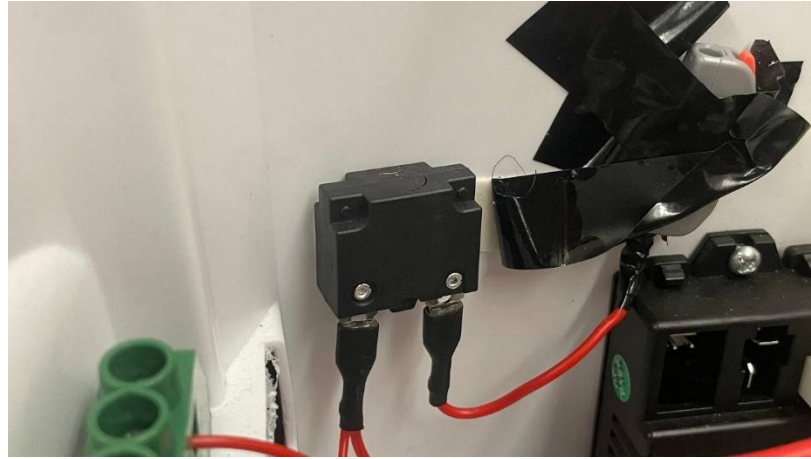


Figure 36 - Fuse



Figure 37 - Safety Switch

The power flow starts from the solar panel, then passes through the MPPT charge controller to charge the battery. From the battery, the power is distributed to the required circuits through the fuse and switch. This structure helped protect the electrical components and provided a simple way to disconnect the power when needed.

The selected battery has a nominal voltage of 12 V and a capacity of 30 Ah. Therefore, the total stored energy of the battery can be estimated as:

$$E = V \times Ah$$

$$E = 12 \times 30 = 360Wh$$

This means the battery can store approximately 360 Wh of energy. The solar panel used in the charging station has a rated power of 100 W. Under ideal sunlight conditions, the charging current can be estimated as:

$$I = \frac{P}{V}$$

$$I = \frac{100}{12} = 8.33A$$

Based on this value, the ideal charging time for the 30 Ah battery is:

$$t_{charging} = \frac{30Ah}{8.33A} = 3.6 \text{ hours}$$

Therefore, the **ideal charging time is approximately 3.6 hours**. However, the actual charging time is expected to be longer because solar panel output changes with sunlight intensity, panel angle, temperature, wiring losses, and MPPT efficiency. This calculation was used only as an estimated reference for the solar charging station performance.

The operating time of the prototype vehicle can also be estimated using the battery energy and the total system power consumption:

$$t = \frac{E}{P_{Load}}$$

The estimated power consumption of the prototype includes the main electrical components used during operation. The two DC motors consume approximately 60 W in total, while the laptop used for ROS2 processing consumes approximately 40 W. In addition, the sensors and Arduino consume approximately 10 W. Therefore, the total estimated load power becomes:

$$P_{Load} = 60 + 40 + 10 = 110W$$

Using the previously calculated battery energy of 360 Wh:

$$t_{Discharging} = \frac{360Wh}{110W} = 3.27 \text{ hours}$$

This means the prototype vehicle **can operate for approximately 3.3 hours before requiring recharge** under average operating conditions. However, the actual operating time may vary depending on the vehicle speed, terrain conditions, motor load, sensor usage, and processing load on the laptop. This calculation was used only as an estimated reference for the operating performance of the prototype vehicle.

The energy during charging can be expressed as:

$$E = P \times t$$

Assuming constant ideal charging power from the solar panel, the charging energy becomes:

$$E_c(t) = 100t$$

where $E_c(t)$ is in Wh and t is in hours. The battery gains 100 Wh per hour, reaching 360 Wh at $t = 3.6$ hours.

Similarly, during discharging, the battery loses energy according to the total load power. Since the load power is estimated as 110 W, the remaining battery energy during operation can be expressed as:

$$E_d(t) = 360 - 110t$$

where $E_d(t)$ is the remaining battery energy in Wh and t is the operating time in hours. This means the battery loses 110 Wh of energy every hour while supplying the prototype load. At $t = 3.27$ hours, the battery energy reaches zero.

Thus, the charging curve increases linearly from 0 to 360 Wh in 3.6 h, while the discharging curve decreases linearly from 360 Wh to 0 in 3.27 h. These graphs represent ideal behavior and assume constant power; in practice, variations in solar irradiance, MPPT efficiency, battery behavior, wiring losses, and motor load cause non-linear performance.

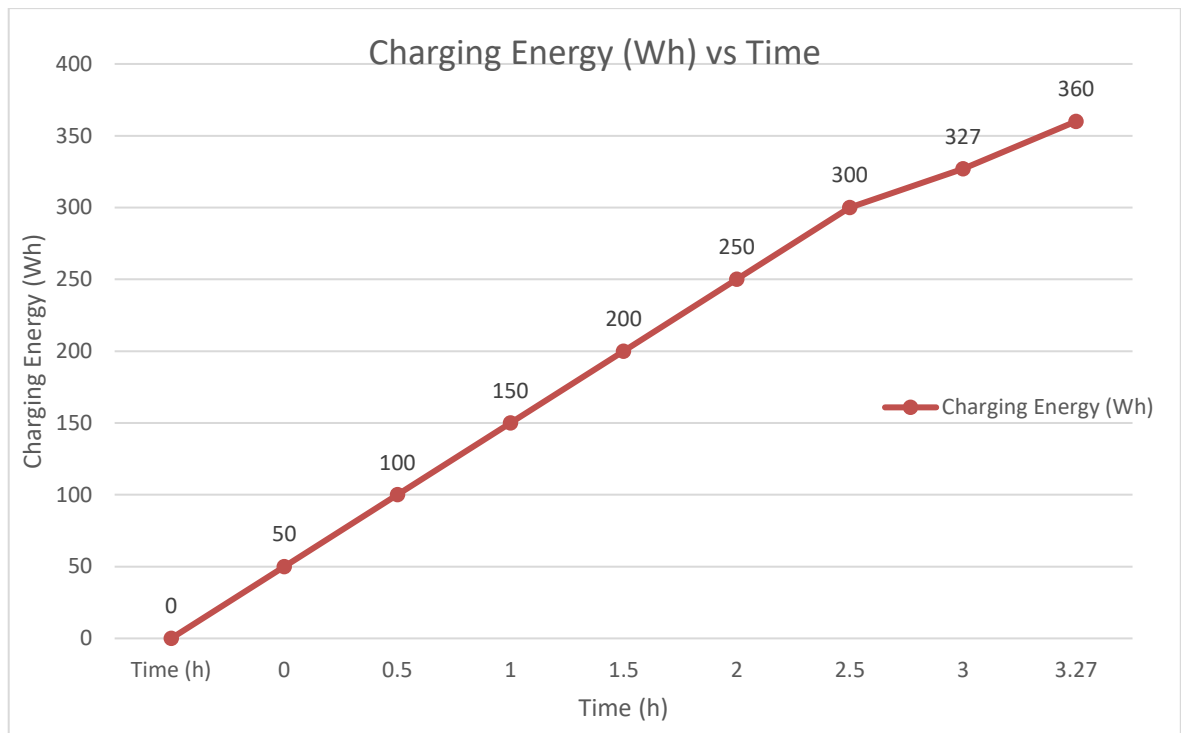


Figure 38 - Charging Energy vs Time

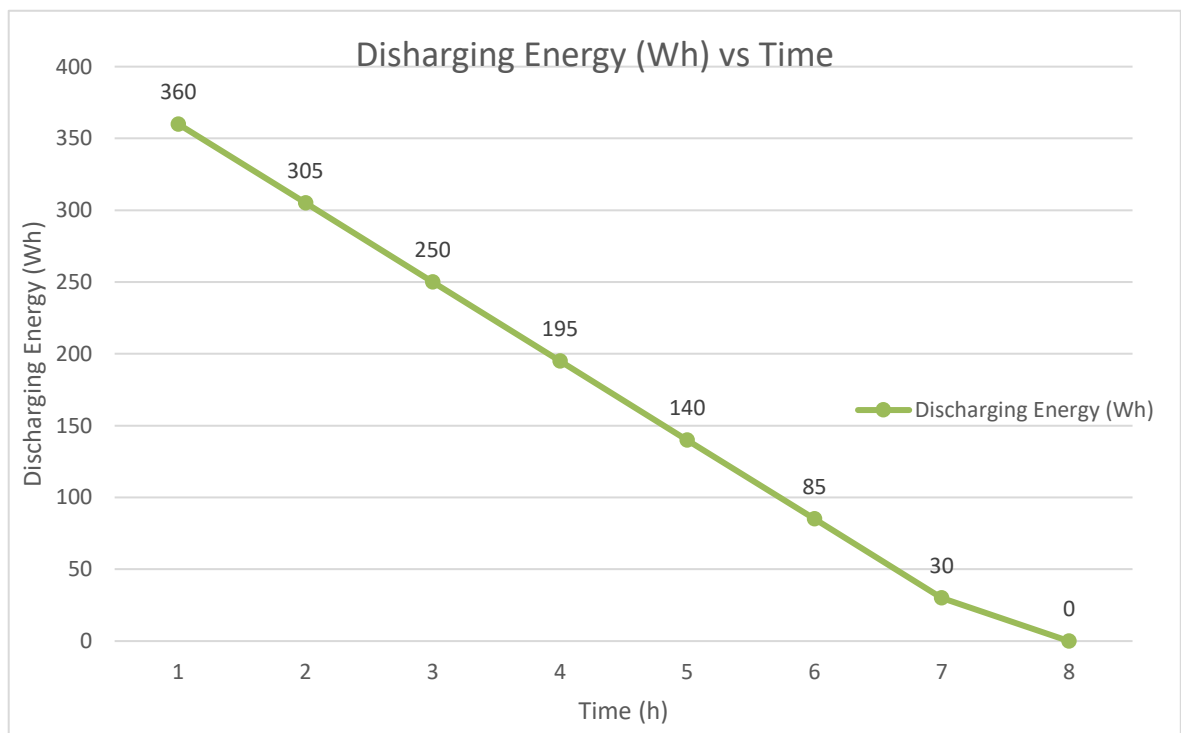


Figure 39 - Discharging Energy vs Time

4.2. SOFTWARE IMPLEMENTATION

4.2.1. ROS2 Architecture and Nodes

The software system was implemented using ROS2 humble on the laptop. ROS2 was used because it allows the vehicle software to be divided into smaller programs called nodes, where each node is responsible for a specific task. This made the system easier to develop, test, and debug compared to writing one large program for the whole vehicle.



Figure 40 - ROS2 Humble

The main package used for the final implementation was my_bot2. This package contains the main launch files, configuration files, saved maps, RViz setup, URDF model, simulation worlds, and custom Python nodes used to operate the vehicle. The project folder also includes other supporting packages, such as my_mapper for map generation, slidar_ros2 for the LiDAR driver, and rf2o_laser_odometry for estimating vehicle movement from LiDAR data. The overall package organization is shown in Figure 41.

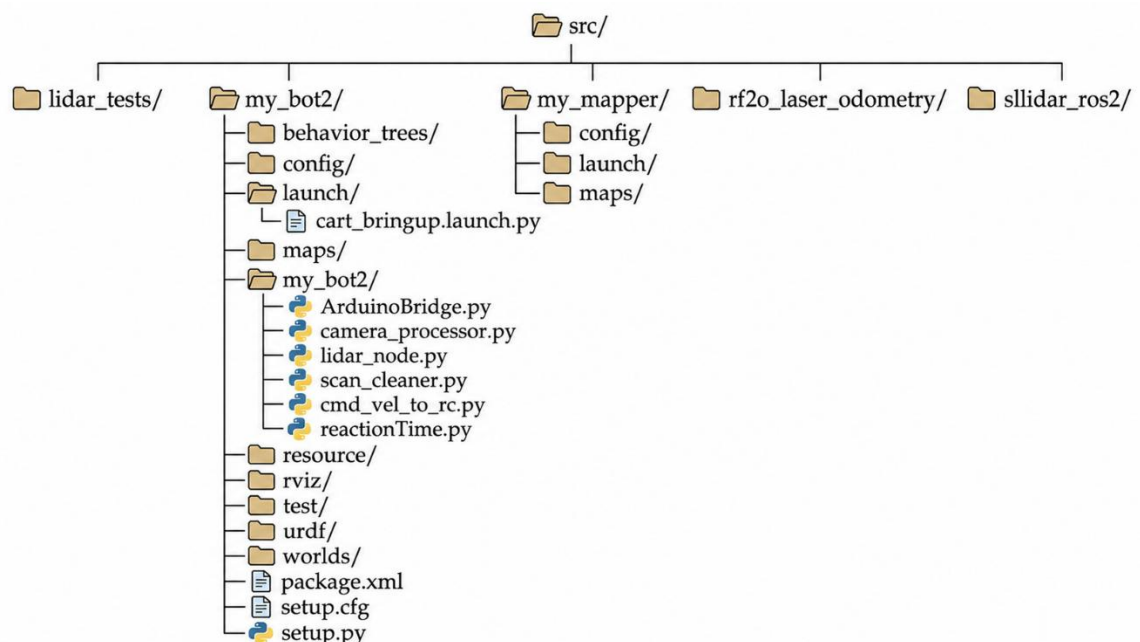


Figure 41 - Files Hierarchy

The main launch file used to start the system was `cart_bringup.launch.py`. This launch file brings up the required ROS2 nodes together instead of running each part manually. It starts the LiDAR node, Arduino bridge node, camera node, TF transforms, RF2O laser odometry, Nav2 navigation stack, and the web interface and RViz. The launch file also loads the Nav2 configuration file `nav2_params_Ackermann.yaml`, which contains the navigation parameters for the Ackermann-style vehicle.

The software flow starts from the sensors. The LiDAR publishes scan data, which is used for mapping, obstacle detection, and odometry. The camera node processes the video feed and detects humans. The navigation system then uses the map, robot position, and sensor data to calculate a safe path. Finally, the movement command is sent to the Arduino, which controls the steering servo and the two DC motors.

Before applying the system to the physical prototype, the vehicle was tested in **Gazebo** and **RViz**. Gazebo was used to simulate the robot model and its movement in a virtual environment, while RViz was used to visualize the map, robot frames, LiDAR scan, navigation goal, and planned path. This simulation stage helped check the robot model, map, frames, navigation parameters, and path planning behavior before connecting the software to the real hardware. It also reduced the risk of damaging the prototype and allowed the team to debug the navigation system step by step.

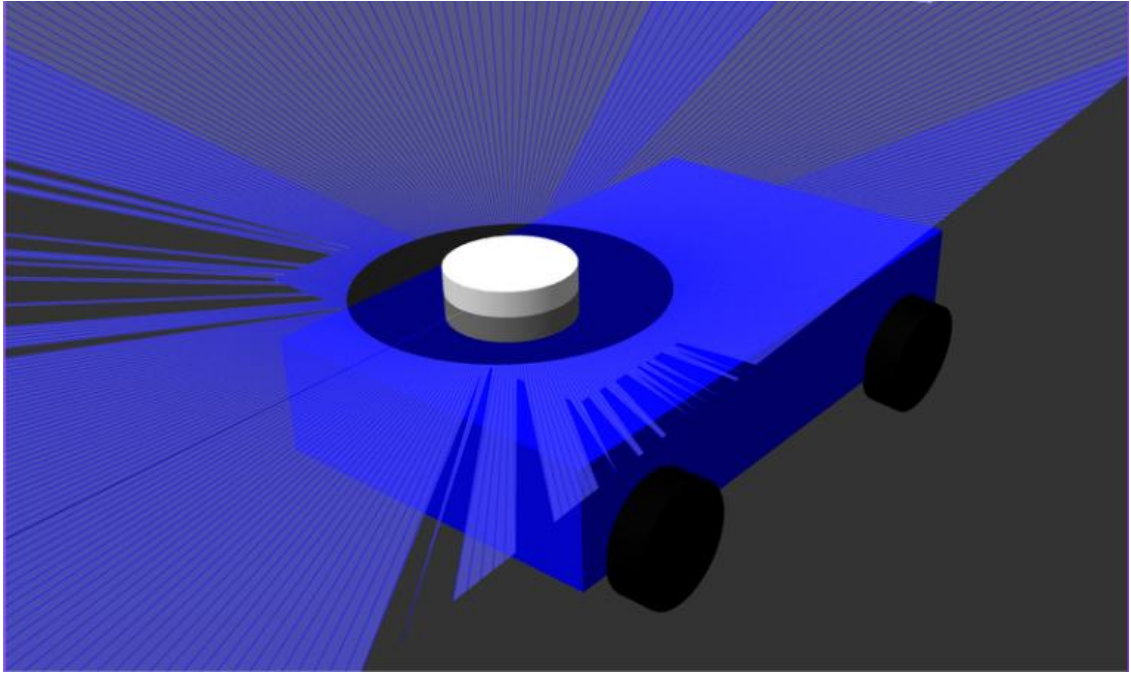


Figure 42 - URDF Robot in Gazebo

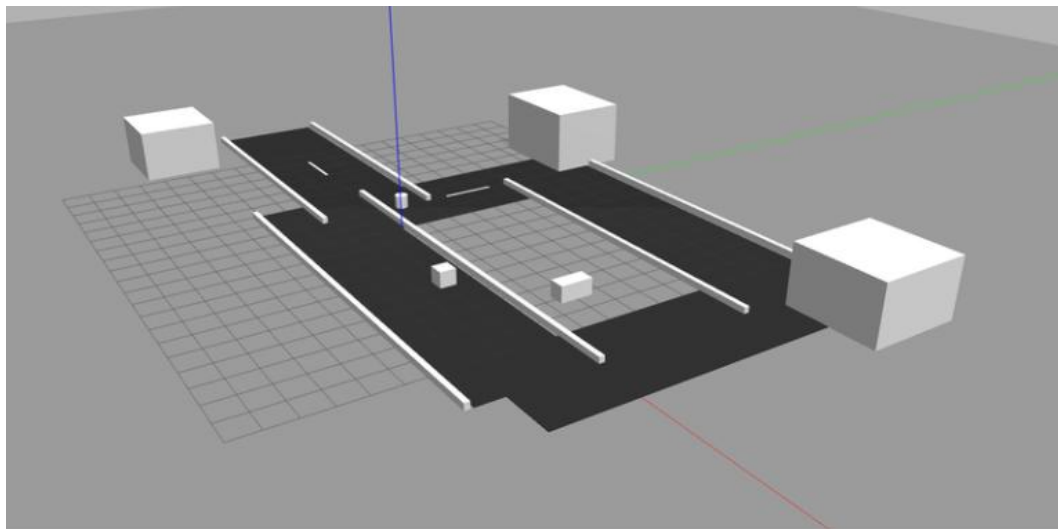


Figure 43 - Gazebo Test Map

Figure 42 and Figure 43 show the simulation and mapping process used before testing the system on the physical prototype. Figure 44 shows the map visualization in Gazebo, where the generated map, robot position, navigation goal, and planned path were displayed to check that the Nav2 system was working correctly. Figure 45 shows the LiDAR scanning process, where the 2D LiDAR detected the surrounding walls, obstacles, and free spaces to create the map of the environment. These two figures helped confirm that the robot could scan the area, build a usable map, localize itself, and plan a path before the software was connected to the real vehicle hardware.

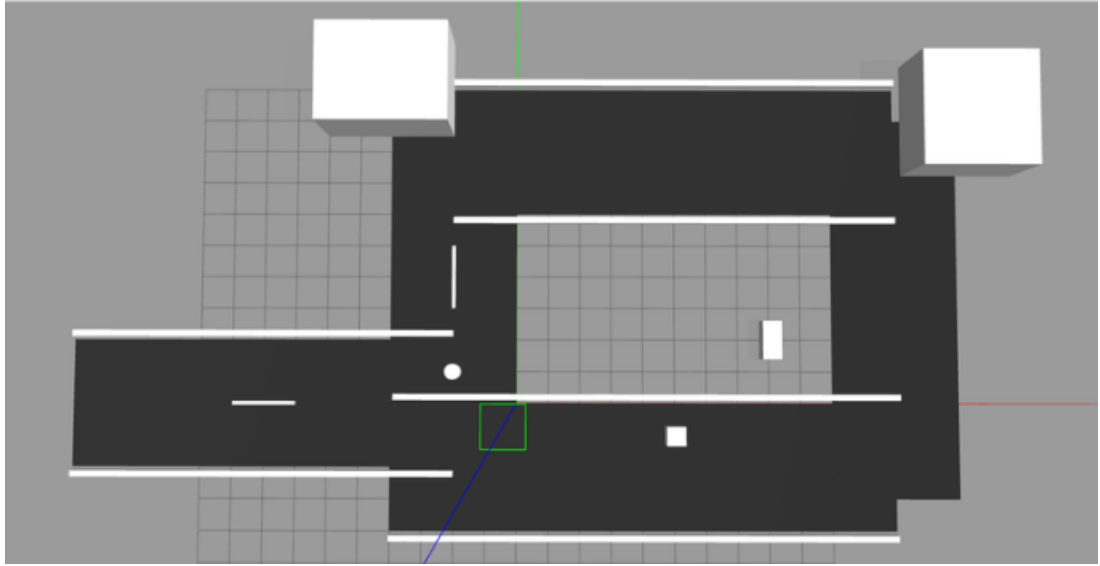


Figure 44 - Gazebo Map Visualization before Laser Scanning



Figure 45 - Rviz Map Visualization After Laser Scanning

4.2.2. *Sensor Integration*

The sensor integration focused on connecting the LiDAR and wide-angle camera to the ROS2 system running on the laptop. The LiDAR was used as the main sensor for map creation, obstacle detection, and localization. The camera was used for human detection to support the safety function of the vehicle.

The LiDAR data was published as scan data in ROS2. This scan data was used by the mapping system to create the map and by the navigation system to understand the surrounding environment. The LiDAR data was also used by the RF2O laser odometry node to estimate the vehicle movement.

The camera was integrated through a camera processing node. This node reads the camera feed and processes the image to detect humans. When a human is detected with confidence greater than 75%, the system sends a stop signal. This stop signal is used to prevent the throttle from continuing, which improves the safety of the vehicle during operation.

To make the sensor data meaningful in ROS2, static transforms were added between the vehicle base, LiDAR, and camera frames. These transforms tell ROS2 where each sensor is mounted on the vehicle. This is important because Nav2 needs to understand the sensor positions correctly in order to interpret obstacles, localization, and navigation data.

The map was created before navigation using the my_mapper package. The LiDAR scanned the testing area, and the mapping system generated a 2D map. After the map was saved, it was loaded later by Nav2 and used as the reference map for localization and path planning.

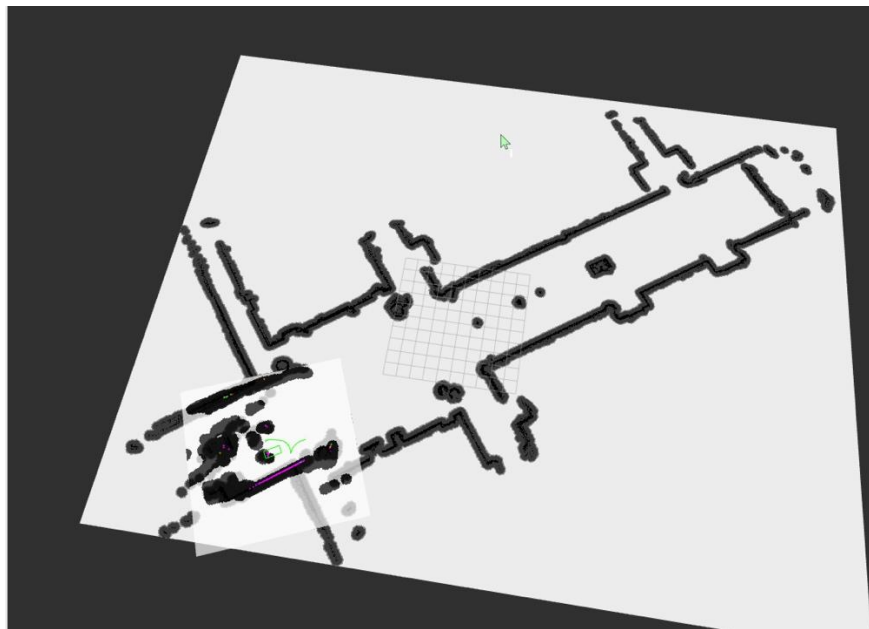


Figure 46 - Map Created Using my_mapper

4.2.3. System Communication

The system communication was designed to connect the high-level ROS2 software running on the laptop with the low-level hardware controlled by the Arduino. The laptop handled navigation, mapping, sensor processing, human detection, and decision-making, while the

Arduino controlled the steering servo and the two DC motors through the BTS motor driver. Additionally, giving feedback to laptop from the optical flow sensor.

The communication flow begins with either the website interface or RViz. In autonomous mode, the user selects the desired destination, and this goal is sent to Nav2. Nav2 uses the saved map, localization data, LiDAR data, and odometry feedback from the optical flow sensor to plan a path from the current vehicle position to the selected destination. After planning the path, Nav2 publishes movement commands on the `/cmd_vel` topic. This topic contains the required forward speed and turning direction.

The `ArduinoBridge.py` node receives the `/cmd_vel` commands and converts them into values that can be used by the Arduino. The forward speed command is converted into a throttle value for the DC motors, while the turning command is converted into a PWM signal for the steering servo. These converted values are then sent to Arduino through serial communication. The Arduino applies commands to the BTS motor driver and the steering servo, which allows the vehicle to move forward, stop, and steer left or right.

The safety stop path is also included in the communication flow. The camera processing node analyzes the camera feed to detect humans. If a human is detected with confidence higher than the selected threshold, the camera node publishes a stop signal on `/camera_stop_signal`. This signal is received by `ArduinoBridge.py`, which sends a stop command to the Arduino and prevents the throttle from continuing. This makes the camera safety system directly connected to the vehicle motion control. Figure 47 shows the complete communication flow between the ROS2 nodes, website interface, sensors, Arduino, and vehicle actuators.

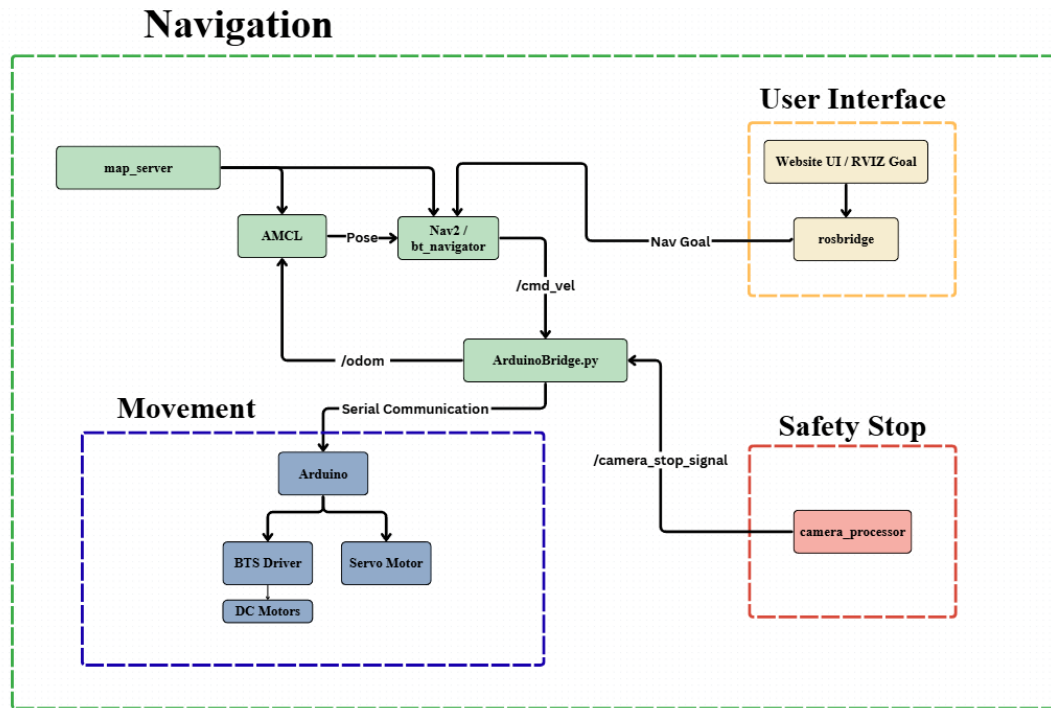


Figure 47 - System Communication Flowchart

The website interface was developed to make the system easier for the user to operate. Instead of sending commands only from ROS2 tools, the user can use the website to choose the desired destination from the map or from predefined locations. After the destination is selected, the website sends the goal to the navigation system, and Nav2 starts planning the path. Figure 48 shows the destination selection interface used by the user to choose the target location.

The website also includes a manual control page. This page was mainly used during testing to verify the throttle, steering, and stop commands before running the full autonomous mode. Through this interface, the user can send simple movement commands such as forward, backward, left, right, and stop. This helped confirm that the Arduino, BTS motor driver, motors, and steering servo were responding correctly. Figure 49 shows the manual drive interface used during testing.

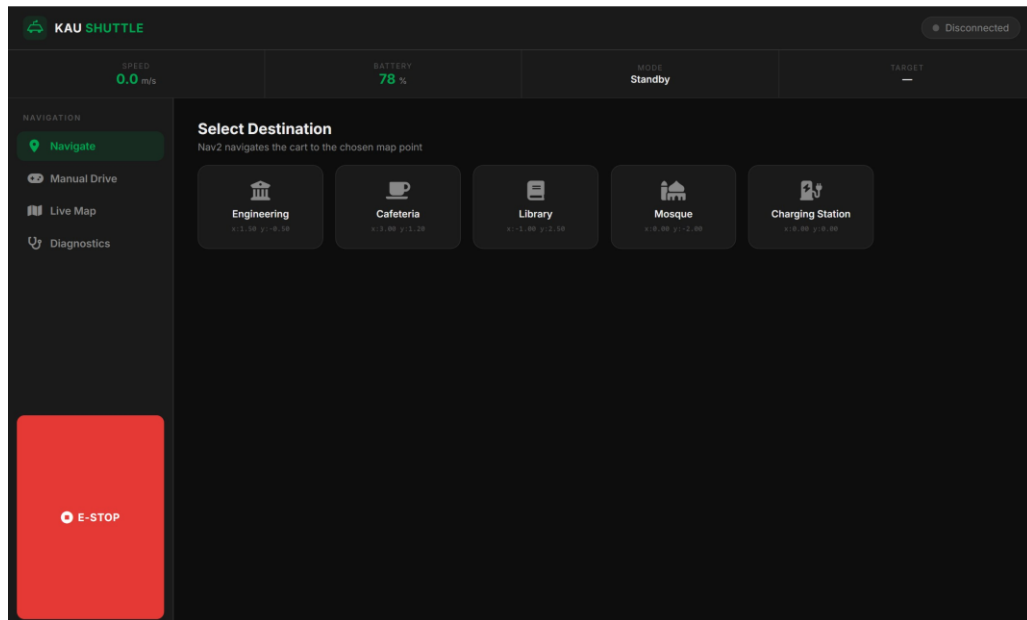


Figure 48 - Destination Selection Interface

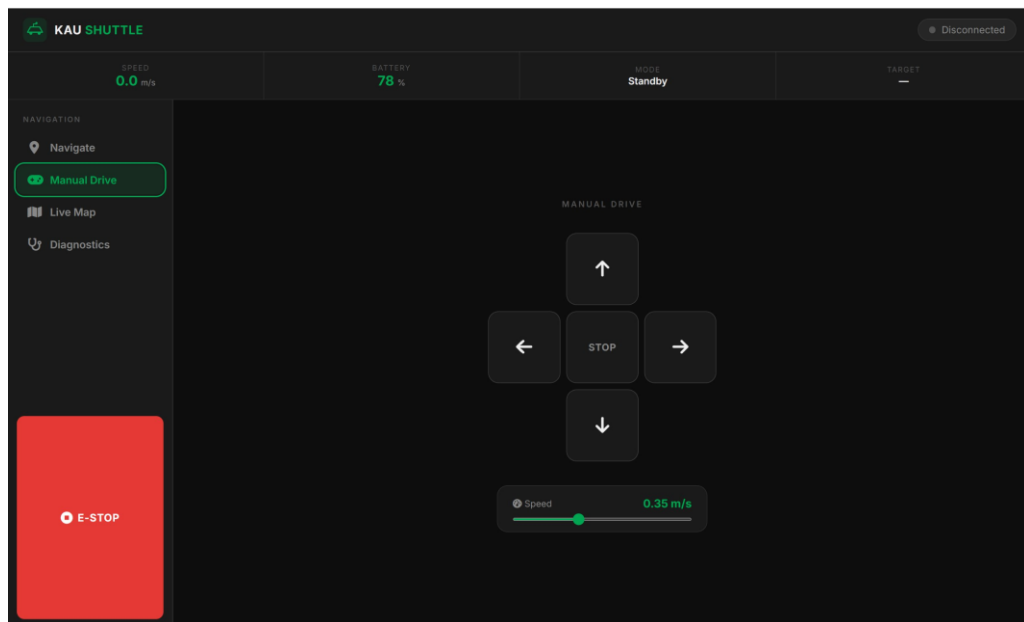


Figure 49 - Manual Drive Interface

4.2.4. Nav2 Navigation Algorithm

Nav2 was used as the main navigation framework for moving the vehicle from its current position to the selected destination. In simple terms, Nav2 receives a goal point, checks the map, plans a path, avoids occupied areas, and publishes movement commands. These commands are then sent to the Arduino to move the real vehicle.

The navigation process starts with the map. First, the LiDAR was used with the mapping package to create a 2D map of the testing area. The generated map shows free areas, occupied

areas, and unknown areas. After the map was created and saved, it was used by Nav2 as the reference map.

Next, the vehicle must know its position inside the map. This is done through localization. The map server loads the saved map, while AMCL estimates the position of the vehicle by comparing the real-time optical flow sensor data supported by the LiDAR scan with the saved map. This allows the vehicle to know where it is without using GPS.

After localization, the user selects the desired destination. During early testing, the goal was sent through RViz to check the navigation behavior. Later, the website was used to let the user choose the destination from the map. Once Nav2 receives the goal, the planner server calculates a path from the current vehicle position to the destination. Then, the controller server follows the path by generating velocity commands on `/cmd_vel`.

After simulation testing, the same navigation structure was applied to the physical vehicle. The simulated robot was replaced by the real prototype, the real LiDAR provided scan data, RF2O provided odometry, the camera provided human detection, and the Arduino bridge converted `/cmd_vel` into throttle and steering commands.

4.2.5. Software Calculations

To connect the ROS2 navigation commands with the physical actuators, the `/cmd_vel` output was converted into throttle and steering values. The linear velocity command was mapped to the motor driver range from -255 to +255, while the angular velocity command was mapped to the calibrated servo PWM range between 1200 μ s and 2100 μ s. This ensured that the vehicle responded to Nav2 commands while staying within safe actuator limits.

4.2.5.1. Throttle Command Mapping

Nav2 publishes movement commands on `/cmd_vel`, while the Arduino controls the motors using a throttle range from -255 to +255. So, the linear velocity command can be mapped to the motor command using:

$$Throttle = \frac{v}{v_{max}} \times 255$$

Where v = desired linear velocity from Nav2

v_{max} = maximum allowed vehicle velocity

4.2.5.2. Steering Command Mapping

The steering command from Nav2 can be converted into a PWM signal for the servo. Since the calibrated steering limits were 1200 μs for maximum left and 2100 μs for maximum right, the steering PWM can be mapped between these two limits.

$$PWM_{Steering} = PWM_{center} + \left(\frac{w}{w_{max}} \right) \times PWM_{range}$$

Where:

$$PWM_{center} = \frac{1200 + 2100}{2} = 1650 \mu s$$

$$PWM_{range} = \frac{2100 - 1200}{2} = 450 \mu s$$

So:

$$PWM_{steering} = 1650 + \left(\frac{\omega}{\omega_{max}} \right) \times 450$$

4.2.5.3. Human Detection Threshold

For the camera safety system, the detection result is accepted only if the human confidence is greater than **75%**.

$$Stop = \begin{cases} 1, & Confidence > 75\% \\ 0, & Confidence \leq 75\% \end{cases}$$

This means the throttle is stopped only when the camera detects a human with confidence higher than **75%**. This prevents weak or uncertain detections from stopping the vehicle unnecessarily.

4.3. IMPLEMENTATION TRIALS AND TESTING

Before the full system integration, each component was tested independently to make sure it worked correctly and safely. This step was important because it allowed the team to identify wiring issues, wrong signal values, or mechanical limits before connecting all subsystems together. The main trials focused on the throttle system, steering system, sensors, and communication between the laptop and Arduino.

4.3.1. Throttle and Steering Calibration Trials

The throttle and steering systems were calibrated separately before final integration. For the steering system, Arduino code was used to test different PWM signal values and define safe servo limits. This was done to prevent the servo from exceeding the mechanical range and damaging the steering mechanism or the 3D-printed gears. After testing, the steering limits were set to 1200 μs for maximum left steering and 2100 μs for maximum right steering. These values provided the best steering range while keeping the mechanism safe.

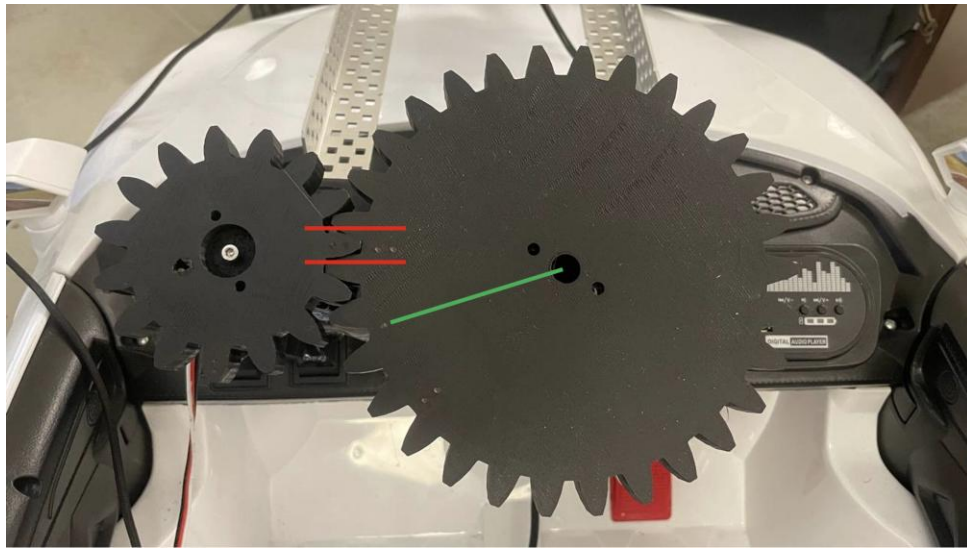


Figure 50 - Maximum Left Steer (PWM = 1200 μs)

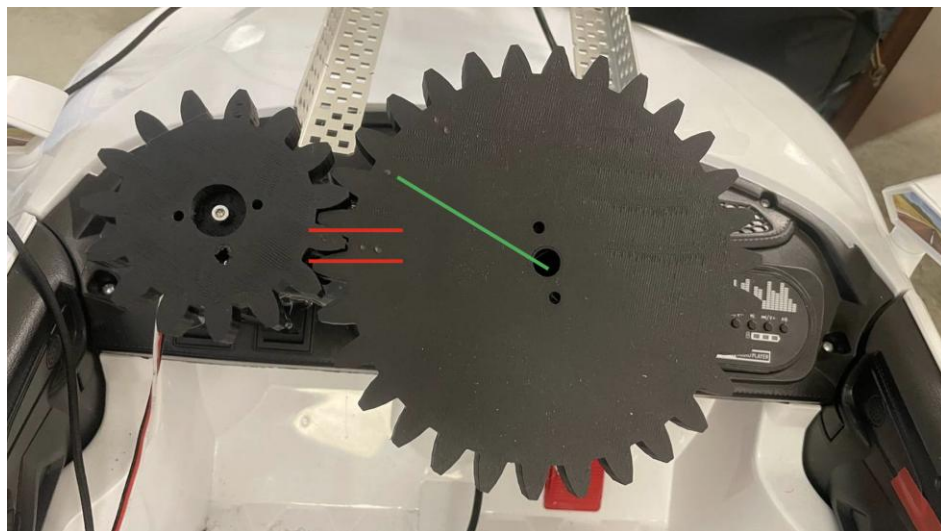


Figure 51 - Maximum Right Steer (PWM = 2100 μs)

For the throttle system, the two DC motors were tested through the BTS motor driver using Arduino commands from -255 to +255. Negative values controlled reverse direction, positive values controlled forward direction, and zero stopped the motors. The full motor

range was tested to confirm smooth response before connecting the throttle system to the laptop control logic.



Figure 52 - Throttle System Calibration

Figure 53 shows the serial monitor output during calibration. The printed values helped verify that the Arduino was receiving the desired inputs correctly and sending the correct commands to the servo motor and BTS motor driver.

```
--- System Ready ---  
Enter command format: THROTTLE,STEERING  
Example: 200,1900 (Forward, Right)  
Example: -150,1100 (Reverse, Left)  
Example: 0,1500 (Stop, Center)  
SUCCESS -> Throttle: 250 | Steering PWM: 1500  
SUCCESS -> Throttle: 0 | Steering PWM: 1800  
SUCCESS -> Throttle: -250 | Steering PWM: 1300  
SUCCESS -> Throttle: 0 | Steering PWM: 1600  
SUCCESS -> Throttle: 0 | Steering PWM: 1500
```

Figure 53 - Arduino Serial Monitor Output During Calibration

4.3.2. Sensor Testing Trials

After confirming that the LiDAR and camera were working individually, each sensor was tested with the throttle system to check the system response. The LiDAR was first combined with the throttle control to verify that the system could detect obstacles while the vehicle was moving. At this stage, the obstacle avoidance algorithm was not fully implemented, so the test focused on confirming that the LiDAR could detect an object and print the detection

status. This helped verify that the LiDAR data could be used later for stopping and avoidance decisions.

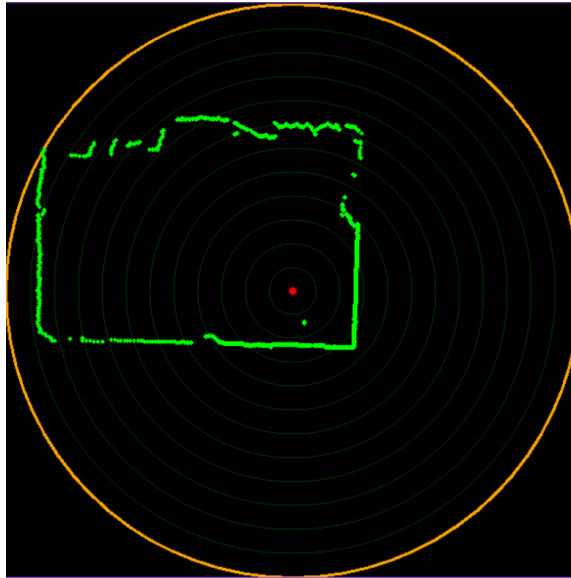


Figure 54 - LiDAR Test

The camera was then tested with the throttle system for human detection. The camera feed was processed by the software to detect humans, and when a human was detected with a confidence level greater than 75%, the throttle command was stopped. This test confirmed that the camera detection output could directly affect the vehicle motion and support the safety requirement of stopping when a human is detected.



Figure 55 - Camera Test

4.4. FINAL ASSEMBLY OF THE PROTOTYPE

After testing each subsystem separately, all components were assembled into the final prototype vehicle. The laptop was used to run the ROS2 system, while the Arduino controlled the steering servo, BTS motor driver, DC motors, and optical flow sensor feedback. The LiDAR and wide-angle camera were fixed using 3D-printed mounts, and the steering servo with the gear mechanism was installed near the steering shaft.

The wiring and power connections were checked and secured before final testing to ensure stable operation. Figure 56 shows the completed prototype after assembling the whole systems



Figure 56 - FINAL ASSEMBLY OF THE PROTOTYPE

CHAPTER – 5 VALIDATION EXPERIMENTS

This chapter includes experiments conducted to validate the final implemented prototype and to verify that it satisfies the project objectives and design specifications defined in Chapter 1. The final prototype used a laptop running ROS2 software as the main processing unit, an Arduino Uno as the low-level controller, a BTS7960 motor driver for the DC motors, a 2D LiDAR sensor, a wide-angle camera, and a solar charging station consisting of a solar panel and an MPPT charge controller. During testing, the measured values included vehicle speed, vehicle stopping distance, vehicle reaction time, destination arrival success, stop-decision accuracy, solar charging efficiency, and destination accuracy. The measured results were then compared with the design specifications; multiple trials are conducted. **Each** trial is marked as either a **PASS** if it was inside the required range, **PARTIAL PASS** if the error was up to 10%, or **FAIL** if the error was more than 10%.

5.1. TESTING METHODOLOGY

The validation methodology used for each performance parameter followed five main steps: identifying the parameter to be tested, preparing the prototype and test tools, conducting the test, recording the measured data, and comparing the result with the corresponding requirement from Chapter 1.

5.2. AUTONOMOUS SYSTEM TESTING

5.2.1. *Vehicle speed Test*

Objective:

This test was conducted to verify the maximum prototype speed is within the required vehicle speed range and to know the range needed to be measured in the stopping distance test.

Requirement/ target:

the required vehicle maximum speed range is 0 to 25 km/h

Setup and tools:

The test was performed using the final implemented prototype, which consisted of a laptop running ROS2, an Arduino Uno, and a BTS7960 motor driver controlling the DC motors. A meter to test the distance travelled, Timer to clock the time taken to cover the distance.

Procedure and Results:

1. A straight test path of 3 m was prepared.
2. The prototype vehicle was placed at a fixed starting point each time.
3. The vehicle was commanded to move forward at maximum speed.
4. The time taken to travel the 3 m distance was recorded.
5. The speed was calculated using distance divided by time, then converted from m/s to km/h.

Table 26 : Recordings of the Speed Test

Trial	Time (s)	Speed (m/s)	Speed (km/h)	Status
1	4.52	0.664	2.39	PASS
2	4.49	0.668	2.41	PASS
3	4.57	0.656	2.36	PASS
4	4.46	0.673	2.42	PASS
5	4.54	0.661	2.38	PASS



Figure 57: Starting Point for the Car



Figure 58: Three Meter Path

Analysis:

The measured speed was between 2.36 km/h and 2.42 km/h, with an average speed of about 2.39 km/h. The vehicle stayed inside the required speed range of 0 to 25 km/h. The results were close to each other in all five trials, which means the speed was stable during testing. Based on this, **the vehicle speed test matches our design specifications.**

5.2.2. Object detection and Reaction time Test

Objective:

This test was conducted to verify that the prototype could detect an object in its path and respond fast enough to stop safely during operation.

Requirement/target:

The obstacle detection sensors must reliably identify objects each time it's tested to reach a 100% accuracy to stop, the reaction time to stop must be in the range of 0 to 0.25 s.

Setup and tools:

The test was performed using the final implemented prototype, which consisted of a laptop running ROS2, an Arduino Uno, a BTS7960 motor driver, a wide-angle camera, and the object detection model used in the final system. The camera-based model was used to detect objects, while the reaction time was taken from the ROS2 terminal output and the detection distance was measured during each trial.

Procedure and Results:

1. The prototype was placed in front of the test object.
2. The object was positioned at a known distance from the prototype.
3. The detection system was started using the final ROS2 camera node.
4. The model confidence and detection distance were recorded for each trial.
5. The reaction time was taken from the terminal output when the stop command was triggered.
6. The measured results were compared with the required target values from Chapter 1.

Table 27: Recordings of the Object Detection and Reaction Time Test

Trial	Model Confidence (%)	Distance (m)	Reaction time	Status
1	95	1.5	0.00010	PASS
2	92	3.8	0.00011	PASS
3	90	7.2	0.00011	PASS
4	81	9.5	0.00012	PASS
5	86	11.7	0.00013	PASS

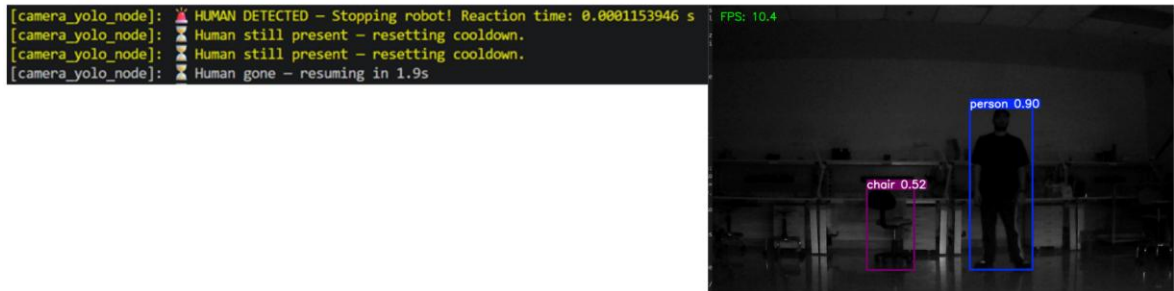


Figure 59: Model confidence and Reaction Time – Third Trial

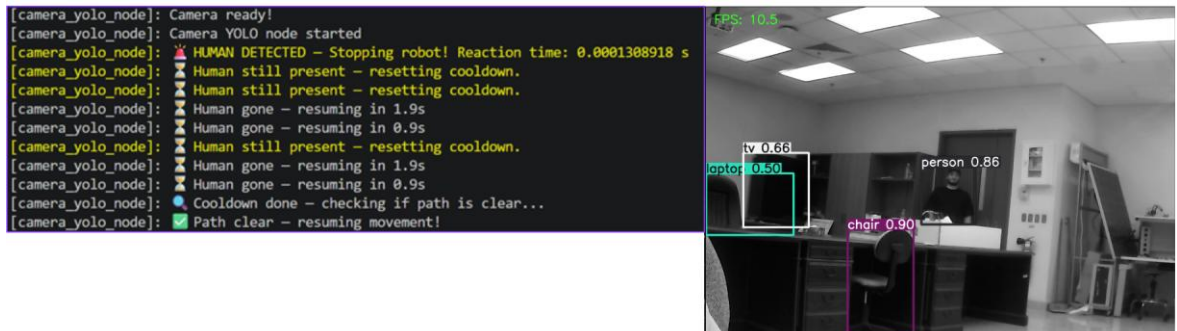


Figure 60: Model confidence and Reaction Time – Fifth Trial

Analysis:

The system successfully detected the human in all five trials and triggered the stop response correctly each time. The recorded reaction times were also below the required limit of 0.25s. Based on these results; **the object detection and reaction time test matches our design specifications**. The distance recorded in the table does not affect the test but mentioned for more details on the test.

5.2.3. Stopping distance Test**Objective:**

This test was conducted to verify that the prototype could stop safely at the specified range after detecting an object while operating at its maximum tested speed.

Requirement/target:

The EV stopping distance at 25 km/h should be in the range of 3.66 m to 5.49 m, while lower speeds require shorter stopping distances. Since the implemented prototype reached a maximum tested speed of about 2.39 km/h, which is below the 5 km/h range listed in Table 1, this test was performed at the prototype's maximum speed to ensure the stopping distance must be in a range from 0 – 0.15 cm.

Setup and tools:

The test was performed using the final implemented prototype, which consisted of a laptop running ROS2, an Arduino Uno, a BTS7960 motor driver, a wide-angle camera, and the stopping logic used in the final system. A measuring tape was used to measure the stopping distance from the point where the stop command was triggered until the vehicle came to a complete stop.

Procedure and Results:

1. A straight test path was prepared.
2. The prototype vehicle was placed at a fixed starting point.
3. The vehicle was commanded to move forward at its maximum tested speed.
4. An object was presented to trigger the stop action.
5. The stopping distance was measured from the detection point until the vehicle fully stopped.
6. The test was repeated for several trials, and the measured values were recorded.

Table 28: Recordings of the Stopping Distance Test

Trial	Stopping Distance (cm)	Status
1	0.13	PASS
2	0.18	PARTIAL PASS
3	0.14	PASS
4	0.17	PARTIAL PASS
5	0.14	PASS

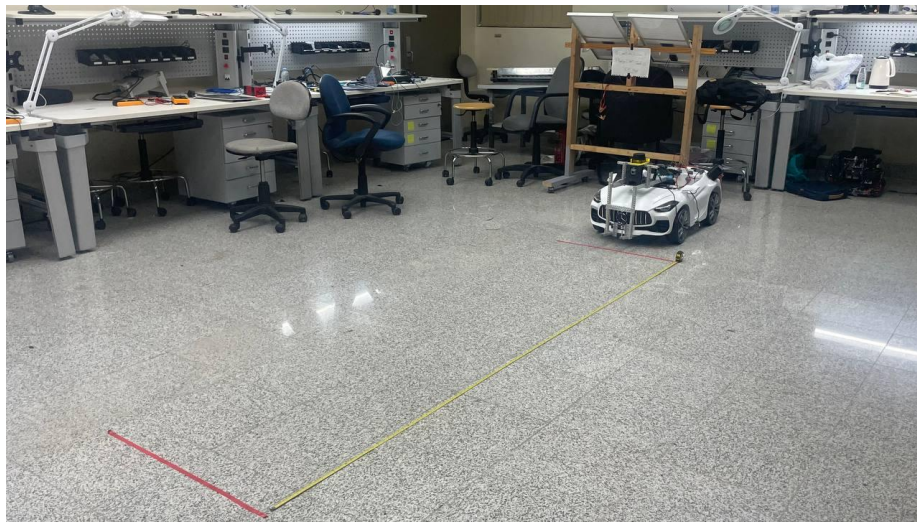


Figure 61: Measured Path to Calculate Stopping Distance

Analysis:

The measured stopping distance ranged from 0.13 m to 0.18 m during the five trials. Since the prototype was tested at a maximum speed of about 2.39 km/h, which is lower than the 5 km/h range from 0 - 0.15 cm, these short stopping distances show that the vehicle was able to stop safely within a very small range. The results were also close to each other, which means the stopping behavior was stable during testing. **Based on these results, the stopping distance test matched the design specification for the implemented prototype.**

5.2.4. Destination accuracy Test

Objective:

This test was conducted to verify that the prototype could reach the selected destination accurately during autonomous movement.

Requirement/target:

The required destination accuracy is 95% to 100%, so the vehicle must reach the exact destination within a 1 m radius related to a selected goal from the map. Which will be marked with a tape to know the exact location that car should reach and the distance from the marker center to Lidar will be the distance that is used to test our specification in the software.

Setup and tools:

The test was performed using the final implemented prototype, which consisted of a laptop running ROS2, an Arduino Uno, a BTS7960 motor driver, a 2D LiDAR, and a wide-angle camera. The destination accuracy test was based on the prototype's position relative to the selected target point on the map.

Procedure and Results:

1. A target destination point was selected on the prepared map and marked in real life.
2. The prototype vehicle was placed inside the map.
3. The autonomous navigation system was started, and the vehicle was commanded to move toward the selected destination.
4. The final stopping position of the prototype was compared with the intended destination point on the map.
5. The distance error between the center of the target point and the final lidar on the vehicle position was recorded for each trial.
6. The measured result was compared with the required 1 m destination radius and the required destination accuracy percentage.

Table 29 - Destination accuracy Test Results

Trial	Out of 1 m Destination Error (m)	Within 1 m Radius	Status
1	0	0.11	PASS
2	0	0.24	PASS
3	0	0.41	PASS
4	0	0.56	PASS
5	0	0.72	PASS



Figure 62: First Destination Marker - Trial 1

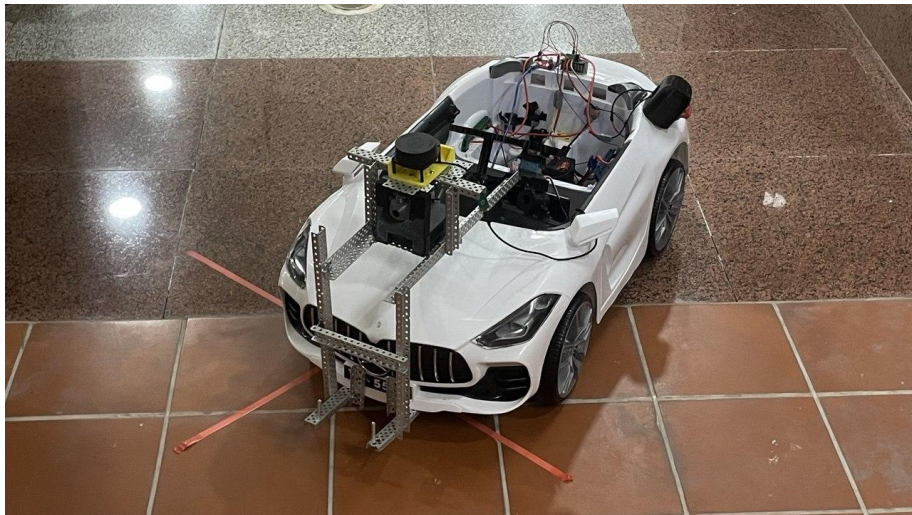


Figure 63: Car at Marker - Trial 1

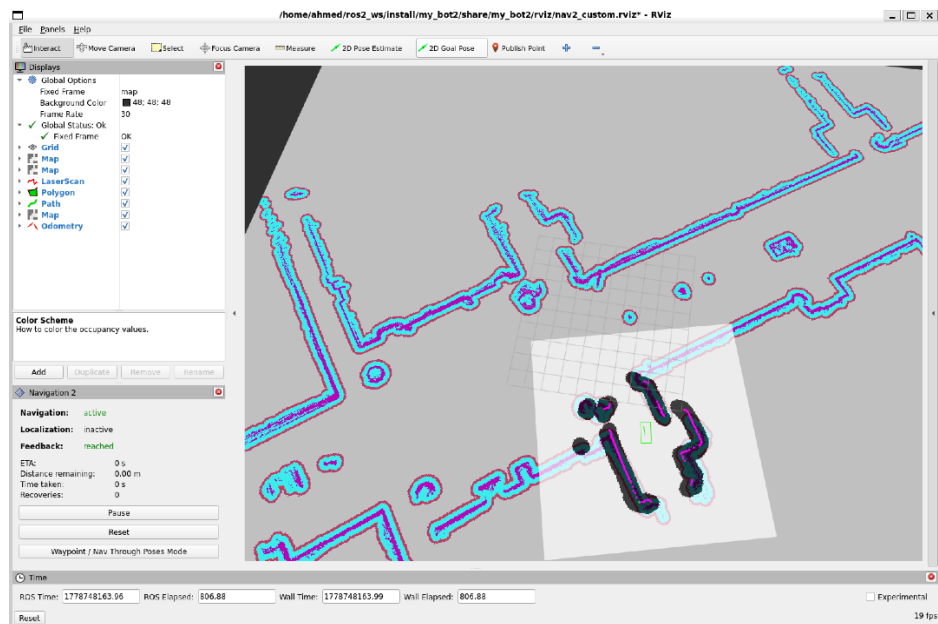


Figure 64: Rviz Visual for car location - Trial 1



Figure 65: Destination Marker - Trial 4



Figure 66: Car at Marker - Trial 4

Analysis:

All five tests passed. The car always reached the right destination within the 1-meter limit. The distance from the exact center went from 0.11 m in test 1 to 0.72 m in test 5. This happened because we changed the Nav2 settings on purpose.

After the first test, we made the distance rule less strict so the car could drive better. If the rule is too strict, the car moves back and forth trying to find the exact spot. By letting the car stop as soon as it enters the 1-meter area, it reaches the destination much more smoothly and still meets the project goals perfectly. **Based on these results, the stopping distance test matched the design specification for the implemented prototype.**

5.3. POWER SYSTEM TESTING

5.3.1. Solar Charging Efficiency

Objective:

To evaluate the efficiency of the solar charge controller by comparing input power from the solar panel to output power delivered to the battery.

Requirement/target:

The charger should achieve an efficiency between 75% and 85% under normal charging conditions.

Setup and tools:

- Solar panel: 18V, 100W
- MPPT charge controller
- Lithium battery: 12V
- ESP32 monitoring circuit
- Two INA3221 current and voltage monitoring sensors
- Custom web-based monitoring dashboard
- Test condition: 2:00 PM sunny day

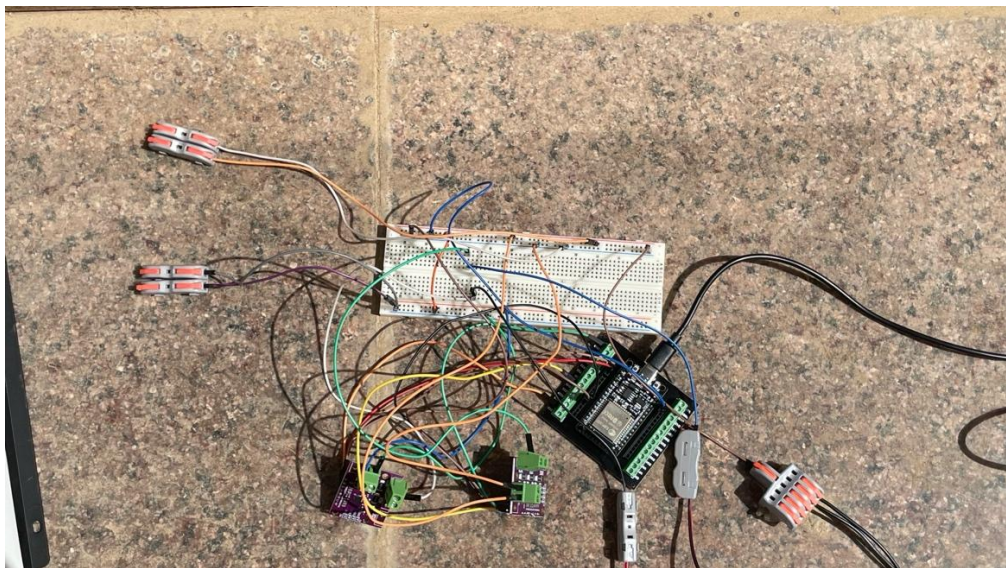


Figure 67 - Monitoring Circuit

Monitoring System Setup:

A custom monitoring circuit was developed using an ESP32 microcontroller and two INA3221 sensors. One INA3221 sensor was connected to the solar panel side to measure

solar voltage and current, while the second INA3221 sensor was connected to the battery charging side to measure battery voltage and charging current.

The ESP32 collected the sensor readings through two separate I2C buses and processed the measurements in real time. A web-based dashboard was created to display:

- Solar voltage, current, and power
- Battery voltage, charging current, and charging power
- System efficiency
- Historical measurement logs
- The ESP32 hosted the website locally over Wi-Fi, allowing the measurements to be monitored from any connected device through a web browser.
- The system also automatically stored the readings in a CSV log file. The ESP32 generated the CSV data from the collected sensor readings and made it available for download through the website dashboard for later analysis and graph generation.

Measurements include:

- Solar voltage (V_{solar})
- Solar current (I_{solar})
- Battery voltage ($V_{battery}$)
- Charging current (I_{charge})



Figure 68 - Charging Connection Setup

Procedure:

1. Connect the solar panel to the MPPT charge controller input.
2. Connect the battery to the MPPT charge controller output.
3. Connect the INA3221 monitoring sensors to the solar and battery sides.
4. Power the ESP32 monitoring system.
5. Open the monitoring dashboard through the ESP32 local IP address.
6. Measure and monitor the solar voltage and current.
7. Measure and monitor the battery voltage and charging current.
8. Record the readings automatically through the CSV logging system.
9. Calculate the input and output power values.
10. Compute the charging efficiency using the equations below.

Equations:

Input Power:

$$P_{in} = V_{solar} \times I_{solar}$$

Output Power:

$$P_{out} = V_{battery} \times I_{charge}$$

Efficiency:

$$\eta = (P_{out} / P_{in}) \times 100\%$$

Results:

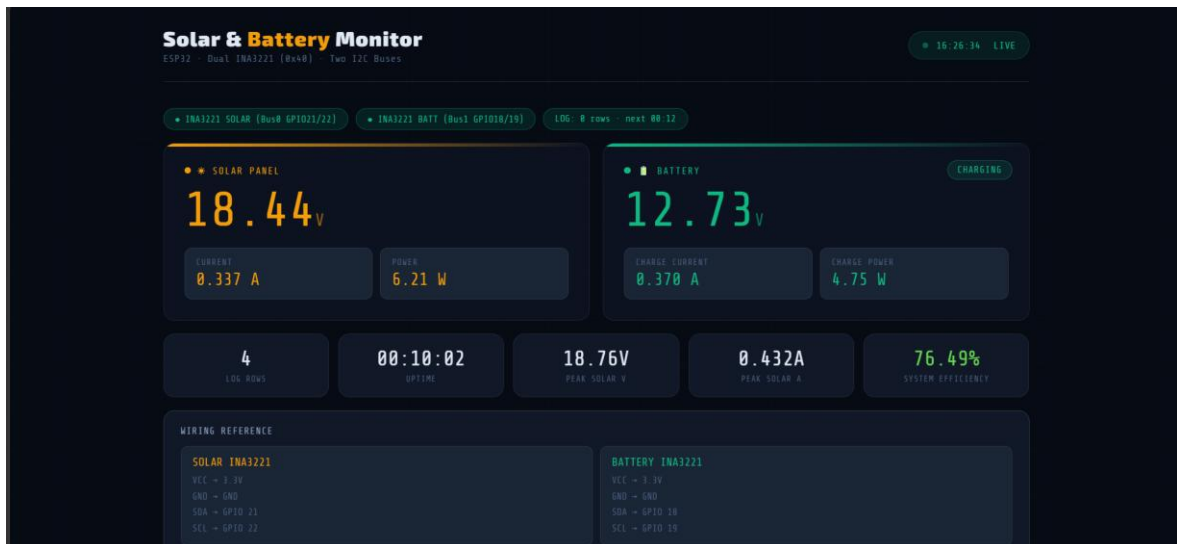


Figure 69: Solar Charging Efficiency Monitoring Dashboard and Measurement Readings

Table 30 - Solar Charging Efficiency Test Results for the MPPT Charge Controller

Time (min)	Solar Voltage (V)	Solar Current (A)	Input Power (W)	Battery Voltage (V)	Charging Current (A)	Output Power (W)	Efficiency (%)
0	18.21	0.284	5.17	12.52	0.311	3.89	75.24
10	18.44	0.337	6.21	12.84	0.370	4.75	76.49
20	18.47	0.405	7.48	13.12	0.448	5.88	78.61
30	18.31	0.468	8.57	13.39	0.520	6.96	81.21
40	18.35	0.411	7.54	13.66	0.459	6.27	83.16
50	18.30	0.331	6.06	13.93	0.364	5.07	83.66
60	18.34	0.231	4.24	14.37	0.246	3.53	83.25

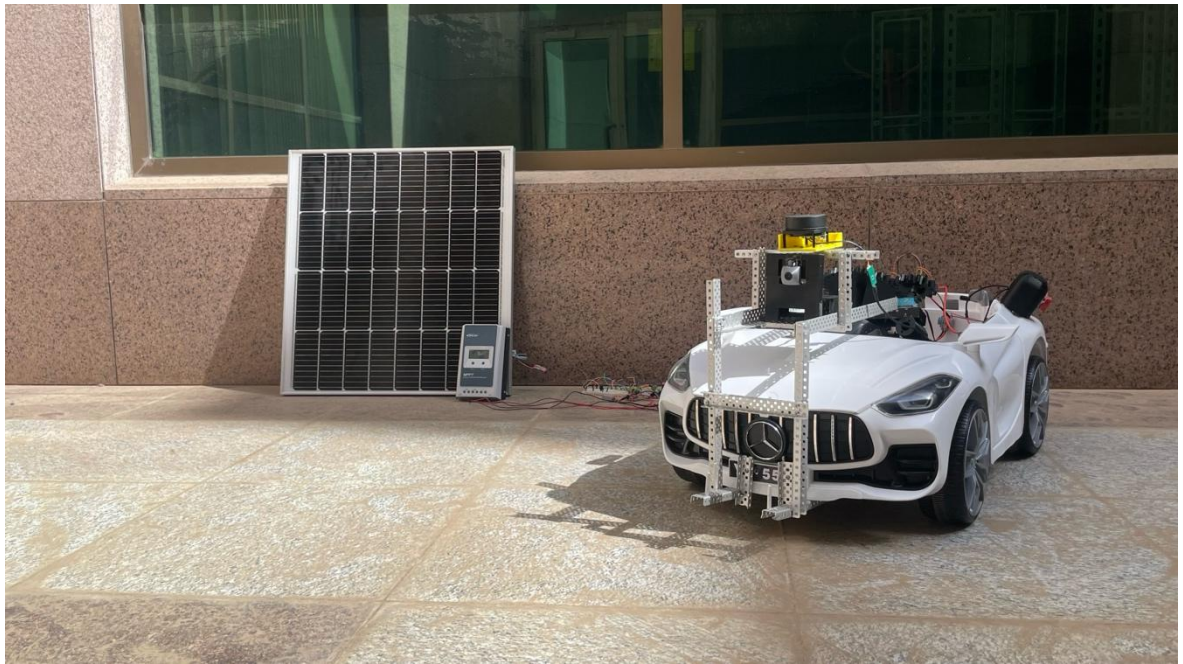


Figure 70 - Charging Station

Analysis:

The calculated efficiency ranged between 75.24% and 83.66%, which satisfies the required target range of 75%–85%. Therefore, the MPPT charge controller achieved a Pass under the tested low-current charging condition.

The monitoring circuit and web dashboard successfully displayed real-time solar and battery measurements while automatically storing the readings in CSV format for analysis. The solar voltage and current fluctuated slightly due to changes in sunlight intensity and panel operating conditions.

The battery voltage increased from 12.52V to 14.37V, indicating successful battery charging. As the battery approached a higher charge level, the charging current decreased slightly, which is normal behavior for lead-acid batteries during the absorption stage.

Some power losses occurred due to heat dissipation, MPPT conversion losses, wiring resistance, battery internal resistance, and minor measurement inaccuracies. Overall, the monitoring system and MPPT charger demonstrated stable and efficient performance throughout the experiment.

CHAPTER – 6 DISCUSSION AND CONCLUSION

6.1 EVALUATION OF SOLUTION

The final prototype was evaluated based on the project objectives, design specifications, validation experiments, and the customer's needs. The customer required an automated electric vehicle prototype supported by a renewable solar charging system, with safe movement and automatic stopping capability. The implemented solution satisfied these needs by combining ROS2/Nav2 navigation, LiDAR sensing, camera-based human detection, Arduino control, and a solar charging station.

The autonomous system met the main requirements for movement, safety, and control. The vehicle operated within the required speed range of 0–25 km/h, detected humans successfully in all trials, and stopped with a reaction time below the required 0.25 s. The stopping distance test also confirmed that the vehicle could stop safely at the tested speed. The navigation objective was achieved using the map-based ROS2/Nav2 system, where the vehicle used LiDAR, optical flow sensor feedback, and the saved map to move toward the selected destination from the website or RViz.

The renewable energy objective was also achieved by implementing the solar charging station using the solar panel, MPPT charge controller, battery, fuse, and main switch. **Overall, the validation results show that the implemented solution met the customer's requirements and the main project specifications.**

6.2 IMPACT OF SOLUTION

This project contributes to renewable energy utilization and autonomous transportation technologies. Environmentally, the use of solar energy reduces dependency on conventional charging methods and supports clean energy applications. Socially, autonomous electric vehicles can improve transportation efficiency and reduce human workload inside campuses and controlled environments.

Economically, the project demonstrates a low-cost prototype that can be scaled into larger systems in the future. From a safety perspective, the integration of LiDAR and computer vision improves obstacle detection and reduces the risk of collisions during vehicle operation.

The project also supports Saudi Arabia Vision 2030 goals related to sustainability, renewable energy adoption, and smart technology development.

6.3 FUTURE WORK

Several improvements can be made to future versions of the project:

- Replace the 2d lidar with 3d lidar for more advance navigation
- Replace the laptop with an embedded onboard computer such as Jetson Orin Nano.
- Add larger solar panels and higher-capacity batteries for longer operation.
- Implement wireless monitoring for battery status.
- Improve obstacle classification accuracy using deep learning models.
- Integrate additional sensors such as IMU and depth cameras.
- Test the system on larger electric vehicles and outdoor environments.

6.4 CONCLUSION

This project presented the design and implementation of an automated electric vehicle powered by solar energy. The main goal was to improve a small electric vehicle by adding self-driving functions and a solar charging system. The project aimed to make the vehicle safer, smarter, and more environmentally friendly.

The project had two main parts: the autonomous system and the solar power system. The autonomous system used ROS2, Nav2, LiDAR, a camera, Arduino, and control parts for steering and throttle. These parts helped the vehicle move, detect obstacles, and stop when needed. The solar power system used a solar panel, MPPT charge controller, battery, fuse, and switched to charge the vehicle using solar energy.

During implementation, the hardware and software were connected into one complete system. Many parts were installed, tested, and adjusted to make sure the vehicle worked

correctly and safely. The final prototype showed that all parts of the system could work together in a good way.

The test results showed that the project achieved its main goals. The vehicle was able to detect obstacles, react to them, and stop within a safe distance. It was also able to move toward selected destinations with acceptable accuracy during the tests.

The solar charging system also gave good results. It showed acceptable charging efficiency under normal conditions, which means the selected solar parts were suitable for the prototype. This proves that solar energy can be used as a useful charging source for small electric vehicles.

This project also had important learning value. It helped in understanding how to combine automation, electronics, sensors, control systems, and renewable energy in one project. It also gave practical experience in building, testing, and improving a real engineering system.

To conclude, **THE PROJECT PROVED THAT IT IS POSSIBLE TO COMBINE AUTONOMOUS DRIVING AND SOLAR ENERGY IN ONE ELECTRIC VEHICLE PROTOTYPE. EVEN THOUGH THE SYSTEM STILL NEEDS SOME FUTURE IMPROVEMENTS, THE PROJECT WAS A SUCCESSFUL STEP TOWARD SMART, SAFE, AND SUSTAINABLE TRANSPORTATION.**

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APPENDIX – A: EVALUATION COMMENTS

A.1 IDENTIFYING THE PROBLEM AND DESIGN REQUIREMENTS

- Revise the objectives for more clarity. Include only the technical constraints. Missing project deliverables. Clarify the project design specifications and include the details for the proposed rehabilitation and performance phases. Check the standards.
- You can make autonomy as one of your objectives.
- Well done.

A.2 CONCEPTUAL DESIGNS

- Comprehensive report.
- Why is Section 1.2 (Problem Statement) placed on a separate page? The previous page contains a large unused blank space. Please adjust the layout to ensure proper use of space and continuous formatting.
- Appendix A is included in the report; however, it does not contain the required Evaluation Comments from the previous evaluation stages. Please add all missing evaluation feedback to this appendix to ensure completeness.

A.3 TERM 1 REPORT AND PRESENTATION

- No comments

A.4 PROGRESS UPDATE

- The details of the crucial components and the control system were not clearly presented.
- No design calculations were provided.
- The validation experiment plan needs to be more specific and clearly aligned with the stated specifications and performance metrics.
- Vague work plan.
- Well done
- Please use the remaining time of the semester wisely to ensure that you complete your

project on time.

- The students have made a very strong effort to deliver good-quality work, and their progress has been very good.

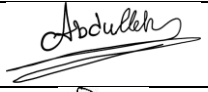
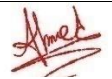
A.5 FINAL REPORT AND PRESENTATION

APPENDIX – B: EFFECTIVE TEAM INTERACTIONS

<i>Team Members</i>					
<i>Name</i>	<i>ID</i>	<i>Specialization</i>	<i>Email Address</i>	<i>Role & Experience</i>	<i>Photograph</i>
Abdullah Binsalman	2236900	Electrical & Computer Engineering (COE)	Abinsalman0011@stu.kau.edu.sa	Control System & integration	
Mohammed Al-Baiti	2250367	Electrical & Computer Engineering (COE)	malbaiti0008@stu.kau.edu.sa	Hardware & Circuit Designer	
Ahmed Al-Harthy	2237094	Electrical & Computer Engineering (COE)	Aalharthy0204@stu.edu.kau.sa	Embedded Systems & Software Developer	

Date:	21/09/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Decide which feature we will add to the car
- Identifying the problem & requirements
- Distribute tasks among the team

Discussion Points:

Discussion on the features to be added such Ai, battery enhancement, and thermal systems

Follow-up of the Last Meeting:

--

Decisions Taken:

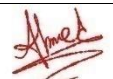
Next meeting will start working on report

Actions to do before next meeting:

Research materials on chapter 1 of the report of section Background, Problem Statement, And Project Objectives,

Date:	24/09/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Review of Project Specifications: Protocol for reviewing the forthcoming project specifications document. - Implementation Strategy: Discussion of the technical implementation plan for the battery switching and monitoring systems. - Report Status Update: Assessment of progress on the "Background," "Problem Statement," and "Project Objectives" sections of the project report.
--

Discussion Points:

- Battery Switching Logic: The team discussed the optimal control strategy for the dual-battery system. - Predictive Maintenance Data Requirements: The team discussed the data acquisition strategy necessary for the predictive maintenance model. - Project Timelines and Deliverables: Firm deadlines for outstanding action items were discussed to ensure project momentum is maintained.

Follow-up of the Last Meeting:

Finalizing sections in chapter 1 of the report
--

Decisions Taken:

Next meeting will finalize the first chapter in this project.

Actions to do before next meeting:

Searching for Design Specifications materials and resources to be used
--

Date:	27/09/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

• Discussion of the first report with the advisor

Discussion Points:

• Review of Chapter 1 • Focus on project objectives
--

Follow-up of the Last Meeting:

• Report 1 completion

Decisions Taken:

• Revise project objectives

Actions to do before next meeting:

• Complete and revise the report • Prepare for presentation in the next meeting
--

Date:	30/09/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Revise the first report again
- Choose a template for the first presentation
- Create structure for the presentation
- Distribute presentation sections

Discussion Points:

- Reviewed report content
- Selected a suitable presentation template
- Structured the presentation
- Assigned sections to team members

Follow-up of the Last Meeting:

- Report revised

Decisions Taken:

- Presentation template selected
- Tasks distributed among team members

Actions to do before next meeting:

- Each member completes their assigned part of the presentation

Date:	24/10/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Revise report 1 feedback

Discussion Points:

- Solar panel dependency
- Add autonomy to the project
- Stopping distance clarification

Follow-up of the Last Meeting:

--

Decisions Taken:

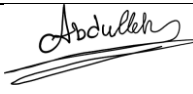
- Using solar panel as main charging source instead of assistance
- Add autonomous mood to the car
- Revise the whole report again

Actions to do before next meeting:

- Get clarification for one of the comments

Date:	25/10/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Discuss adding autonomy to the project
- Start planning for the conceptual design

Discussion Points:

- Determine the needed part
- How big the project will be
- What we can ignore and replace with autonomy

Follow-up of the Last Meeting:

- Confirm solar power using

Decisions Taken:

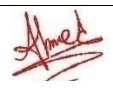
- Determine some essential parts and parts to search about
- See different autonomy options
- May ignore the monitoring system

Actions to do before next meeting:

- Search more about autonomy options

Date:	28/10/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

Discuss different autonomy options
--

Discussion Points:

- ROS option - Ready-made model option - Line following

Follow-up of the Last Meeting:

Delete the monitoring system
--

Decisions Taken:

We need to search more to decide which one will be used

Actions to do before next meeting:

• Search about each one of the options and see their cons and pros	
Date:	30/10/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Discuss the options after the searching

Discussion Points:

- ROS
- Some ready-made models
- Line following

Follow-up of the Last Meeting:

- Done studying about the options

Decisions Taken:

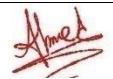
- ROS over kill
- Line following is too simple
- Working on ready-made models

Actions to do before next meeting:

- Search to find the best models and tools to use

Date:	1/11/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

• Pixhawk and its configurations • GPS module
--

Discussion Points:

• Plan to use ready-made Pixhawk configuration for the EV • Choose a suitable GPS module

Follow-up of the Last Meeting:

• Find Pixhawk option

Decisions Taken:

• Search more for the configurations • Choose m8n GPS module

Actions to do before next meeting:

• Find some related configurations
--

Date:	4/11/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Working on baseline design - Revise report 1 for needed changes
--

Discussion Points:

- Cost estimation - Circuit design - Needed parts

Follow-up of the Last Meeting:

Continue working on the report
--

Decisions Taken:

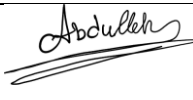
- Determine the cost estimation - Determine the needed parts - Finish circuit diagram

Actions to do before next meeting:

Finishing report 2
--

Date:	7/11/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

Revise report 2 before submission

Discussion Points:

Reading the report and find the mistakes and non-clarify points

Follow-up of the Last Meeting:

Finished report 2

Decisions Taken:

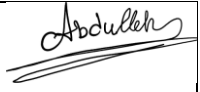
The report is ready to submit

Actions to do before next meeting:

Submit the report

Date:	20/11/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Review Report 2 submission feedback
- Discuss transition from design phase to implementation phase
- Identify the required hardware parts for the prototype
- Divide implementation tasks among team members

Discussion Points:

- The team reviewed the submitted Report 2 and discussed the sections that may need improvement in the next report version.
The team discussed the main components required for the implementation stage, including the solar panel, MPPT charge controller, battery, LiDAR, camera, controller, motor driver, and steering mechanism.
The team discussed how the autonomous system and power system will be integrated together in the final prototype.
Responsibilities were divided between control system, hardware/circuit design, and embedded/software development.

Follow-up of the Last Meeting:

- Report 2 was completed and submitted.

Decisions Taken:

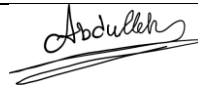
- Start preparing for the implementation phase.
- Use the current baseline design as the main guide for hardware and software work.
- Prepare a final purchasing and component-checking list before starting assembly.

Actions to do before next meeting:

- Review all required parts and check their availability.
- Prepare the implementation schedule.
- Start checking possible mechanical mounting methods for sensors and steering parts.

Date:	04/12/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Finalize implementation plan
- Confirm electrical and mechanical requirements
- Discuss possible changes to the baseline design
- Prepare for ordering parts

Discussion Points:

- The team reviewed the complete system design and checked if all selected components can be practically installed on the prototype vehicle.
- The power system connection was discussed, including the solar panel, MPPT charge controller, battery, fuse, and main safety switch.
- The team discussed the mechanical difficulty of installing the steering actuator and sensor mounts.
- The team discussed whether any changes are needed due to cost, availability, or implementation difficulty.

Follow-up of the Last Meeting:

- Required parts were reviewed and listed.
- Initial implementation schedule was prepared.

Decisions Taken:

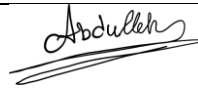
- Keep the solar charging system as a separate charging station instead of permanently mounting the solar panel on the vehicle.
- Add fuse protection and a main switch for safety during testing.
- Continue with the selected autonomous system components.

Actions to do before next meeting:

- Prepare the final bill of materials.
- Start ordering the unavailable parts.
- Begin designing the mechanical mounts for the LiDAR and camera.

Date:	18/12/2025
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Review ordered components
- Discuss mechanical design of sensor mounts
- Discuss software preparation before hardware arrival
- Update project work plan

Discussion Points:

- The team checked the ordered components and confirmed the expected arrival dates.
- The LiDAR and camera mounting positions were discussed to make sure both sensors can cover the required area clearly.
- The team discussed using 3D-printed mounts to make the sensor installation stable.
- The software team started preparing the required environment for ROS2, camera processing, and LiDAR testing.
- Possible problems related to wiring, power distribution, and component protection were discussed.

Follow-up of the Last Meeting:

- Final bill of materials was prepared.
- Main hardware components were selected and ordering started.

Decisions Taken:

- Use 3D-printed mounts for the LiDAR and wide-angle camera.
- Prepare the software environment while waiting for all hardware parts to arrive.

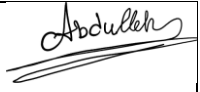
- Keep the implementation plan flexible in case some parts arrive late.

Actions to do before next meeting:

- Continue tracking the ordered parts.
- Start preparing the 3D models for the sensor mounts.
- Install and test the required software packages.

Date:	05/01/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Start official implementation stage
- Confirm parts arrival and remaining missing items
- Begin power system assembly planning
- Assign detailed implementation tasks

Discussion Points:

- The team reviewed the implementation timeline and confirmed the main tasks for January.
- The solar and battery system assembly was discussed, especially the correct connection between the solar panel, MPPT charge controller, battery, fuse, and switch.
- The team discussed how to test each subsystem separately before full system integration.
- The software and hardware teams agreed to test the motor control and sensor systems separately before combining them.

Follow-up of the Last Meeting:

- 3D mount designs were prepared.
- Software environment installation started.
- Ordered parts were being tracked.

Decisions Taken:

- Start with power system assembly and basic safety wiring.
- Test each subsystem separately to reduce integration problems.

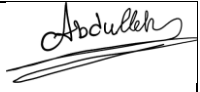
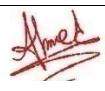
- Document all implementation steps for Chapter 4 of the final report.

Actions to do before next meeting:

- Assemble and test the solar charging connection.
- Prepare the battery, fuse, and safety switch wiring.
- Continue software setup and sensor testing preparation.

Date:	15/01/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Assemble the solar charging system
- Test battery charging and protection circuit
- Review sensor mounting progress
- Discuss initial motor control test

Discussion Points:

- The team assembled the solar charging system using the solar panel, MPPT charge controller, battery, fuse, and safety switch.
- The battery charging process was checked to make sure the MPPT controller regulates the voltage and current correctly.
- The team discussed the importance of keeping the wiring secure during testing.
- The LiDAR and camera mounts were reviewed and prepared for printing.
- The motor driver and Arduino connection were discussed for the next stage of vehicle movement testing.

Follow-up of the Last Meeting:

- Power system wiring was prepared.
- Safety components were added to the circuit plan.

Decisions Taken:

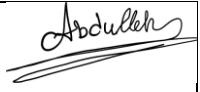
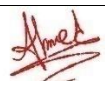
- The solar charging system will be tested separately before connecting it to the complete prototype.
- The battery will act as the main power source during vehicle operation.
- Continue with motor driver and Arduino testing in the next stage.

Actions to do before next meeting:

- Complete the solar charging test.
- Print and install the LiDAR and camera mounts.
- Prepare Arduino code for basic motor movement testing.

Date:	01/02/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Test throttle system
- Test steering servo mechanism
- Connect Arduino with motor driver
- Check basic vehicle movement

Discussion Points:

- The team connected the Arduino to the motor driver and tested the two DC motors at low speed.
- Forward movement and motor direction were checked to confirm correct wiring.
- The steering servo mechanism was tested separately to check the range of motion and mechanical response.
- The team discussed calibration of steering angles to avoid excessive movement.
- Safety during testing was emphasized, especially using the main switch to stop the system quickly if needed.

Follow-up of the Last Meeting:

- Solar charging system was tested separately.
- Sensor mounts were printed and prepared.
- Arduino motor control code was prepared.

Decisions Taken:

- Continue using low-speed testing until the steering and throttle systems become stable.
- Adjust the steering mechanism before full autonomous testing.

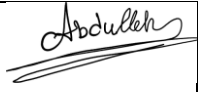
- Record calibration results for the implementation chapter.

Actions to do before next meeting:

- Complete steering calibration.
- Improve motor control code.
- Start testing LiDAR and camera individually.

Date:	20/02/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Test LiDAR sensor
- Test wide-angle camera
- Review ROS2 software setup
- Discuss communication between laptop and Arduino

Discussion Points:

- The LiDAR was connected and tested to confirm that scan data can be received correctly.
- The wide-angle camera was tested to confirm that the video feed is clear and suitable for human detection.
- The team discussed the ROS2 system structure and how sensor data will be processed.
- Communication between the laptop and Arduino was tested to send movement commands from software to hardware.
- The team discussed using the laptop as the main processing unit for the final prototype because it was more practical for testing and debugging.

Follow-up of the Last Meeting:

- Steering calibration was improved.
- Motor control code was tested at low speed.

Decisions Taken:

- Use the laptop to run ROS2 and sensor processing during final testing.
- Use Arduino as the low-level controller for motors and steering.

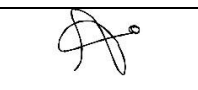
- Continue testing sensors with the throttle system.

Actions to do before next meeting:

- Combine LiDAR data with basic stop logic.
- Test camera-based human detection.
- Improve serial communication between laptop and Arduino.

Date:	12/03/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Integrate sensors with vehicle movement
- Test obstacle detection response
- Test human detection stopping function
- Prepare for full system field testing

Discussion Points:

- The team connected the LiDAR detection output with the throttle control to test automatic stopping when an obstacle is detected.
- The camera system was tested for human detection, and the software was adjusted to stop the vehicle when a human is detected.
- Several trials were performed to check if the vehicle stops correctly during movement.
- The team discussed possible delays between sensor detection and motor stopping.
- The wiring and mounted components were checked to make sure the system is stable before full field testing.

Follow-up of the Last Meeting:

- LiDAR and camera were tested individually.
- Communication between laptop and Arduino was improved.

Decisions Taken:

- Start preparing final validation experiments.
- Use repeated trials to measure speed, reaction time, stopping distance, and destination accuracy.

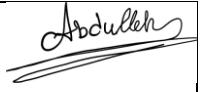
- Record all results clearly for Chapter 5.

Actions to do before next meeting:

- Prepare testing tools and measurement setup.
- Perform full system movement testing.
- Collect photos and data for the report.

Date:	30/03/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Conduct final validation experiments
- Record autonomous system testing results
- Record solar charging efficiency results
- Compare results with project requirements

Discussion Points:

- The team performed the vehicle speed test using a measured straight path and recorded the travel time for several trials.
- Object detection and reaction time tests were performed to verify the system response.
- Stopping distance tests were conducted to check whether the vehicle stops safely after detecting an obstacle.
- Destination accuracy testing was performed to evaluate the autonomous movement performance.
- The solar charging efficiency test was conducted by measuring solar input voltage/current and battery output voltage/current.
- The team discussed how to present the results in tables and compare them with the target requirements.

Follow-up of the Last Meeting:

- Full system integration was completed.
- Testing setup was prepared.

Decisions Taken:

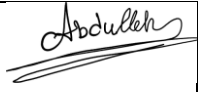
- Use the collected experimental results in Chapter 5.
- Mark each experiment as PASS, PARTIAL PASS, or FAIL based on comparison with the requirements.
- Start writing the discussion and conclusion sections.

Actions to do before next meeting:

- Complete all testing tables and calculations.
- Add testing photos and captions to the report.
- Start preparing final presentation content.

Date:	14/05/2026
Team:	38
Project Title:	Automated vehicle system powered by solar energy

Attendees:

Member 1:	Abdullah Mohammed Binsalman	Signature:	
Member 2:	Mohammed Abdullah Al-Baiti	Signature:	
Member 3:	Ahmed Salem Al-Harthy	Signature:	
Advisor:	Dr. Sultan Alghamdi	Signature:	
Customer:	Dr. Sultan Alghamdi	Signature:	

Agenda:

- Review final report
- Review final presentation
- Confirm demonstration video and prototype documentation
- Finalize appendix sections and project management content

Discussion Points:

- The team reviewed the final report chapters and checked formatting, figures, tables, and captions.
- The implementation chapter was reviewed to make sure it clearly explains the hardware assembly, power system, steering system, throttle system, sensors, and software architecture.
- The validation chapter was reviewed to confirm that all test results are included and explained.
- The team discussed the conclusion and future work, including improving the system with better sensors, embedded onboard computer, larger solar panels, and more advanced navigation.
- The final presentation was reviewed and divided between team members.
- The team checked that the appendices include meeting minutes, project management, and ethical responsibility sections.

Follow-up of the Last Meeting:

- Validation experiments were completed.
- Testing tables, calculations, and figures were added to the report.
- Final presentation draft was prepared.

Decisions Taken:

- The final report is ready for final checking and submission.
- The presentation content is approved after minor edits.
- The team will prepare for the final project demonstration and presentation.

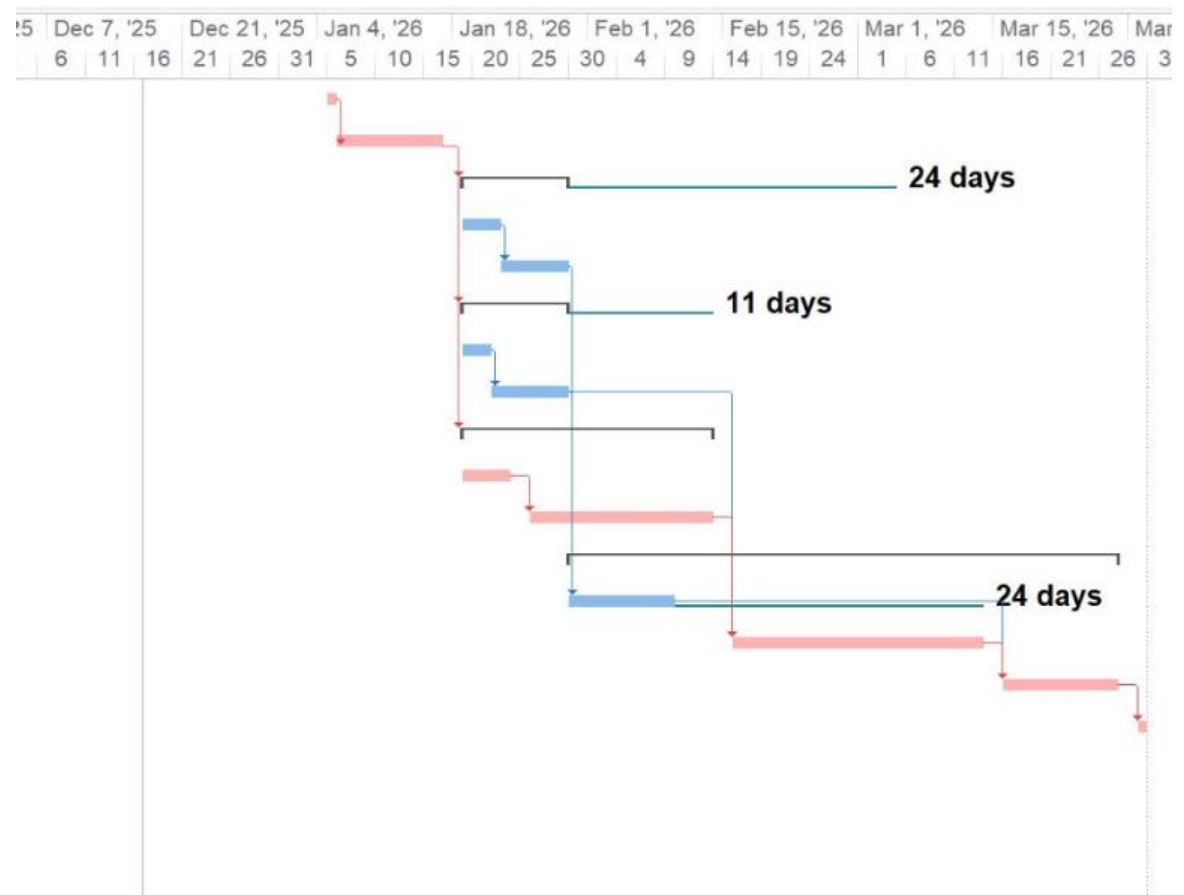
Actions to do before next meeting:

- Submit the final report after final proofreading.
- Practice the final presentation.
- Prepare the prototype and demonstration video for final evaluation.

APPENDIX – C: USE OF PROJECT MANAGEMENT TECHNIQUES

Member Responsibility	ID	Task Name	Start Day	End Day	Predecessors
ALL	T1	Ordering Parts and Arriving	5/1/2026	Day 9	-
M2	T2	Solar & Battery Assembly	Day 10	Day 14	T1
M2	T3	Power Distribution Wiring	Day 15	Day 19	T2
M3	T4	Actuator Mounting & Wiring	Day 10	Day 13	T1
M3	T5	Pixhawk & GPS Calibration	Day 14	Day 20	T4
M1	T6	Jetson Nano & ROS Setup	Day 10	Day 15	T1
M1	T7	Autonomous Logic Coding	Day 16	Day 29	T6
M2 and M3	T8	Physical Integration	Day 21	Day 29	T3, T5
ALL	T9	Software Integration	Day 30	Day 50	T7, T5
ALL	T10	Full System Field Testing	Day 51	30/3/2026	T8, T9

	Tasks	Task Mode	Name	Duration	Start	Finish	Predecessors
1			Start	1 day	Mon 1/5/26	Mon 1/5/26	
2	T1		Buying Compoents and Arriving	9 days	Tue 1/6/26	Fri 1/16/26	1
3			Power Assembly	9 days	Mon 1/19/26	Thu 1/29/26	2
4	T2		Solar and Battery Assembly	4 days	Mon 1/19/26	Thu 1/22/26	
5	T3		Power Distribution Wiring	5 days	Fri 1/23/26	Thu 1/29/26	4
6			Control Assembly	9 days	Mon 1/19/26	Thu 1/29/26	2
7	T4		Actuator Mounting and Wiring	3 days	Mon 1/19/26	Wed 1/21/26	
8	T5		Pixhawk and GPS Setup	6 days	Thu 1/22/26	Thu 1/29/26	7
9			Software Programming	20 days	Mon 1/19/26	Fri 2/13/26	2
10	T6		Jestson Nano and ROS Setup	5 days	Mon 1/19/26	Fri 1/23/26	
11	T7		Autonoumus Coding	15 days	Mon 1/26/26	Fri 2/13/26	10
12			Buliding Prototype	41 days	Fri 1/30/26	Fri 3/27/26	
13	T8		Physical Intergration	7 days	Fri 1/30/26	Mon 2/9/26	8,5
14	T9		Software Intergration	20 days	Mon 2/16/26	Fri 3/13/26	8,11
15	T10		Testing Prototype	10 days	Mon 3/16/26	Fri 3/27/26	13,14
16			Finish	1 day	Mon 3/30/26	Mon 3/30/26	15



APPENDIX – D: RECOGNITION OF ETHICAL AND PROFESSIONAL RESPONSIBILITY

This project follows the **IEEE Code of Ethics**, which requires engineers to prioritize public safety, be honest in reporting results, maintain technical competence, and disclose limitations that may affect system performance.

The main ethical issue in this project is the use of an experimental autonomous vehicle in a shared campus environment. Since the system is intended to detect obstacles and stop automatically, any sensing or control failure could create a safety risk for students and staff. According to the IEEE Code of Ethics, engineers must make decisions consistent with the safety, health, and welfare of the public and must promptly disclose any factors that could endanger people or the environment.

This ethical issue appears in the current prototype because the vehicle was tested at about 2.39 km/h, which is much lower than the 25 km/h design target, and some stopping-distance trials resulted in only a partial pass. In addition, the implemented design differs from the original baseline: the Jetson Orin Nano was replaced with a laptop, the Pixhawk with an Arduino, the ESC with a BTS motor driver, and an optical flow sensor was added. Under the IEEE Code, engineers must be honest and realistic in stating claims based on available data and must disclose limitations clearly.

Based on this analysis, the informed decision is that the prototype should only be operated under supervision and in a controlled environment. A human operator must remain present, the emergency stop must be accessible, and the system should not be presented as fully validated for unrestricted public use until further testing is completed. This decision is consistent with the IEEE requirement to protect public welfare and to undertake engineering tasks only with proper competence and full disclosure of limitations.